AMBA and AXI Connect Example

PWM Control

[*Reference*] ▪

Table of Contents

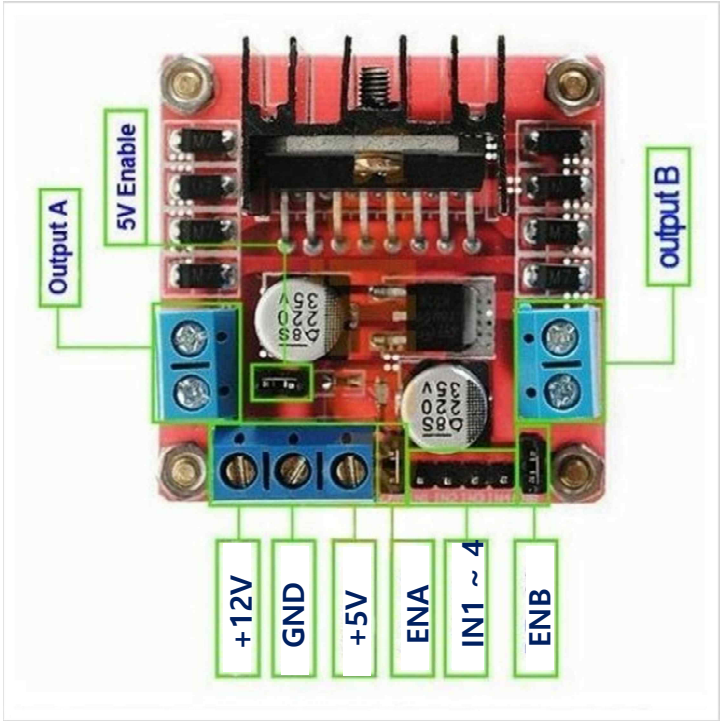
- SoC Peripheral RTC Design Project (프로젝트 최종 계획서 제출)

➢ 모터 구동을 위한 모듈 : *L298N*(모터 드라이버)[*1/2*]

- *L298N* 기능 : 1. DC, 스텝 모터의 방향 및 속도 제어
  2. H-브리지 구조: 모터를 양방향으로 제어가능. 즉, 모터의 회전방향을 제어하거나 정지
  3. 두개의 모터를 독립적으로 제어. 두개의 출력 핀 제공(Output A, Output B)
  4. 주로 PWM신호를 사용하여 모터 속도를 조절

- *L298N* 핀 구성 :

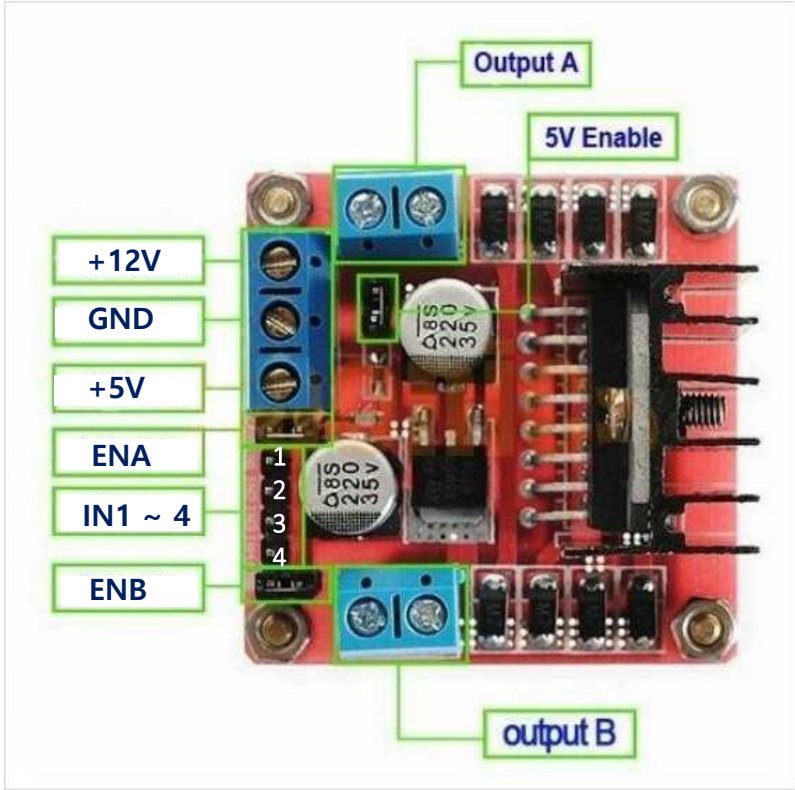
1. *IN1, IN2* : 첫 번째 모터의 회전 방향 제어
2. *IN3, IN4* : 두 번째 모터의 회전 방향 제어
3. *OUTA* : 첫 번째 모터의 출력 핀(모터와 연결)
4. *OUTB* : 두 번째 모터의 출력 핀(모터와 연결)
5. *ENA* : 첫 번째 모터의 속도 제어를 위한 PWM 입력 핀
6. *ENB* : 두 번째 모터의 속도 제어를 위한 PWM 입력 핀
7. *+12V* : 모터 전원 입력(+5V ~ 35V)
8. *+5V* : 내부 논리 회로 전원 입력 (+5V)
9. *GND* : 공통 접지



➢ 모터 구동을 위한 모듈 : *L298N*(모터 드라이버)[*2/2*]

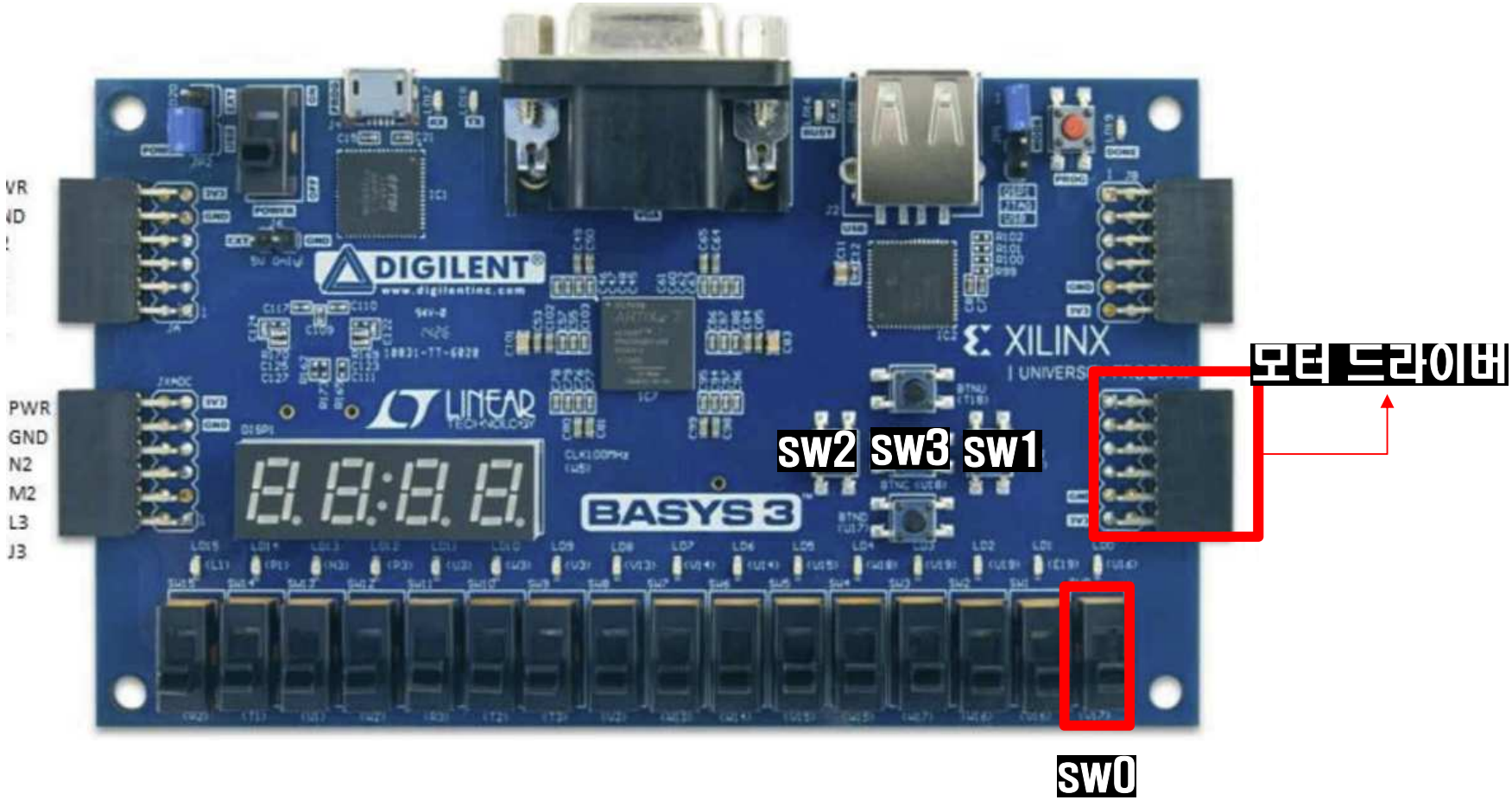
- 모터 구동을 위한 결선

1. ***IN1, IN2***: ***DC MOTER*** 결선(***DC MOTOR***방향제어)
2. ***OUTA***: ***DC MOTER*** 결선 (***MOTOR*** 출력 핀)
3. ***ENA*** : ***DC MOTER*** 결선(속도제어)
4. ***+12V*** : 외부전원 +5V인가
5. ***+5V*** : 결선X
6. ***GND*** : 외부전원과 ***BASYS3***을 공통 접지



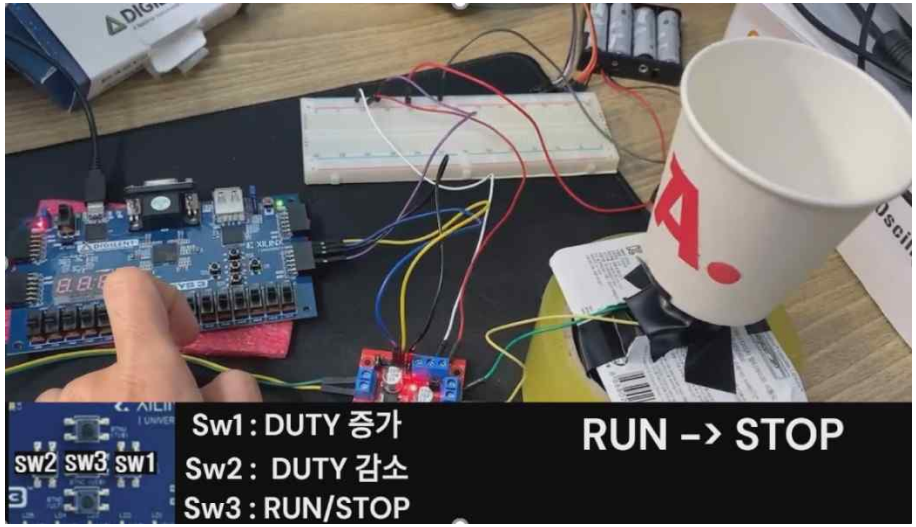
➢ Design Goal

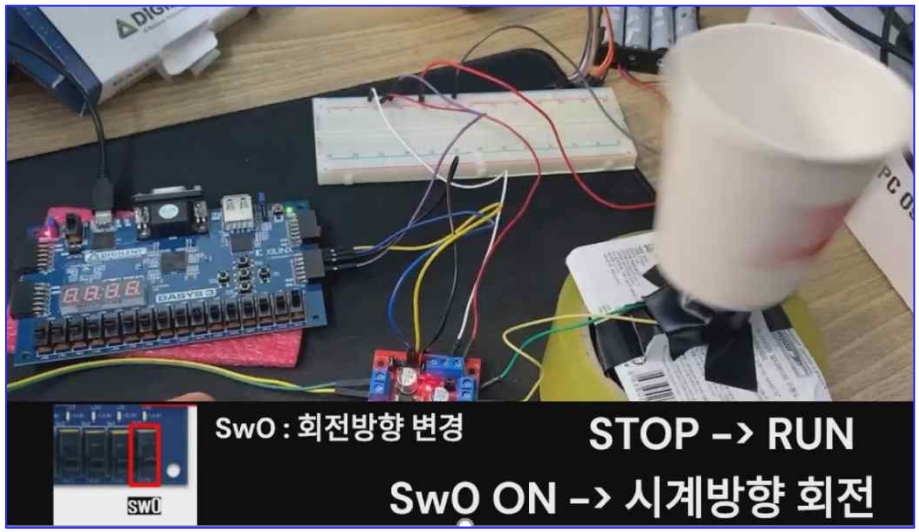
- GPIO를 활용한 PWM Control 실습

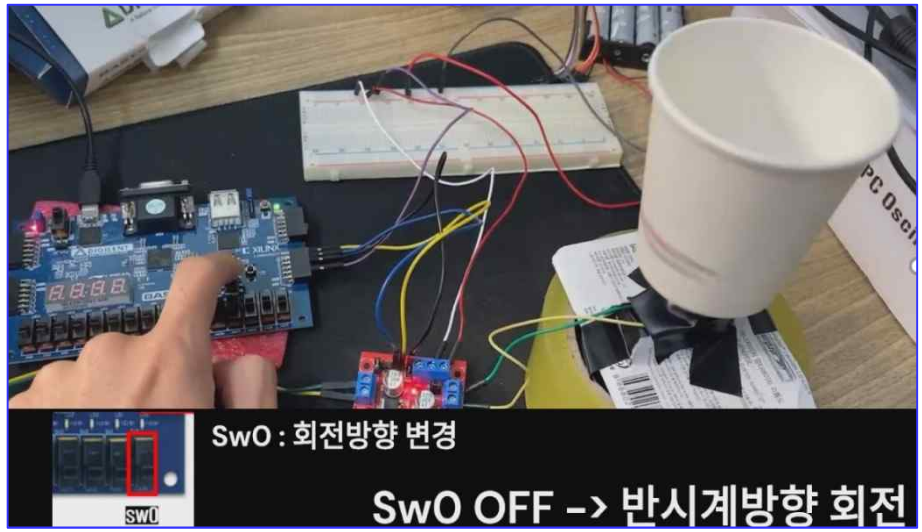


- SW0 : 기본상태에서 0b01, 토글일때 0b10
- SW1 : PWM 듀티비 증가 SW2 : PWM 듀티비 감소 SW3 : RUN / STOP

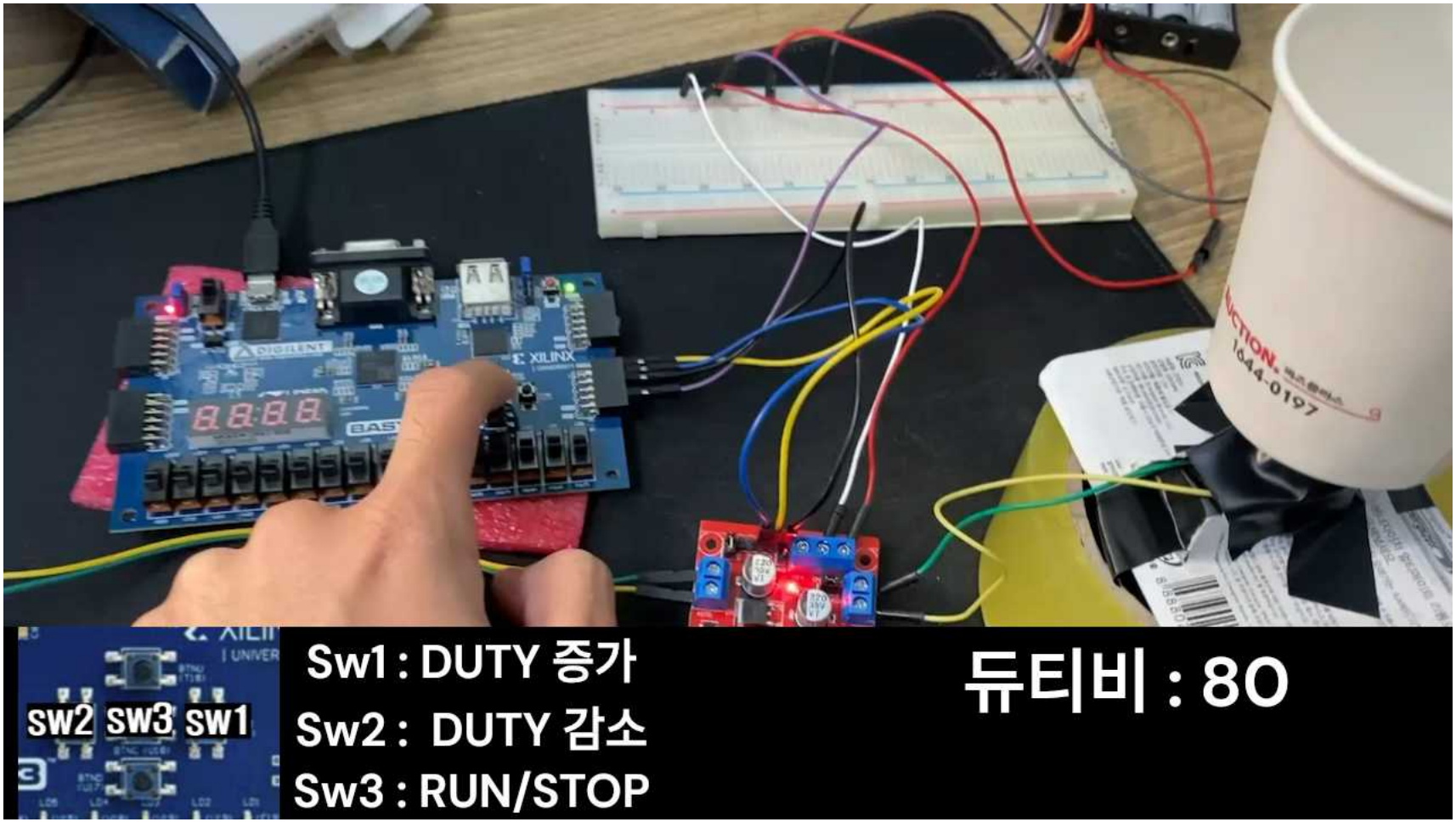
➢ *AXI Connect ➔ PWM Control ➔ Program Device and Implementation ➔ Check The Result : Image*







- *AXI Connect ➔ PWM Control ➔ Program Device and Implementation ➔* ***Check The Result : Video***

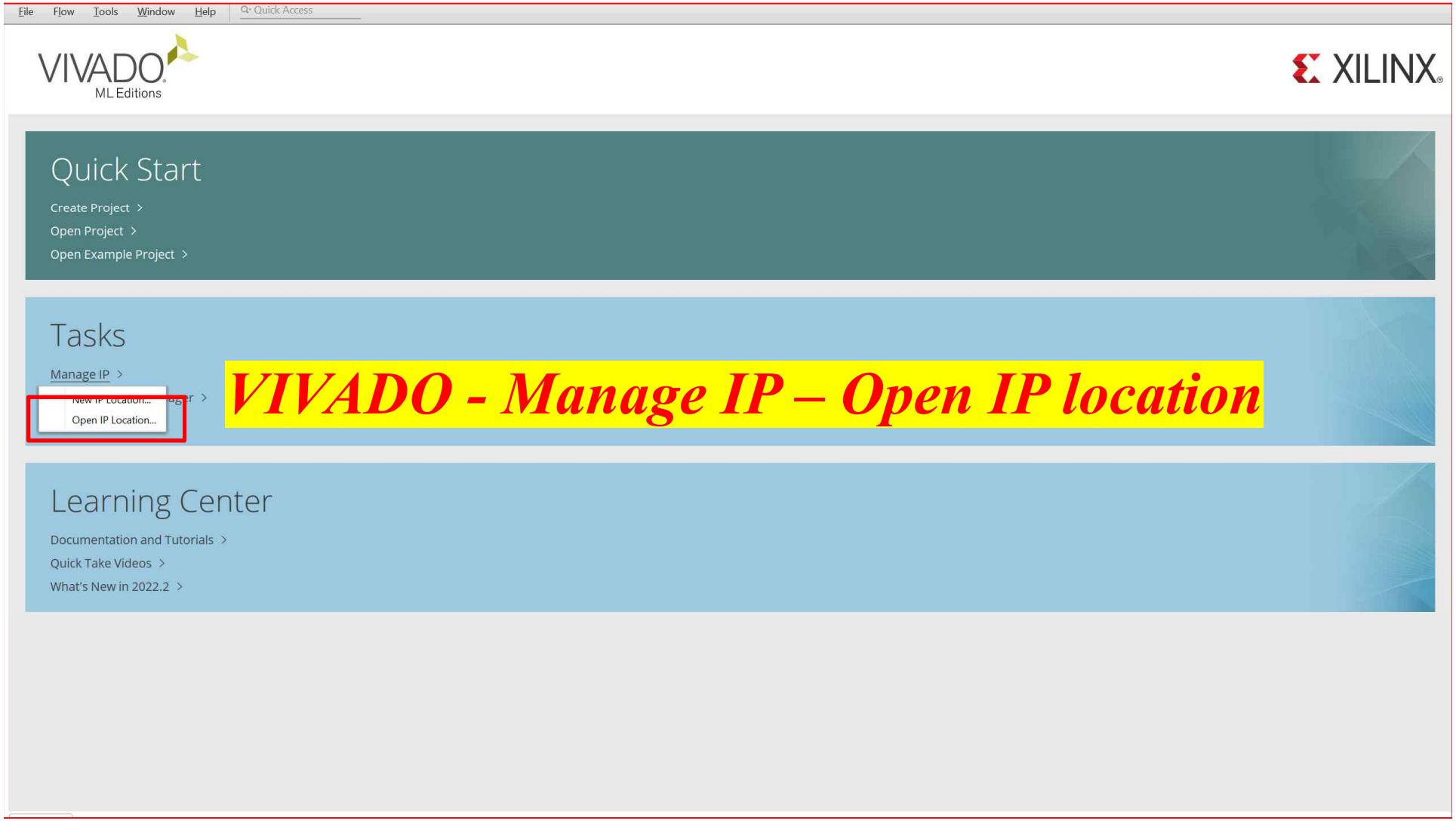


AMBA and AXI Connect Example

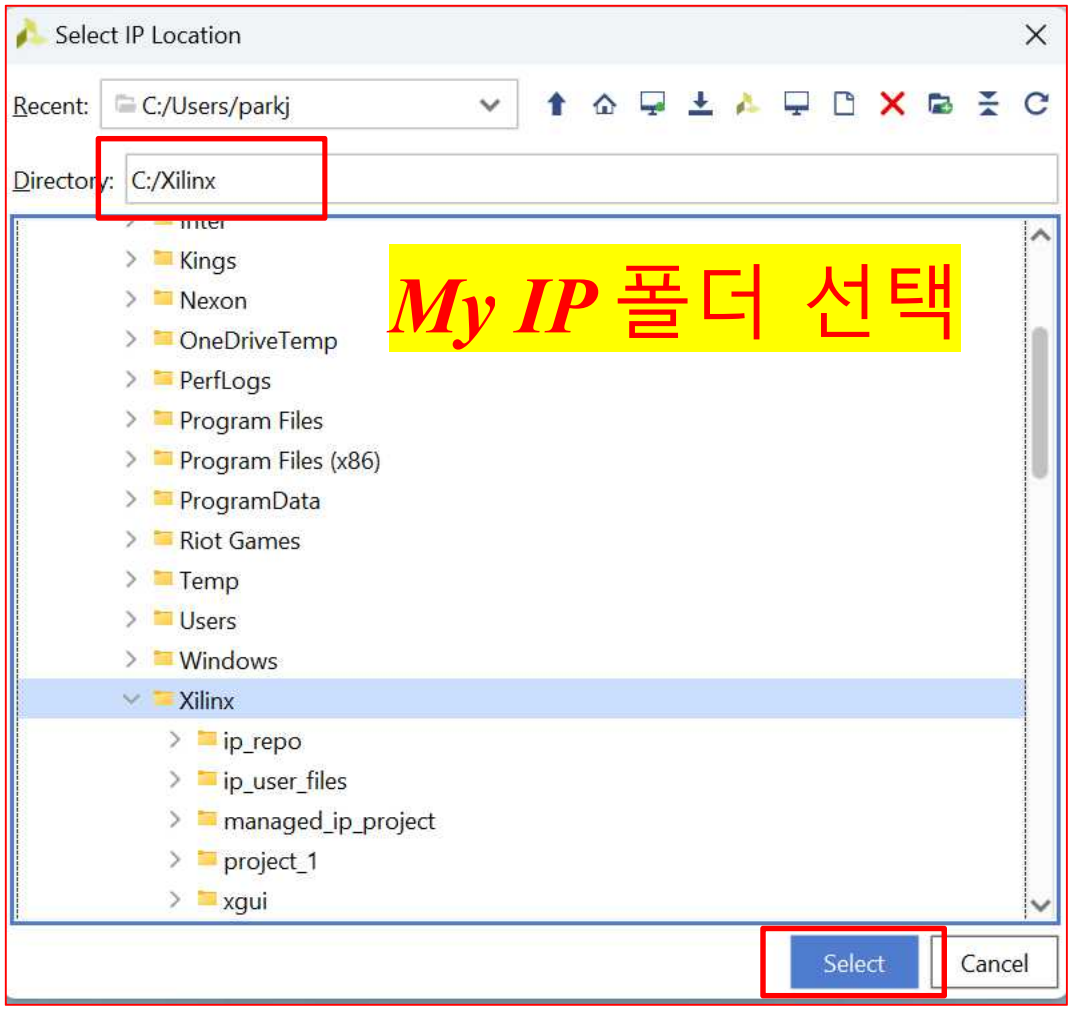
PWM Control

[Create IP]

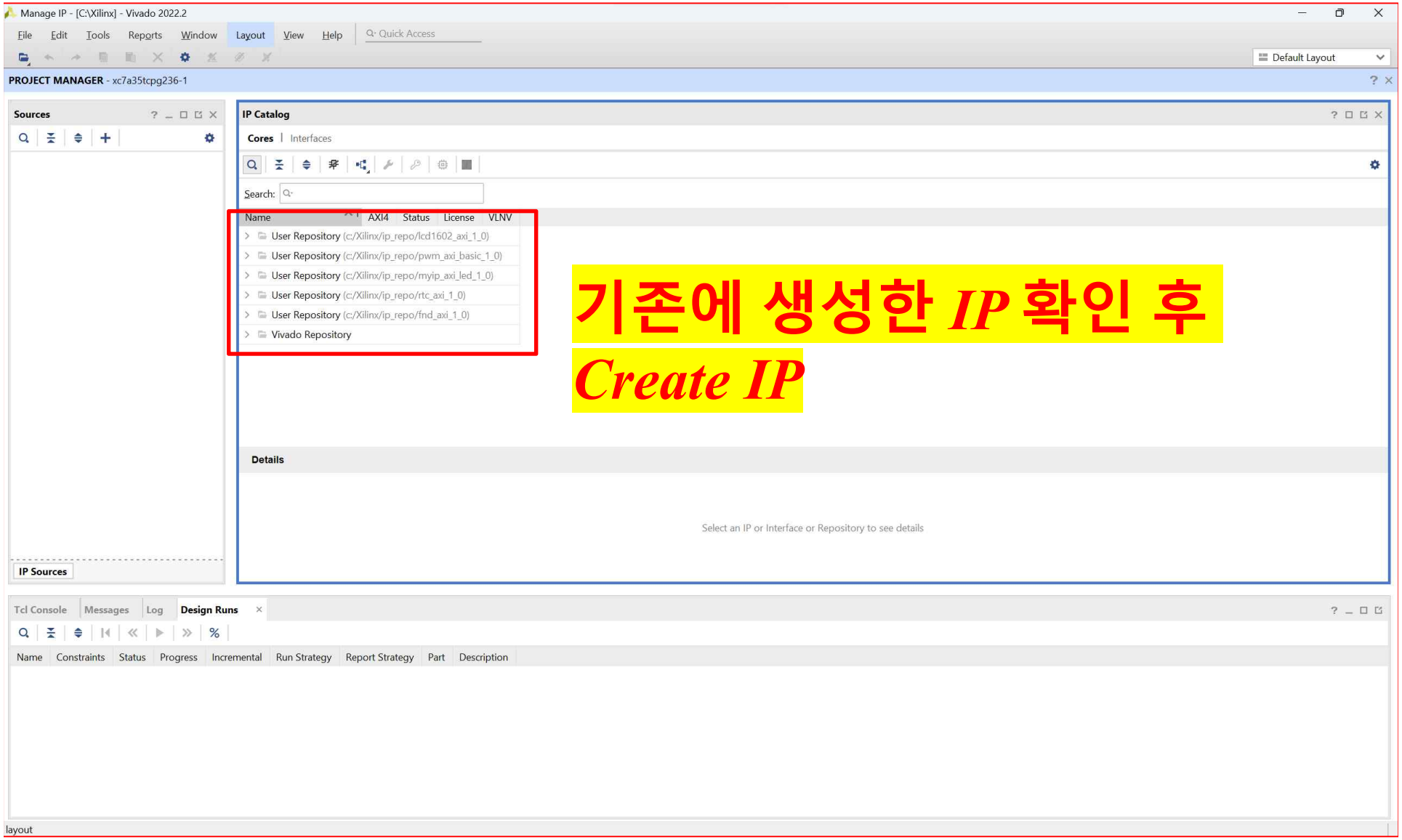
- ***AXI Connect ➔ PWM Control***



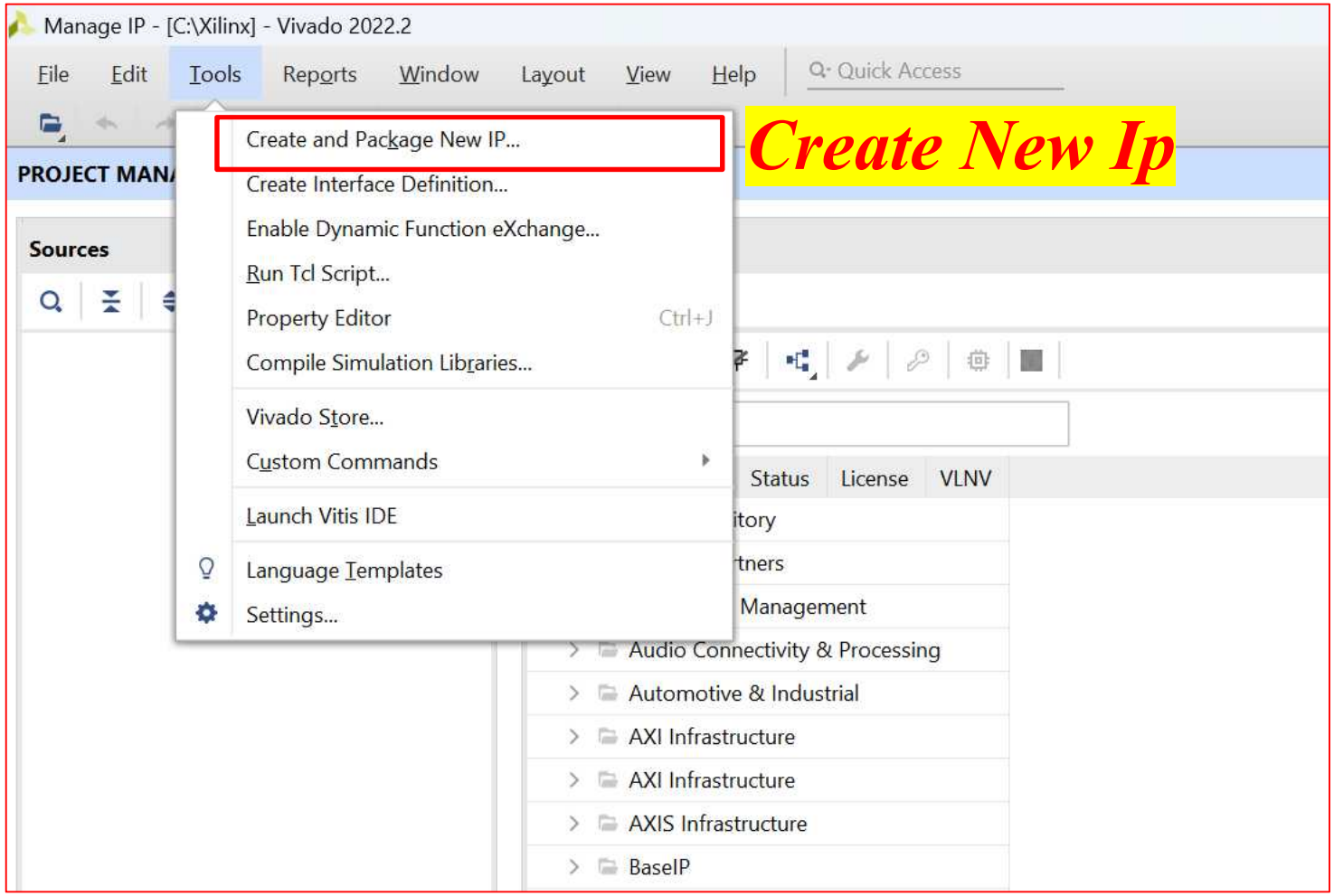
- AXI Connect ➔ PWM Control



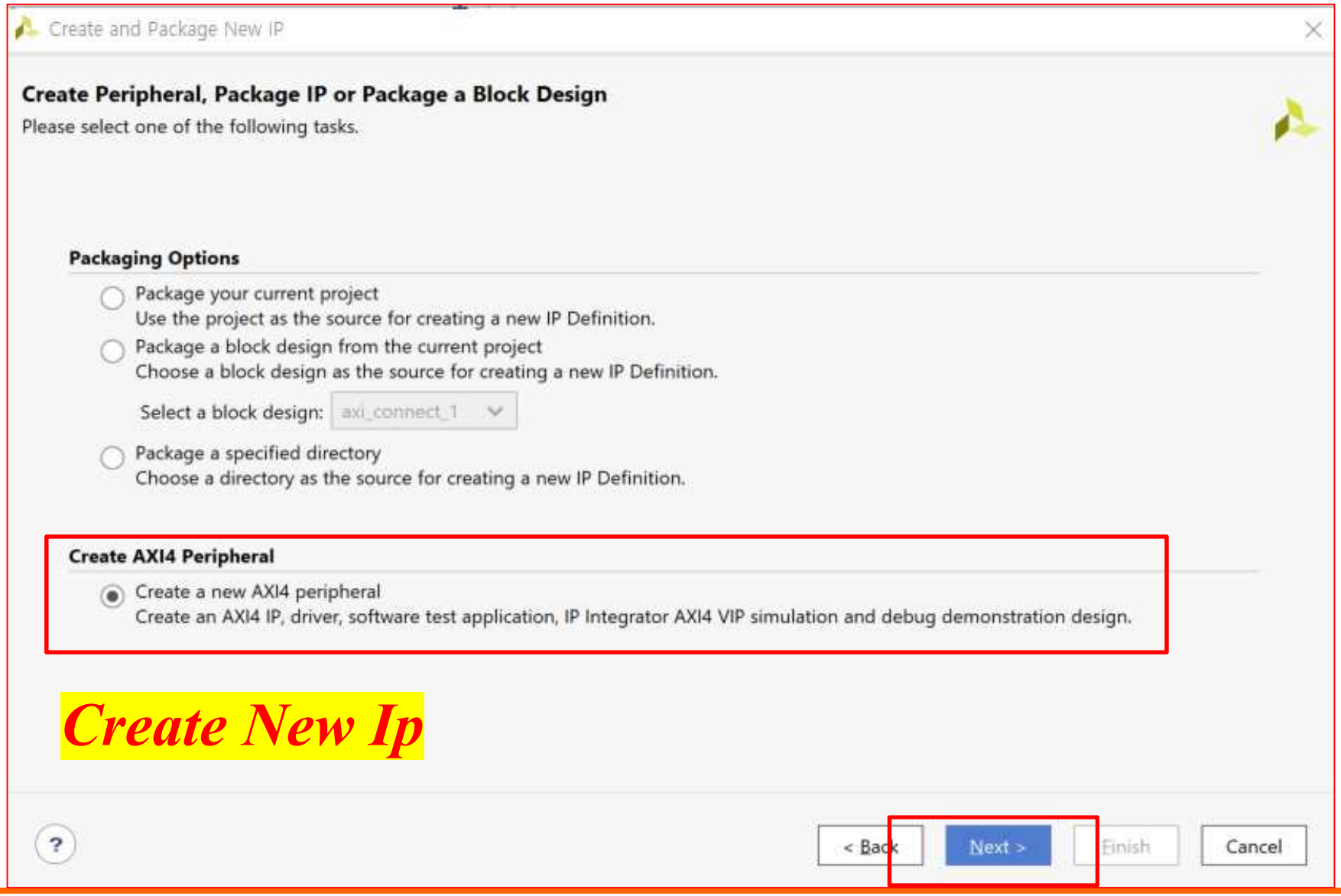
➢ *AXI Connect ➔ PWM Control*



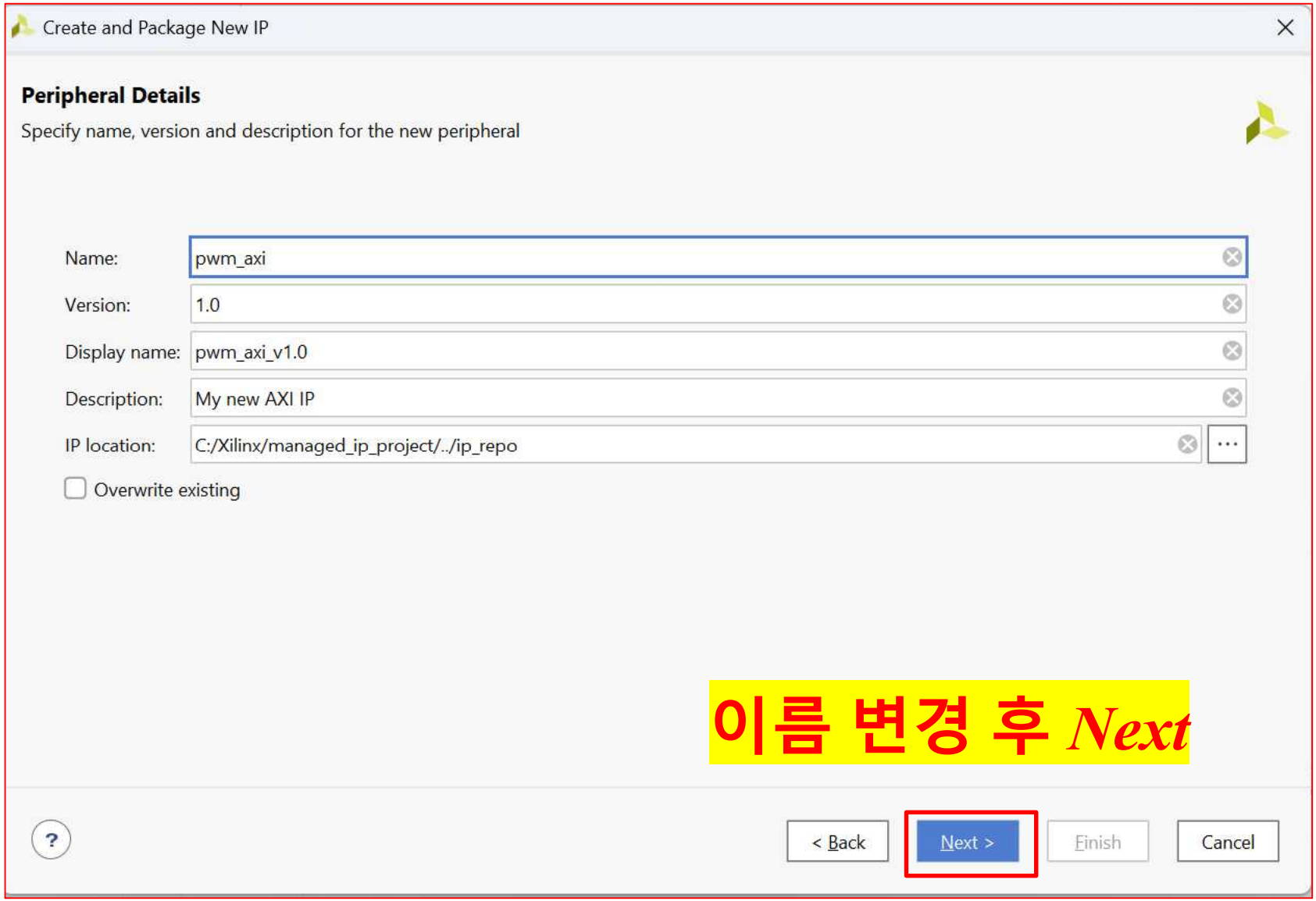
- ***AXI Connect ➔ PWM Control***



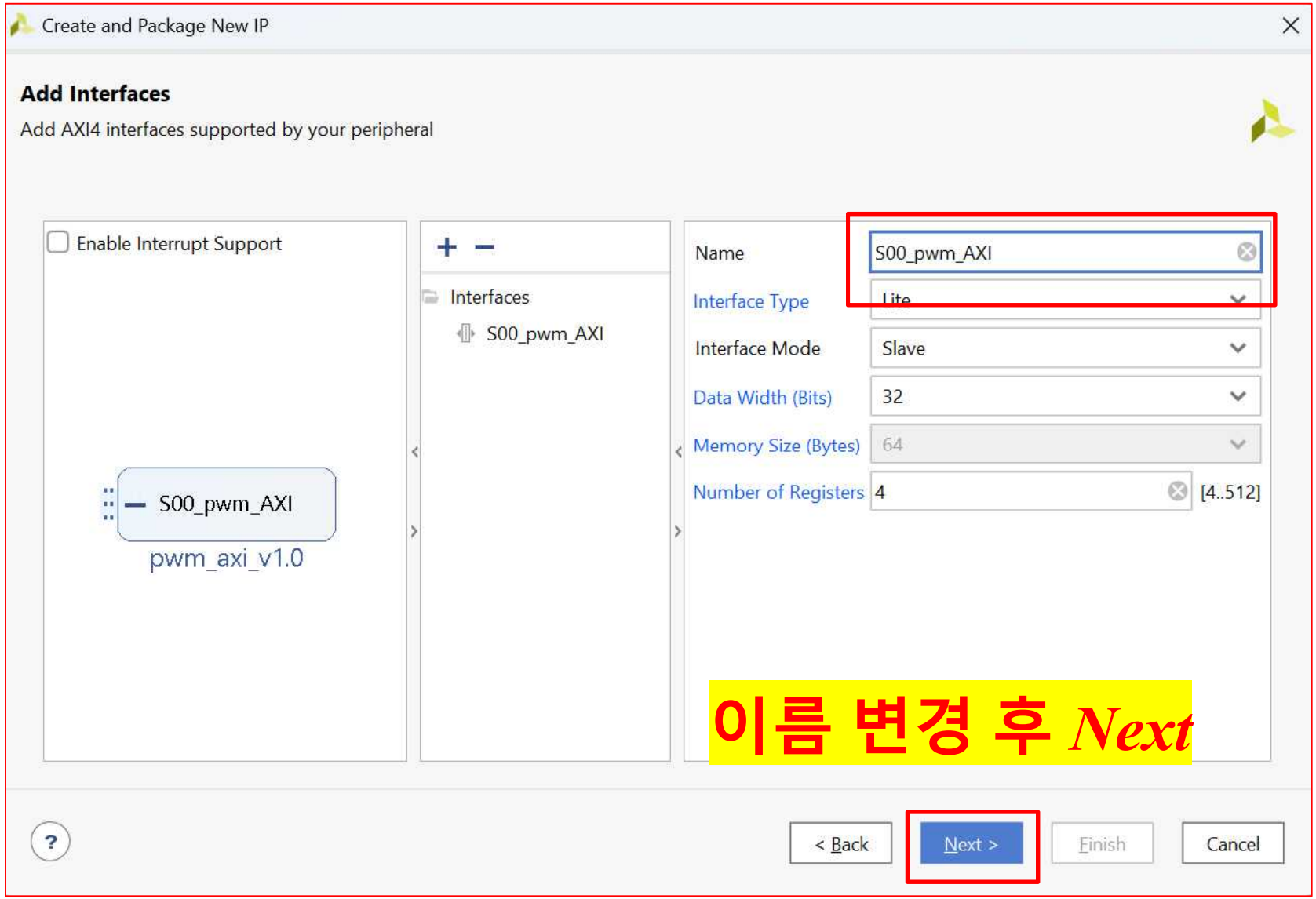
- AXI Connect ➔ PWM Control



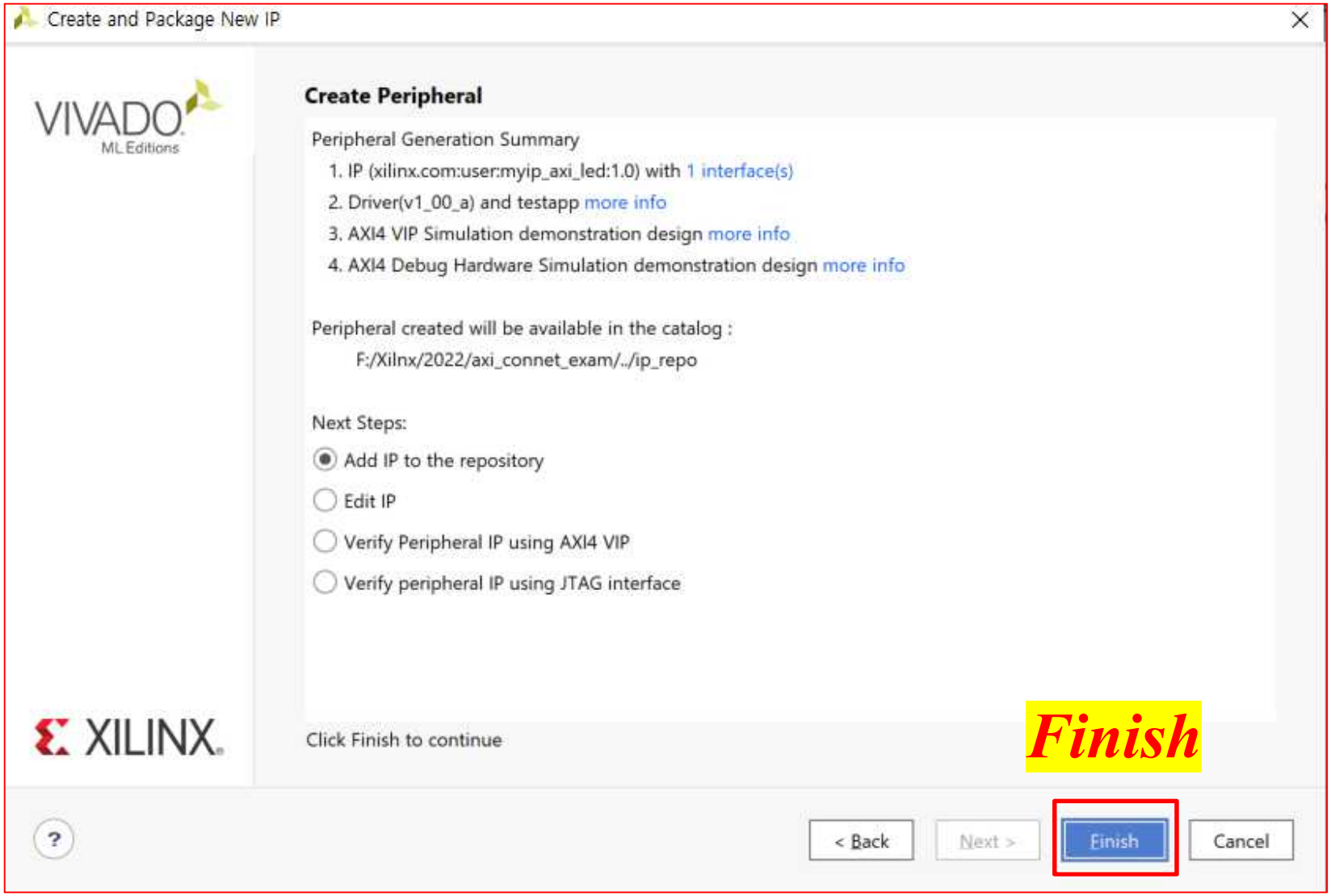
➢ AXI Connect ➔ PWM Control



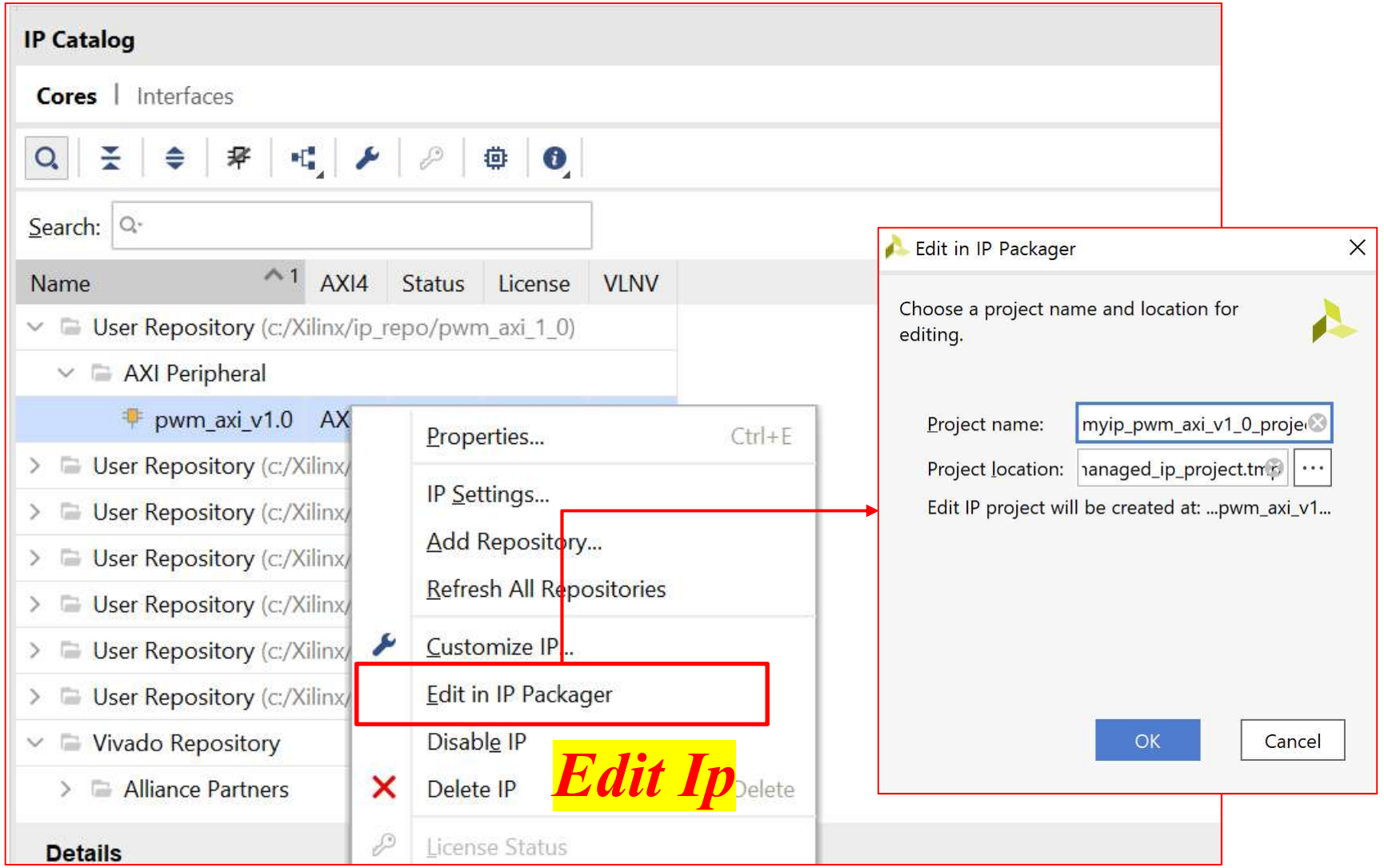
➢ AXI Connect ➔ PWM Control



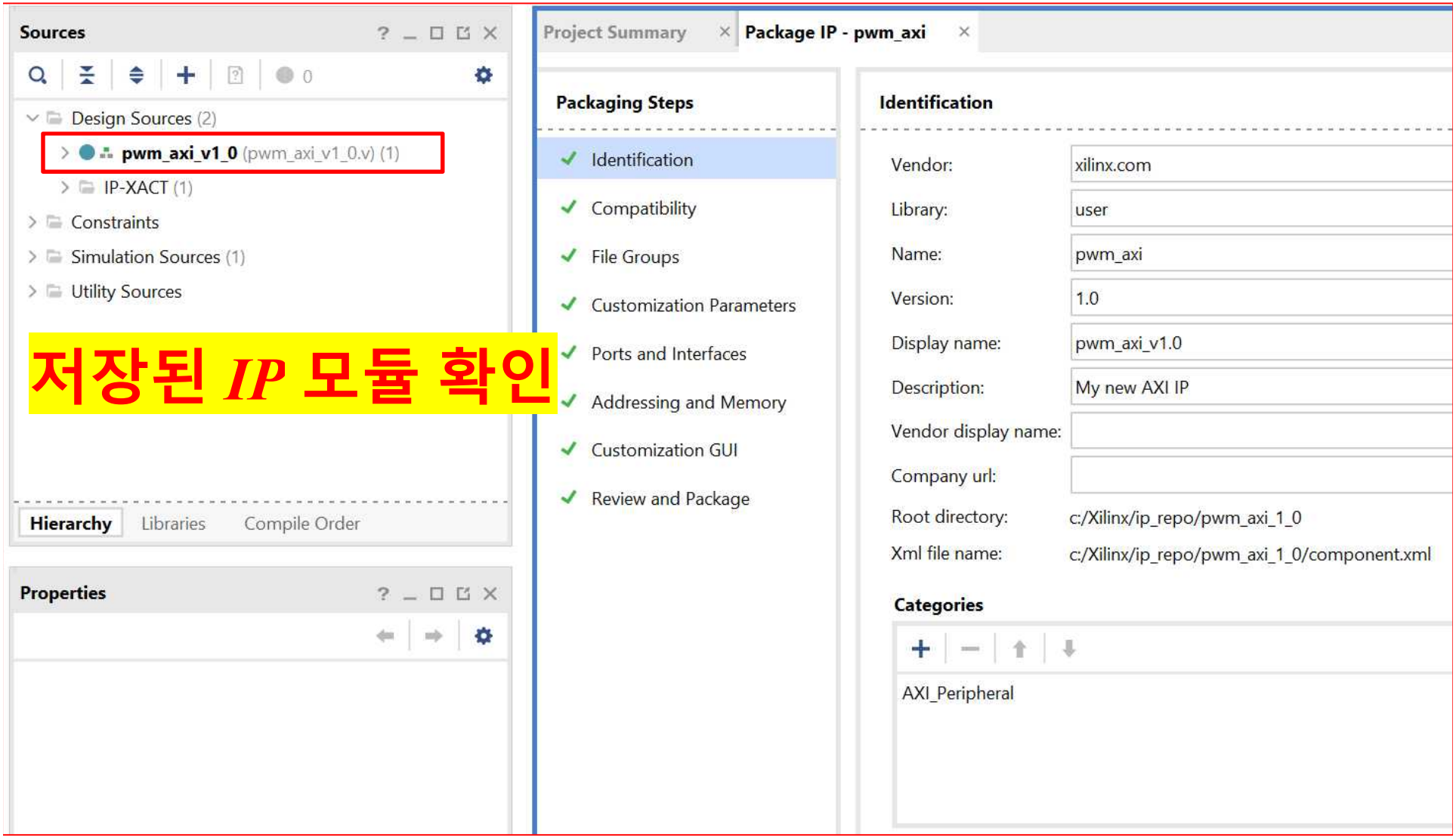
➢ AXI Connect ➔ PWM Control



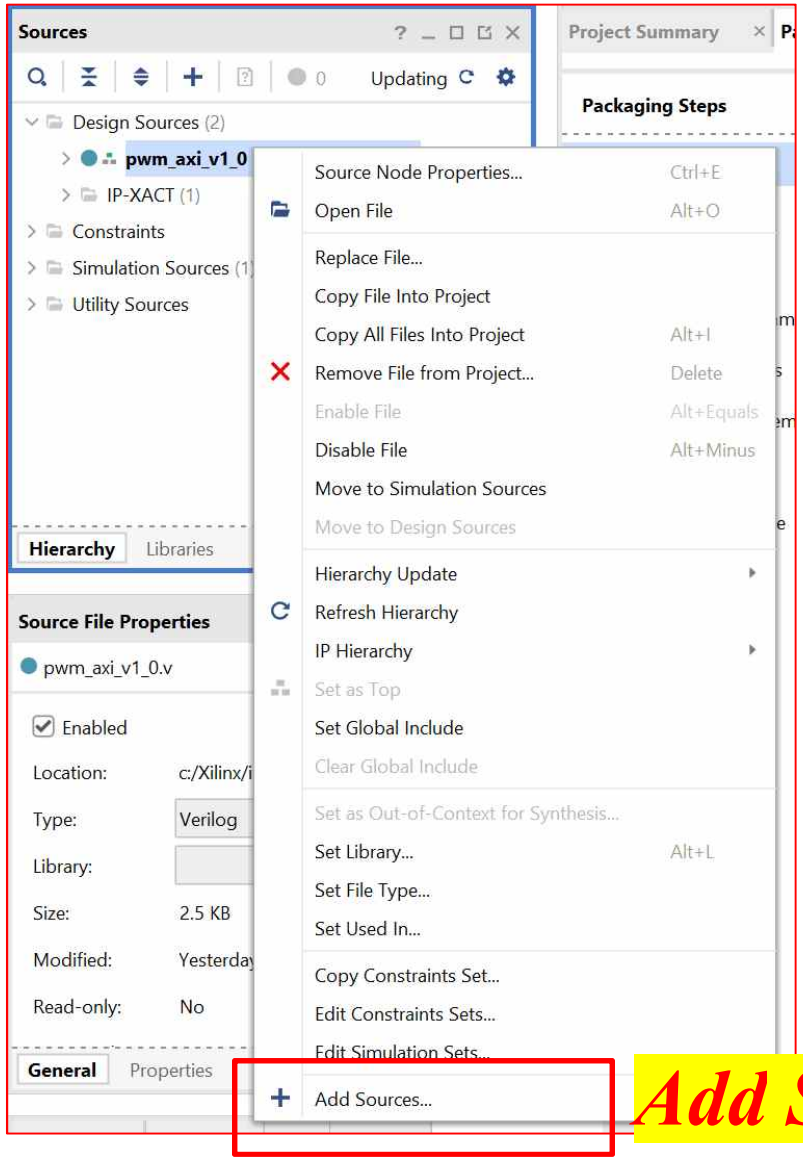
➢ AXI Connect ➔ PWM Control



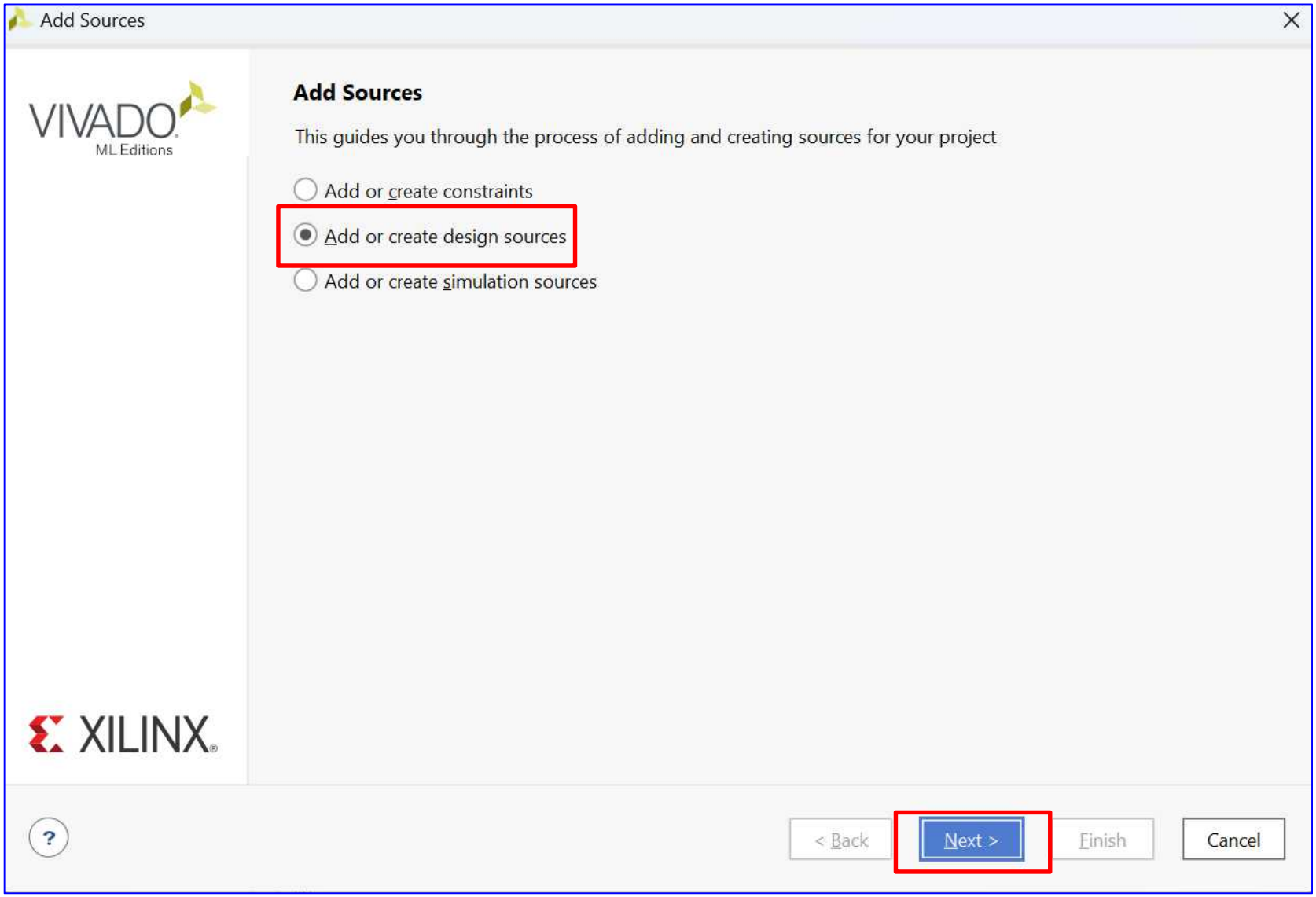
➢ AXI Connect ➔ PWM Control



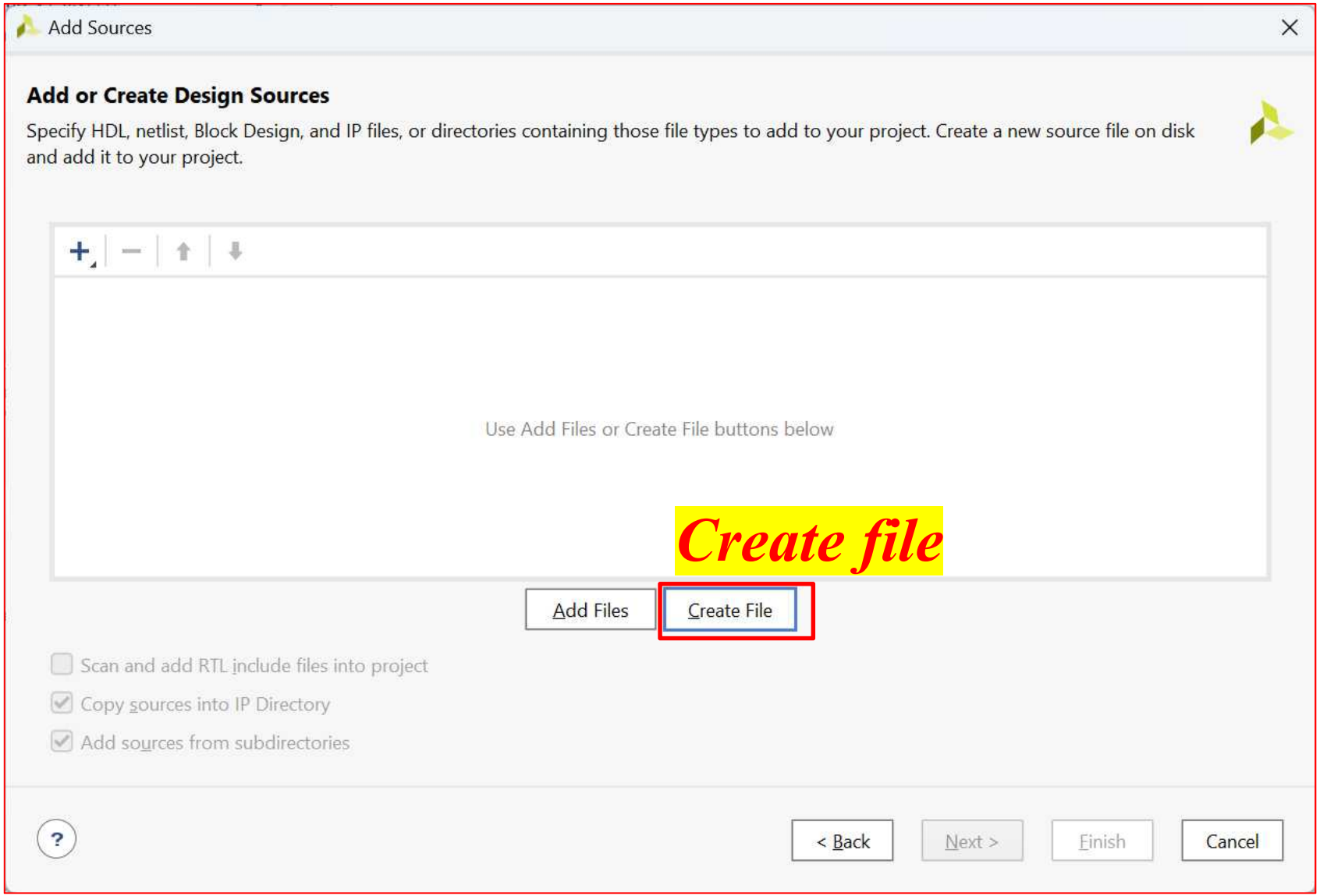
➢ AXI Connect ➔ PWM Control



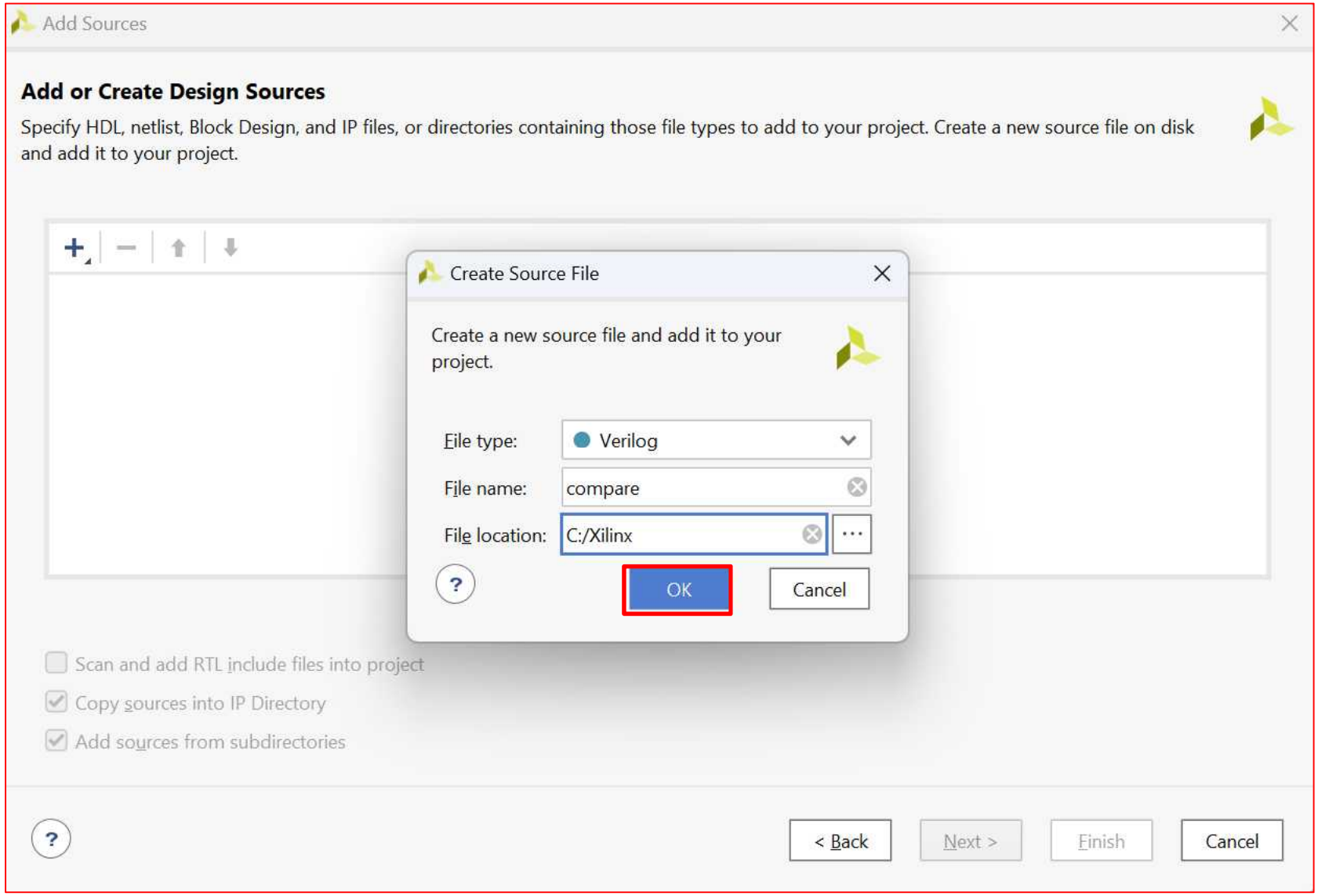
- ***AXI Connect ➔ PWM Control***



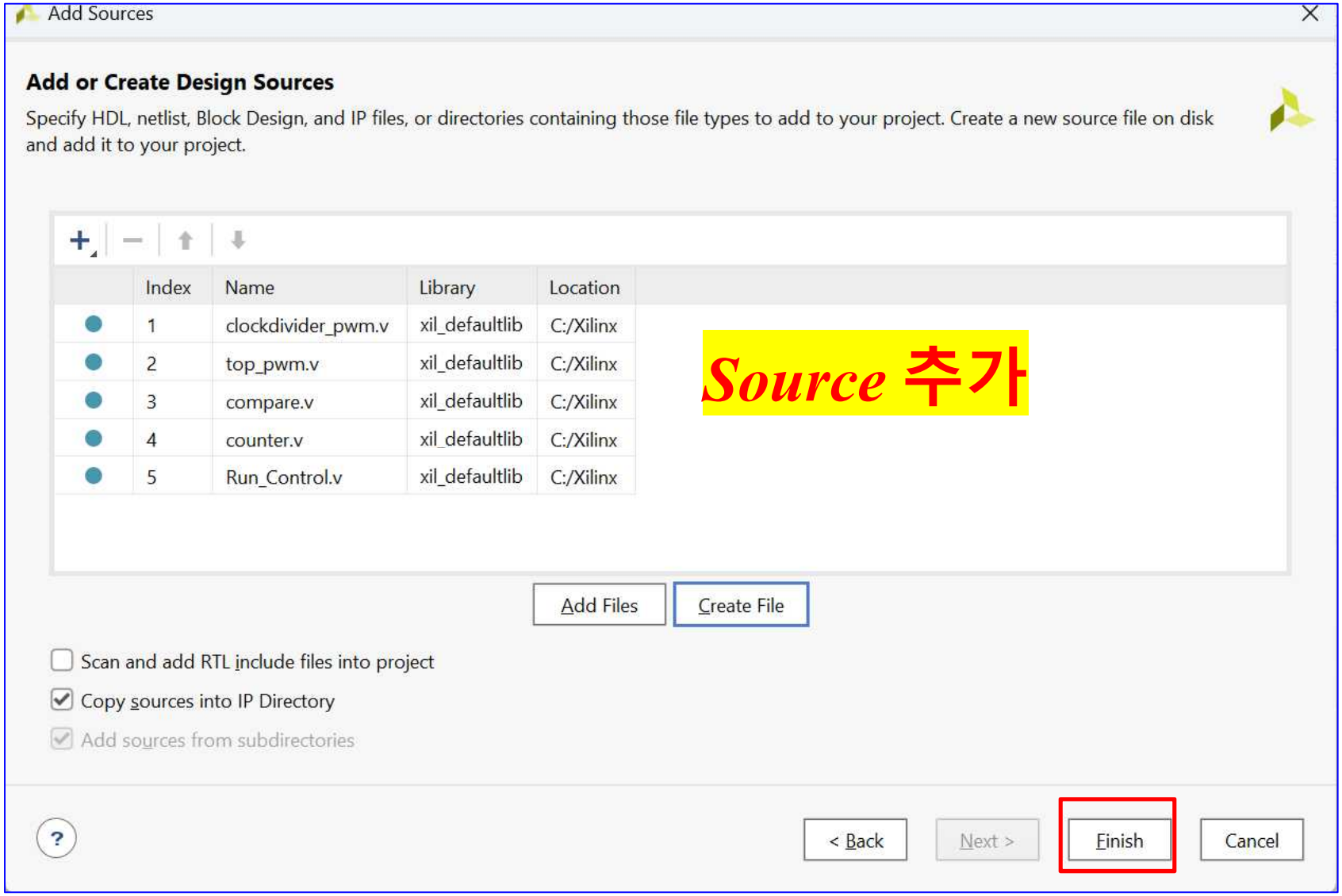
➢ AXI Connect ➔ PWM Control



➢ AXI Connect ➔ PWM Control



➢ AXI Connect ➔ PWM Control



- AXI Connect ➔ PWM Control
  ➔ top_pwm

```
1.  `timescale 1ns / 1ps
2.  module top_pwm(
3.  reset, clk, i_ocr, i_en, i_dir, o_dir,
    o_motor_pwm, w_counter, w_clkout
4.  );
5.  input reset;
6.  input clk;
7.  input [7:0] i_ocr;
8.  input i_en;
9.  input [1:0] i_dir;
10. output [1:0] o_dir;
11. output o_motor_pwm;
12. output [7:0] w_counter;
13. output w_clkout;
14. clockdivider_pwm U0 (
15.     .inclk(clk),
16.     .reset(reset),
17.     .clk_out(w_clkout)
18. );
```

```
19. counter U1 (
20.     .inclk(w_clkout),
21.     .reset(reset),
22.     .out_counter(w_counter)
23. );
24.   wire [7:0] w_counter;
25.   wire [7:0] i_ocr;
26.   wire [1:0] i_dir;
27.   wire [1:0] o_dir;
28.  compare U2 (
29.     .count(w_counter),
30.     .i_ocr(i_ocr),
31.     .i_en(i_en),
32.     .out_pwm(o_motor_pwm)
33. );
33. Run_Control U3(
34.   .i_direction(i_dir),
35.   .o_direction(o_dir),
36.   .i_en(i_en)
37.   );
38. endmodule
```

- AXI Connect ➔ PWM Control
  - ➔ clockdivider_pwm

```
1.  `timescale 1ns / 1ps
2.  module clockdivider_pwm (
3.      input  inclk,
4.      input  reset,
5.      output reg  clk_out
6.  );
7.    reg [25:0] cnt = 0;
8.    always @(posedge inclk, negedge reset) begin
9.      if (~reset) begin
10.       cnt <= 0;
11.       clk_out= 0;
12.     end else begin
13.       if (cnt == (49)) begin
14.         cnt <= 0;
15.         case(clk_out)
16.         1'b0 : clk_out <= 1;
17.         1'b1 : clk_out <= 0;
18.         endcase
19.     end else begin
20.         cnt <=  cnt + 1;
21.       end
22.     end
23.   end
24. endmodule
```

- AXI Connect ➔ PWM Control
  ➔ counter

axi_pwm_exam.xpr

```
1.  `timescale 1ns / 1ps
2.  module counter (
3.      input inclk,
4.      input reset,
5.      output [7:0] out_counter
6.  );
7.  reg [7:0] count=0;
8.  assign out_counter = count;
9.   always @(posedge inclk, negedge reset) begin
10.   if (~reset)    begin
11.         count <= 0;
12.      end else begin
13.
14.          if(count>= 99) begin
15.             count <= 0;
16.          end else begin
17.             count = count + 1;
18.               end
19.            end
20. end
21. endmodule
```

- AXI Connect ➔ PWM Control

  ➔ compare

```
1.  `timescale 1ns / 1ps

2.  module compare(
3.  count, i_ocr, out_pwm, i_en
4.      );
5.
6.  input [7:0] count;
7.  input [7:0] i_ocr;
8.  input i_en;
9.  output out_pwm;

10. reg pwm= 0;

11. assign out_pwm = pwm;
```

```
12. always @(*) begin

13.  if(i_en) begin  // i_en == 1

14.   if(count < i_ocr) begin
15.         pwm = 1'b1;
16. end else begin
17.         pwm = 1'b0;
18. end

19.  end else begin  // i_en == 0
20.           pwm = 1'b0;
21.        end
22.    end
23. endmodule
```

- AXI Connect ➔ PWM Control
  ➔ Run_Control

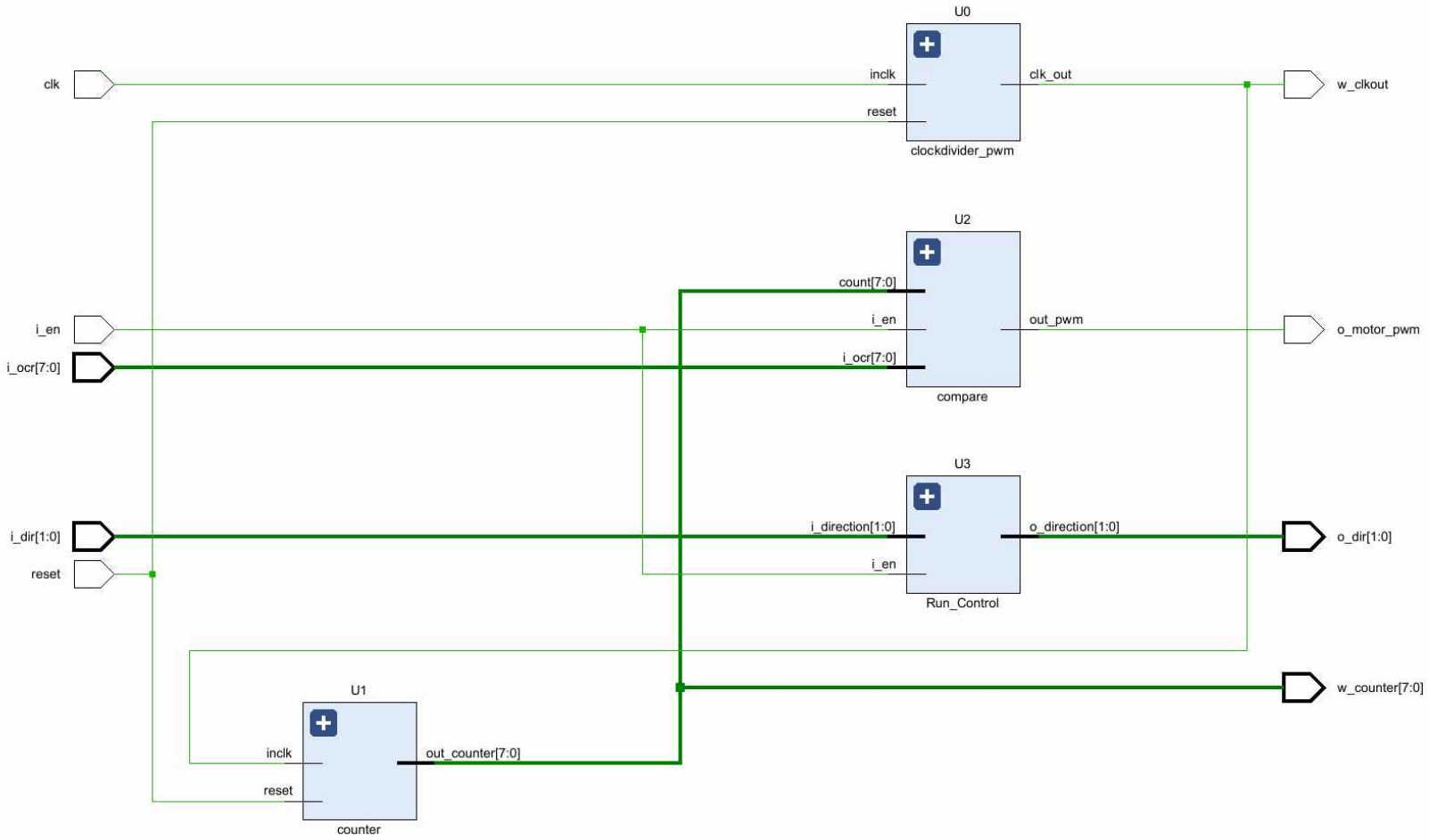
```
1.  `timescale 1ns / 1ps
2.  module Run_Control(
3.      input [1:0] i_direction,
4.      input i_en,
5.      output [1:0] o_direction
6.      );
7.
8.      reg [1:0] r_dir;
9.      assign o_direction = r_dir;
10.
11.     always @(*) begin
12.     if(i_en==1'b1)
13.         case(i_direction)
14.             2'b00 : r_dir = 2'b00;
15.             2'b01 : r_dir = 2'b01;
16.             2'b10 : r_dir = 2'b10;
17.             2'b11 : r_dir = 2'b00;
18.             default : r_dir = 2'b00;
19.     endcase
20.      else
21.          r_dir = 2'b00;
22.
23.  end
24.  endmodule
```

- AXI Connect ➔ PWM Control
  ➔ top_pwm

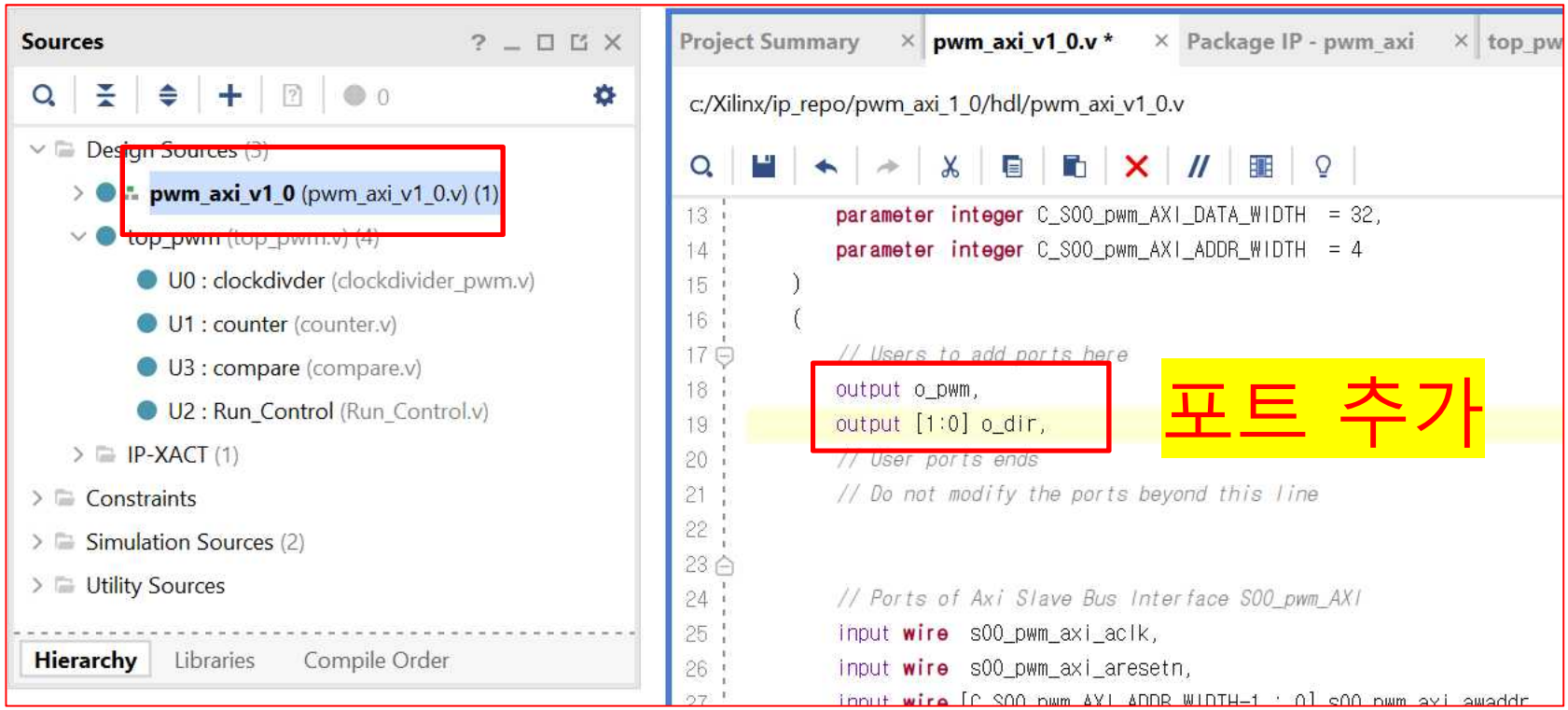
```
1.  `timescale 1ns / 1ps
2.  module top_pwm(
3.  reset, clk, i_ocr, i_en, i_dir, o_dir,
    o_motor_pwm, w_counter, w_clkout
4.  );
5.  input reset;
6.  input clk;
7.  input [7:0] i_ocr;
8.  input i_en;
9.  input [1:0] i_dir;
10. output [1:0] o_dir;
11. output o_motor_pwm;
12. output [7:0] w_counter;
13. output w_clkout;
14. clockdivider_pwm U0 (
15.     .inclk(clk),
16.     .reset(reset),
17.     .clk_out(w_clkout)
18. );
```

```
19. counter U1 (
20.     .inclk(w_clkout),
21.     .reset(reset),
22.     .out_counter(w_counter)
23. );
24.   wire [7:0] w_counter;
25.   wire [7:0] i_ocr;
26.   wire [1:0] i_dir;
27.   wire [1:0] o_dir;
28.  compare U2 (
29.     .count(w_counter),
30.     .i_ocr(i_ocr),
31.     .i_en(i_en),
32.     .out_pwm(o_motor_pwm)
33. );
33. Run_Control U3(
34.   .i_direction(i_dir),
35.   .o_direction(o_dir),
36.   .i_en(i_en)
37.   );
38. endmodule
```

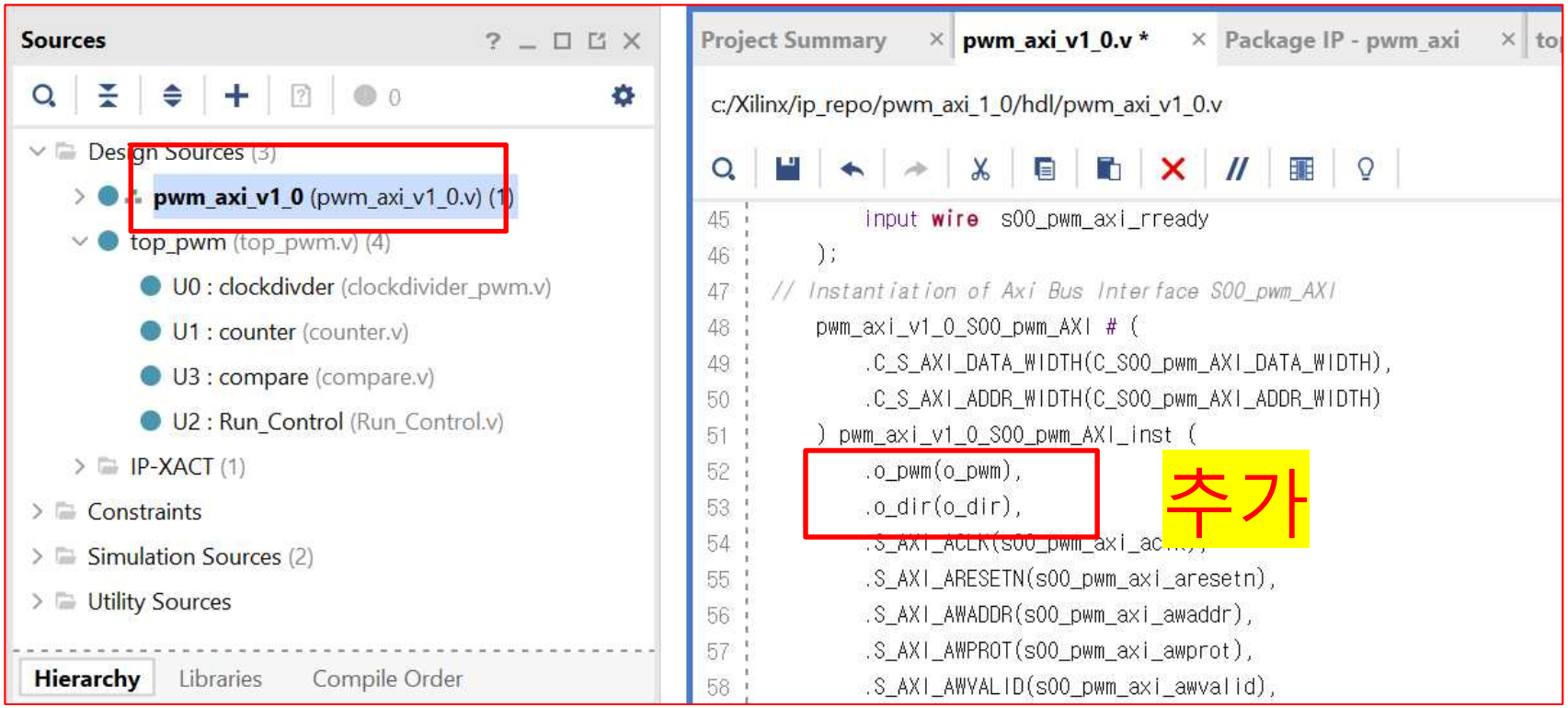
- *AXI Connect → PWM Control → top_pwm : RTL*



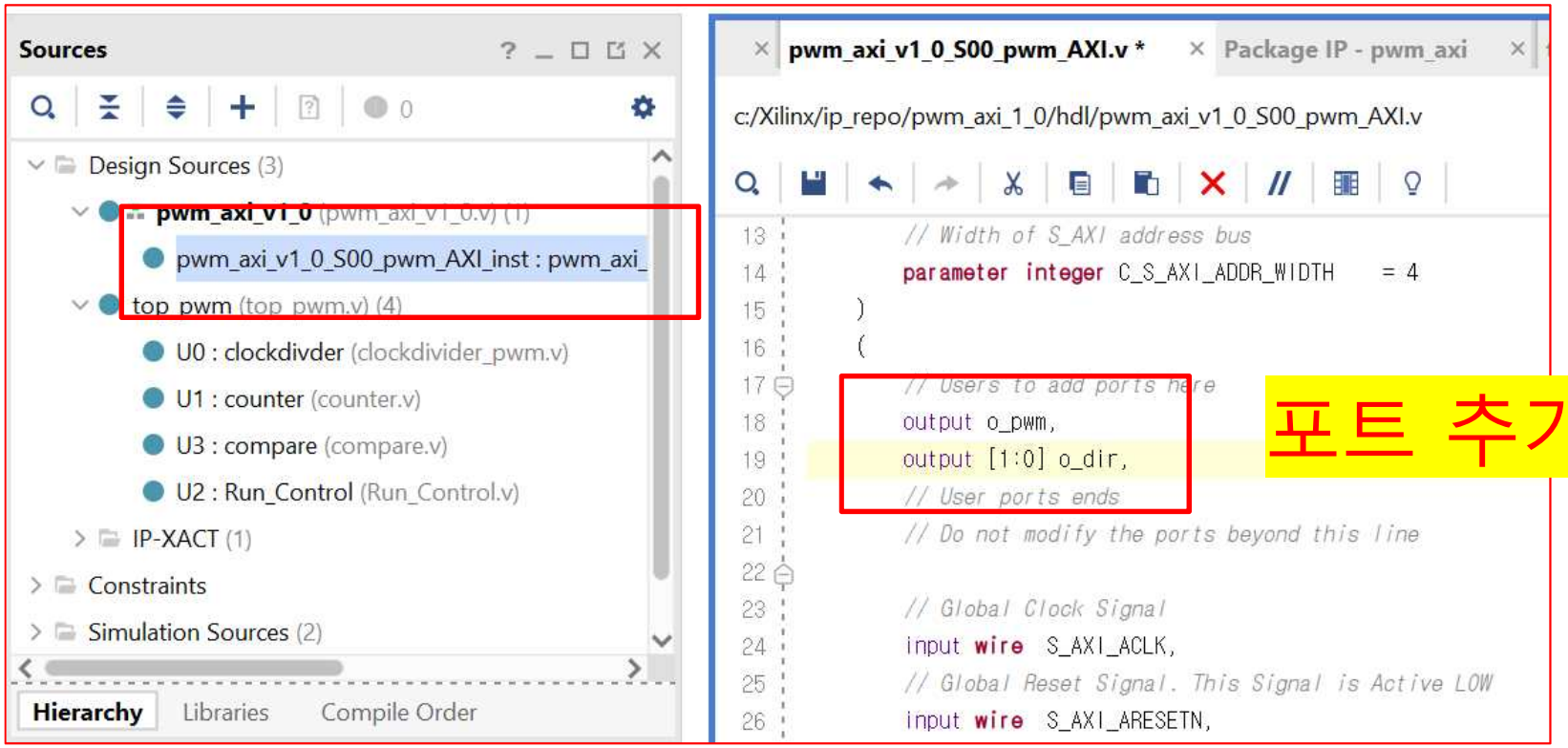
➢ AXI Connect ➔ PWM Control



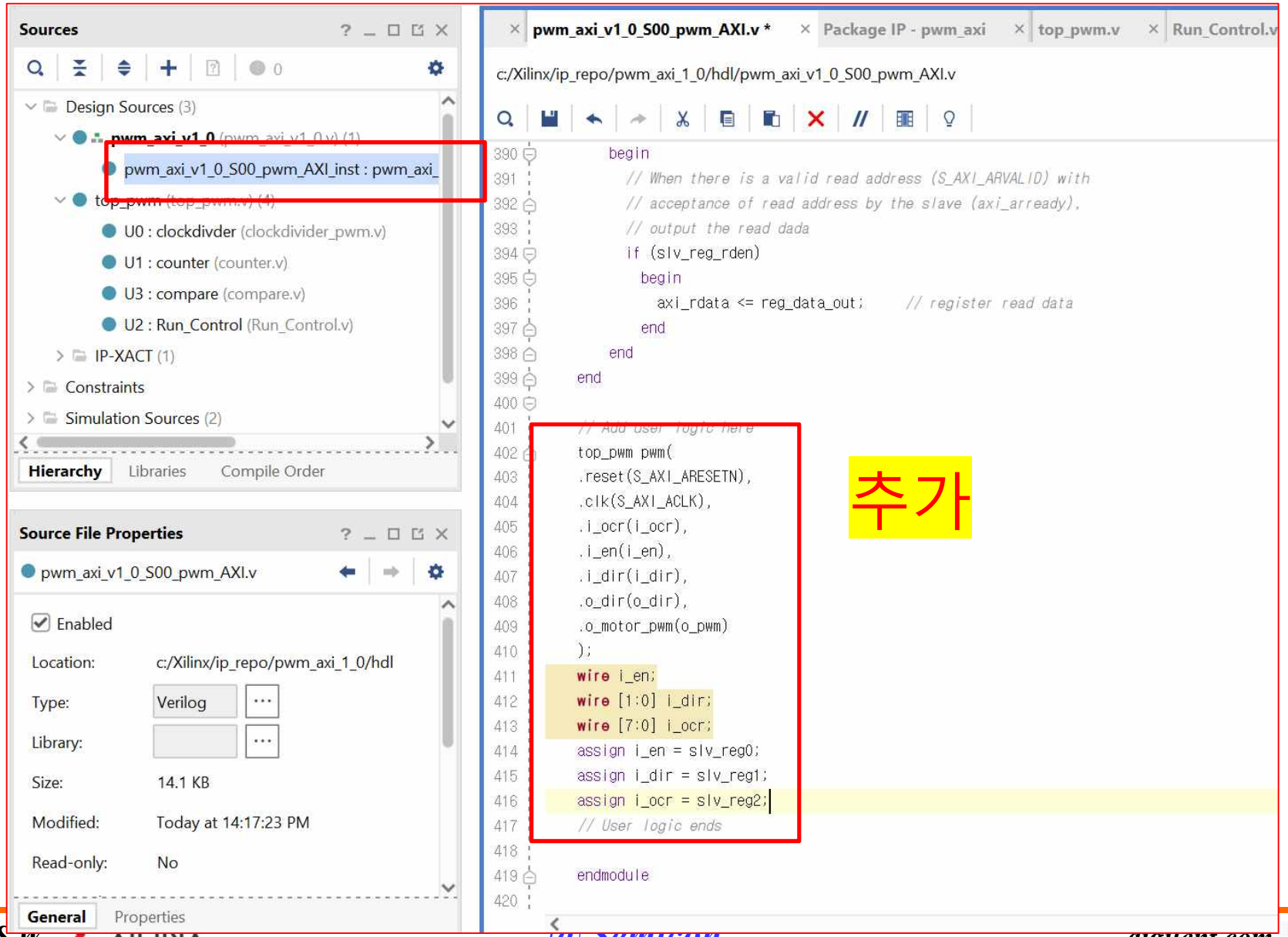
➢ AXI Connect ➔ PWM Control



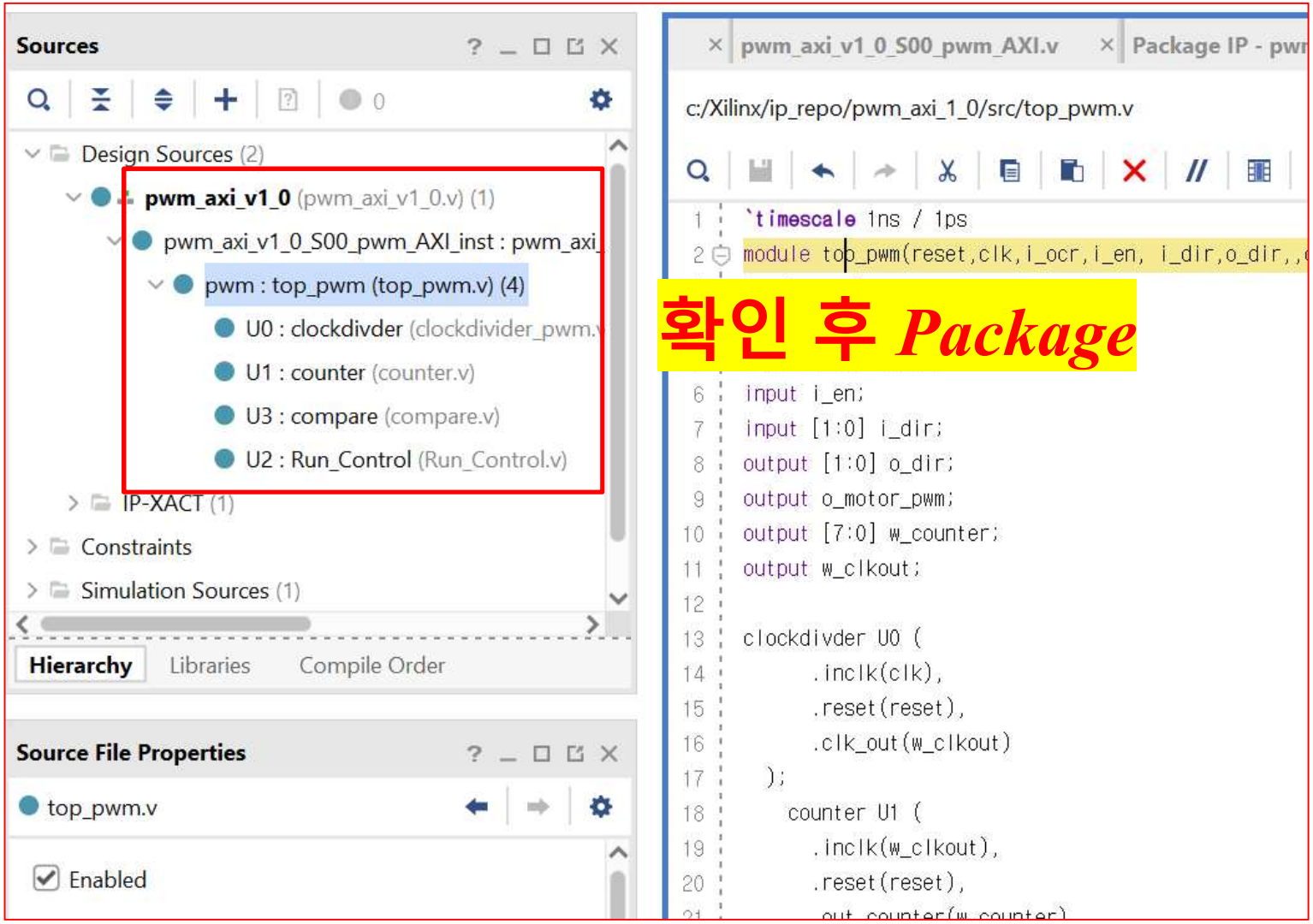
➢ AXI Connect ➔ PWM Control

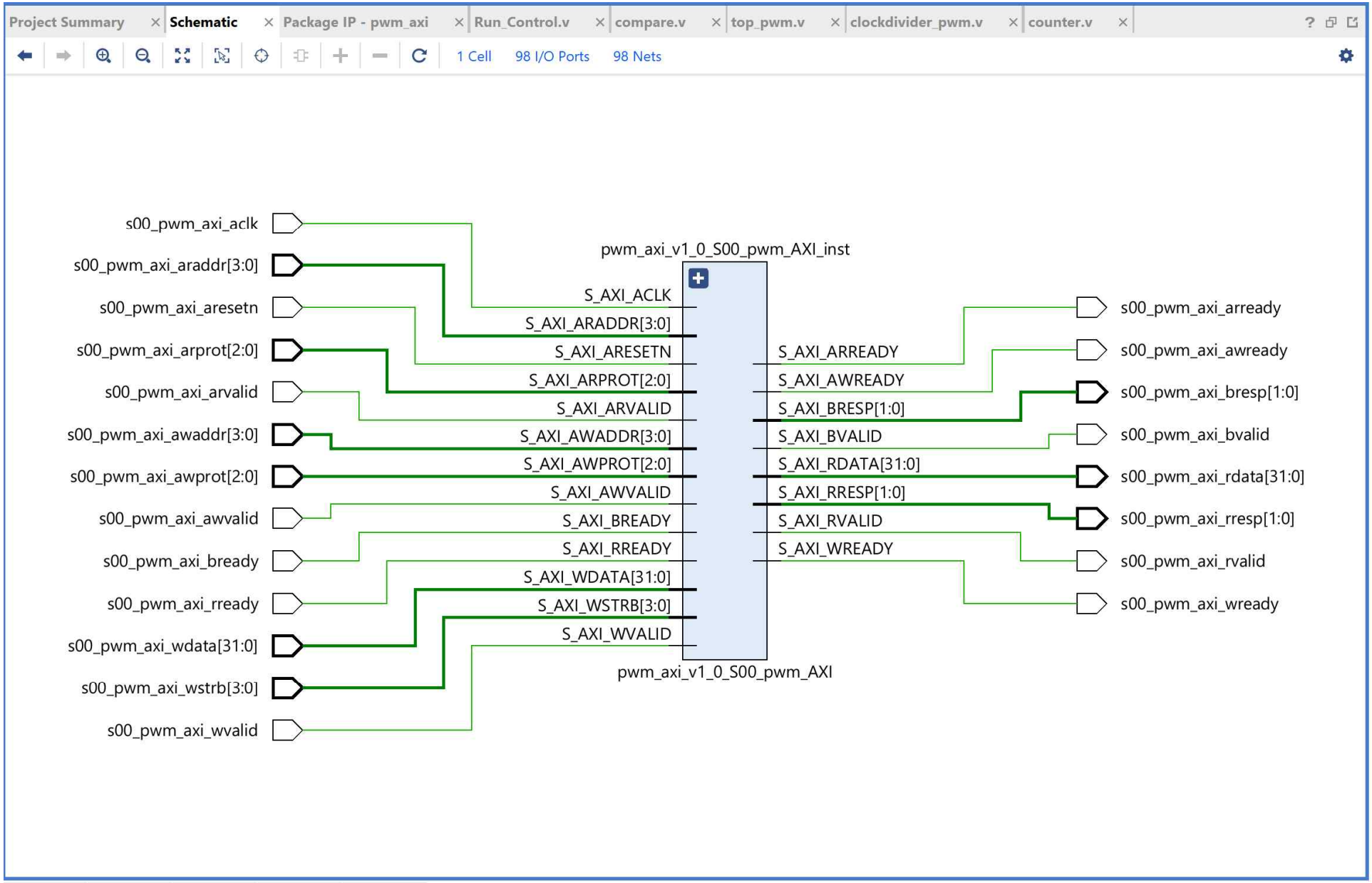


➢ AXI Connect ➔ PWM Control



➢ *AXI Connect ➔ PWM Control*



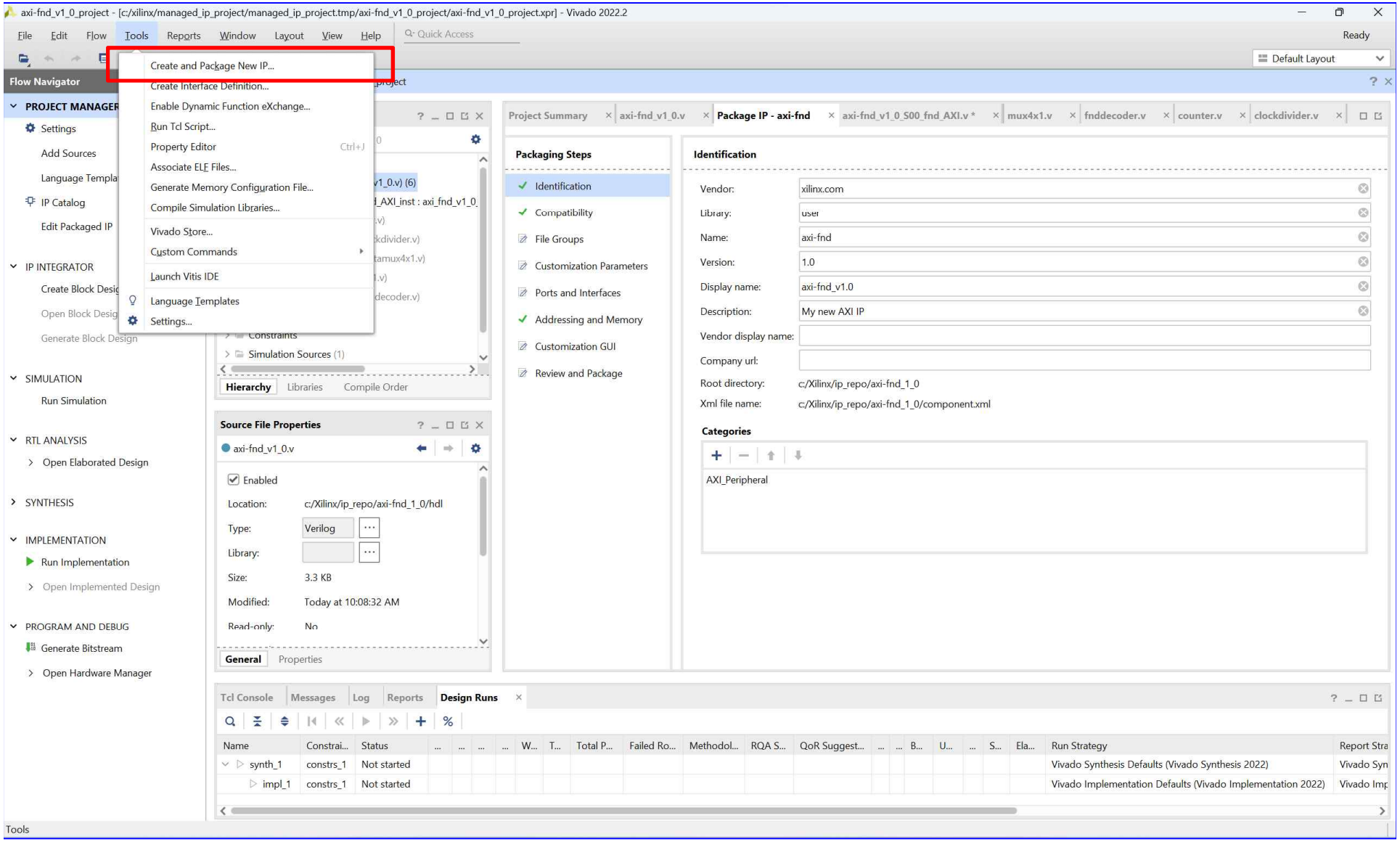

Project Summary
Schematic
Package IP - pwm_axi
Run_Control.v
compare.v
top_pwm.v
clockdivider_pwm.v
counter.v
1 Cell
98 I/O Ports
98 Nets
s00_pwm_axi_aclk
s00_pwm_axi_araddr[3:0]
s00_pwm_axi_aresetn
s00_pwm_axi_arprot[2:0]
s00_pwm_axi_arvalid
s00_pwm_axi_awaddr[3:0]
s00_pwm_axi_awprot[2:0]
s00_pwm_axi_awvalid
s00_pwm_axi_bready
s00_pwm_axi_rready
s00_pwm_axi_wdata[31:0]
s00_pwm_axi_wstrb[3:0]
s00_pwm_axi_wvalid
pwm_axi_v1_0_S00_pwm_AXI_inst
S_AXI_ACLK
S_AXI_ARADDR[3:0]
S_AXI_ARESETN
S_AXI_ARPROT[2:0]
S_AXI_ARVALID
S_AXI_AWADDR[3:0]
S_AXI_AWPROT[2:0]
S_AXI_AWVALID
S_AXI_BREADY
S_AXI_RREADY
S_AXI_WDATA[31:0]
S_AXI_WSTRB[3:0]
S_AXI_WVALID
S_AXI_ARREADY
S_AXI_AWREADY
S_AXI_BRESP[1:0]
S_AXI_BVALID
S_AXI_RDATA[31:0]
S_AXI_RRESP[1:0]
S_AXI_RVALID
S_AXI_WREADY
pwm_axi_v1_0_S00_pwm_AXI
s00_pwm_axi_arready
s00_pwm_axi_awready
s00_pwm_axi_bresp[1:0]
s00_pwm_axi_bvalid
s00_pwm_axi_rdata[31:0]
s00_pwm_axi_rresp[1:0]
s00_pwm_axi_rvalid
s00_pwm_axi_wready
Tcl Console
Messages
Log
Reports
Design Runs

AMBA and AXI Connect Example

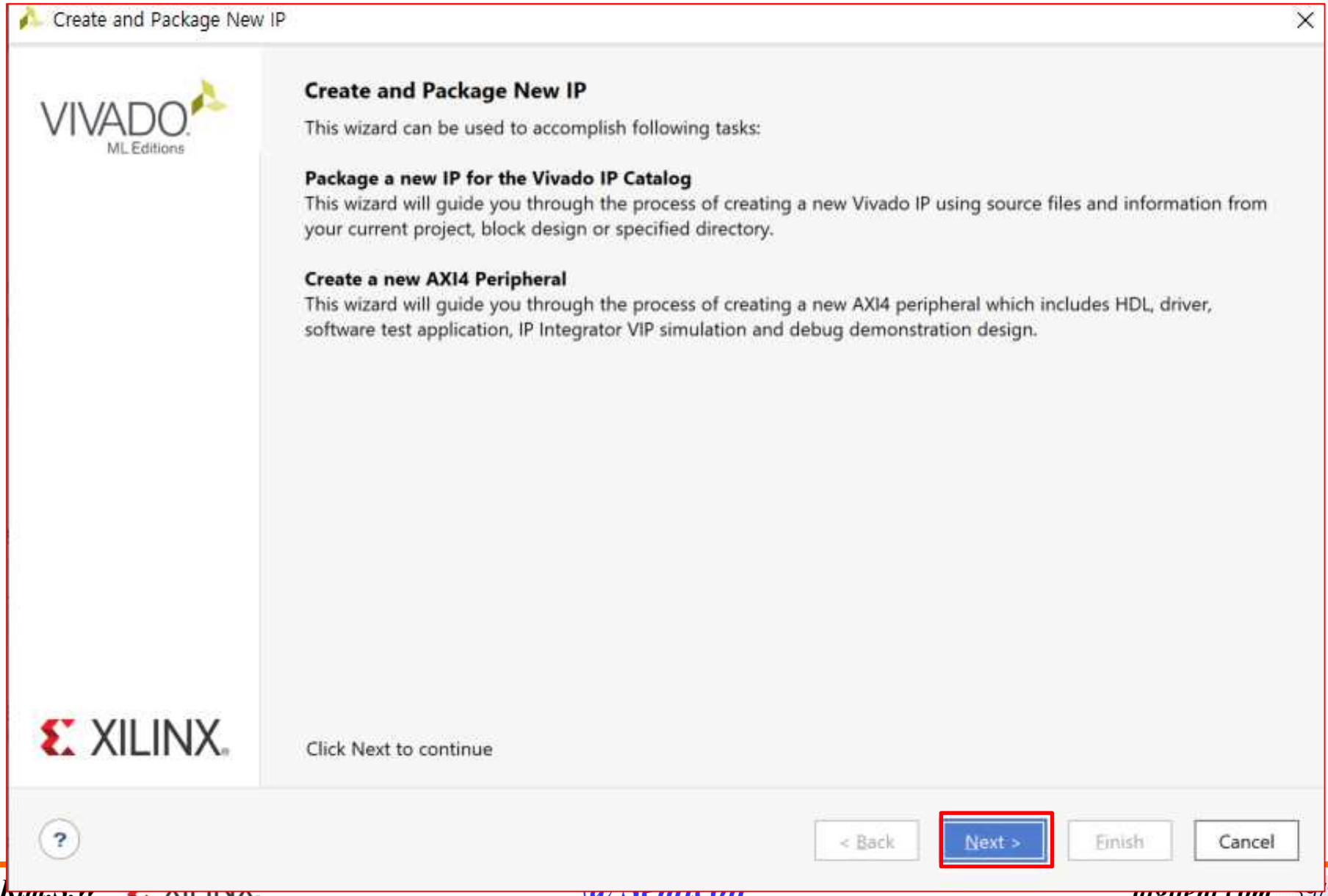
PWM Control

[Create and Package New IP]

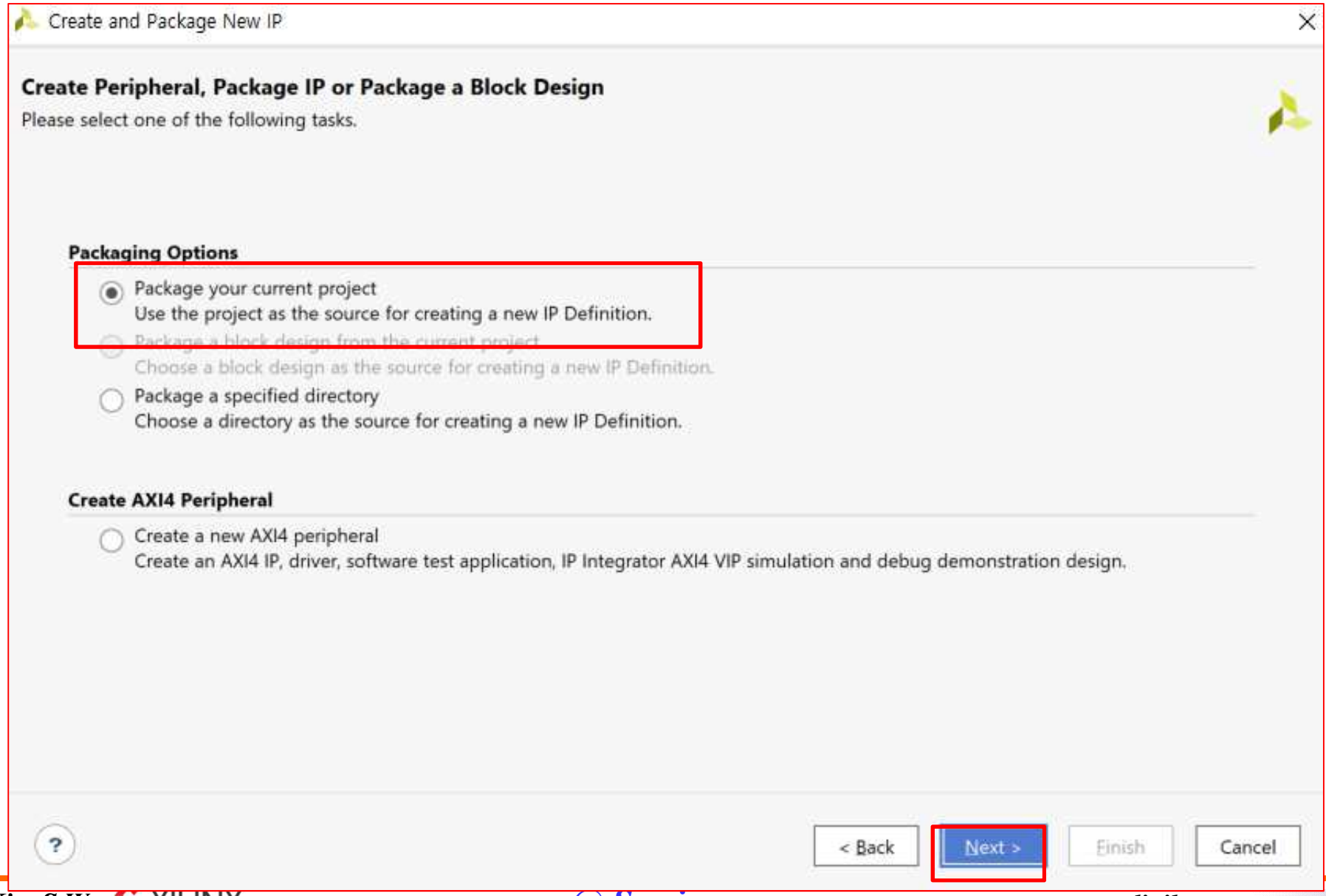
➢ AXI Connect ➔ PWM Control



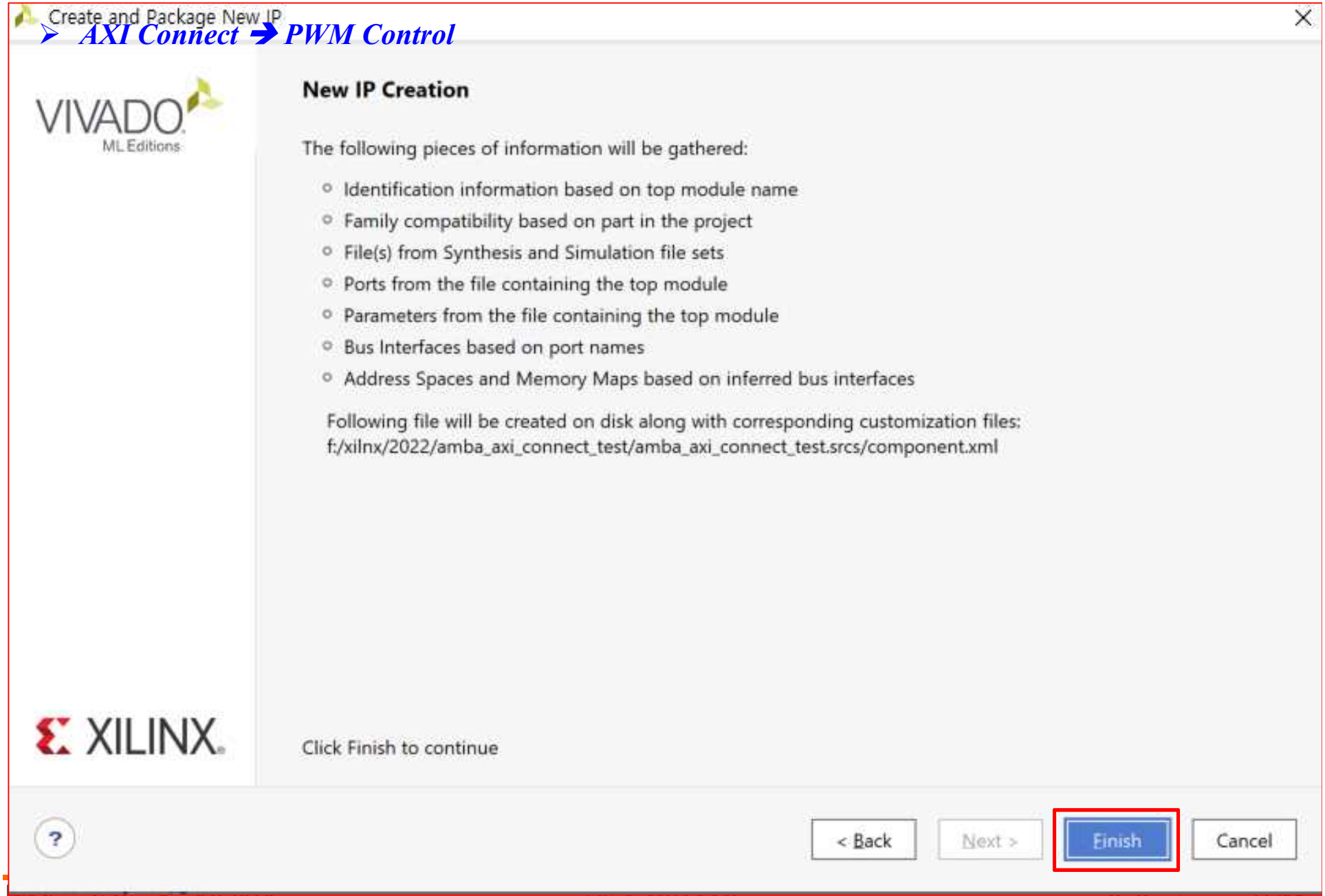
- ***AXI Connect ➔ PWM Control***



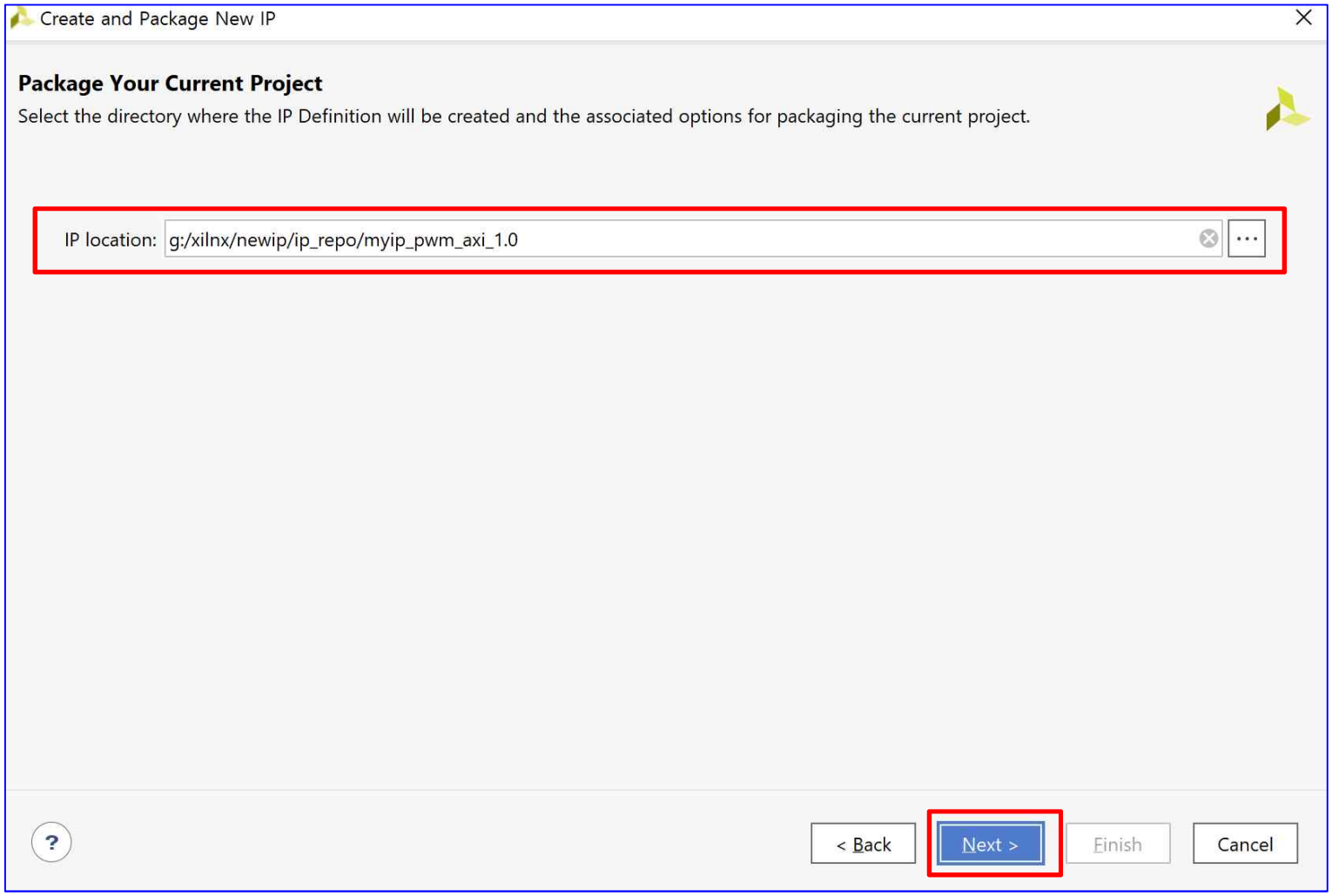
- AXI Connect ➔ PWM Control



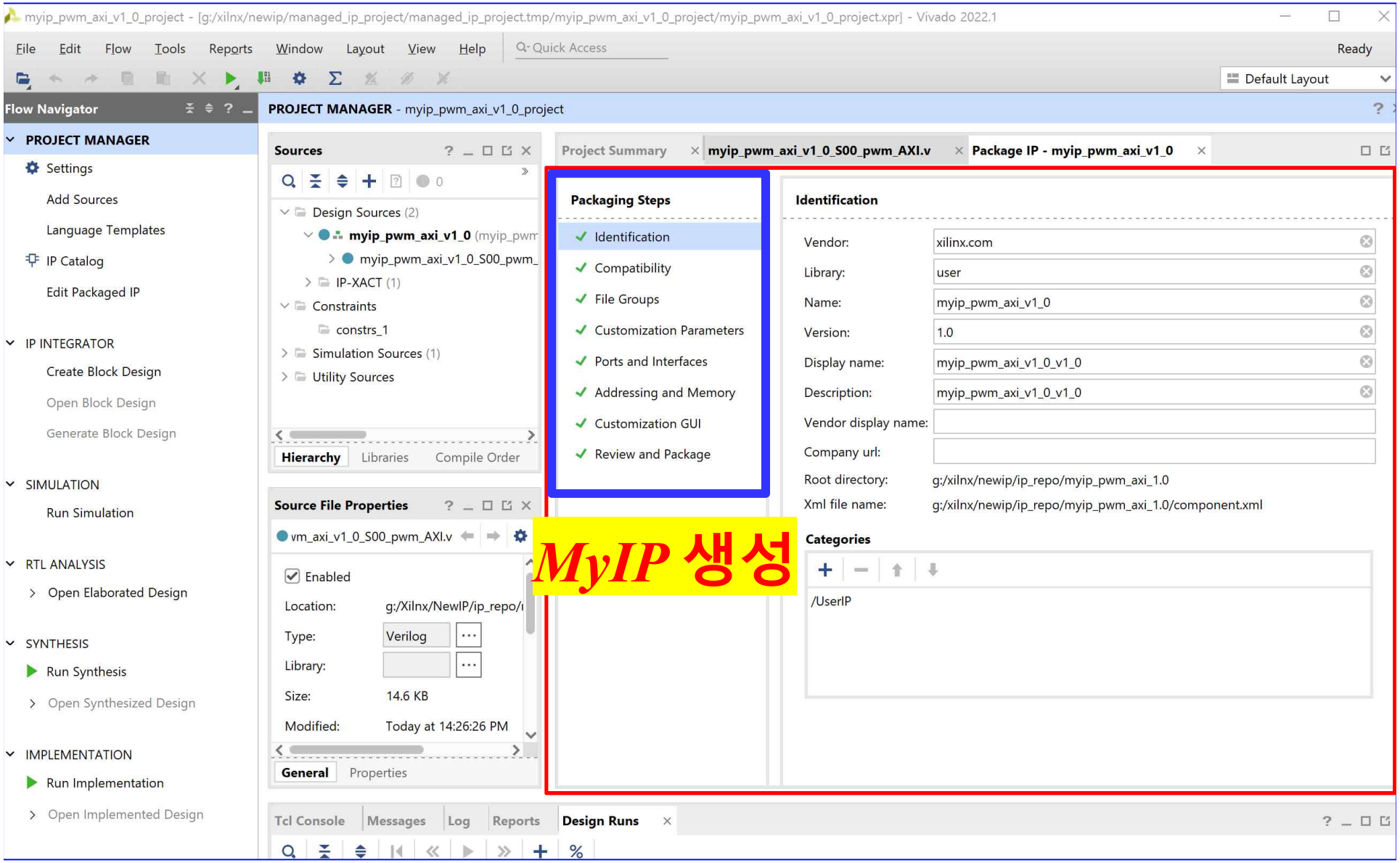
➢ AXI Connect ➔ PWM Control



➢ AXI Connect ➔ PWM Control



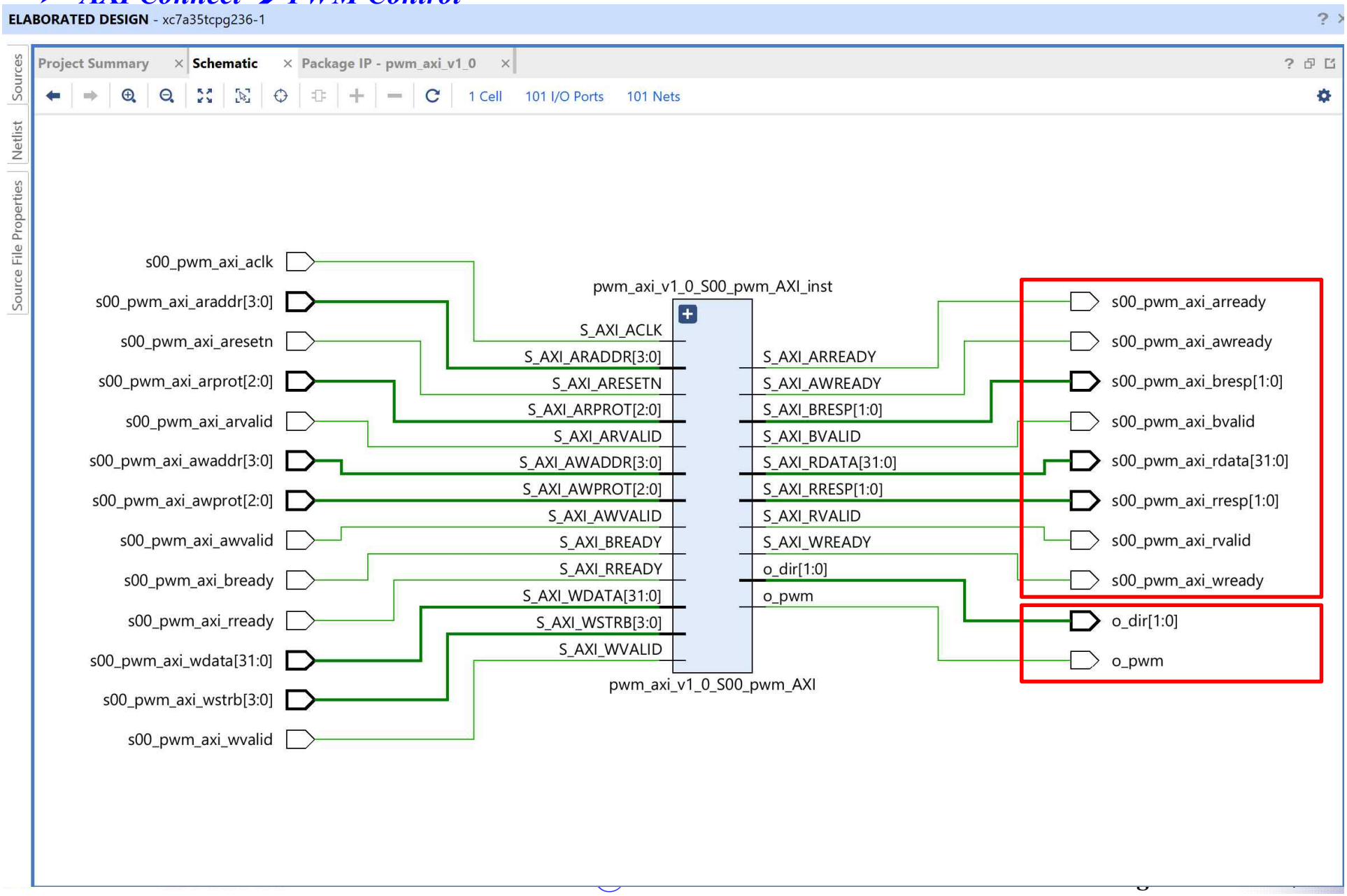
➢ AXI Connect ➔ PWM Control



AMBA and AXI Connect

axi_pwm_exam.xpr

➢ AXI Connect ➔ PWM Control



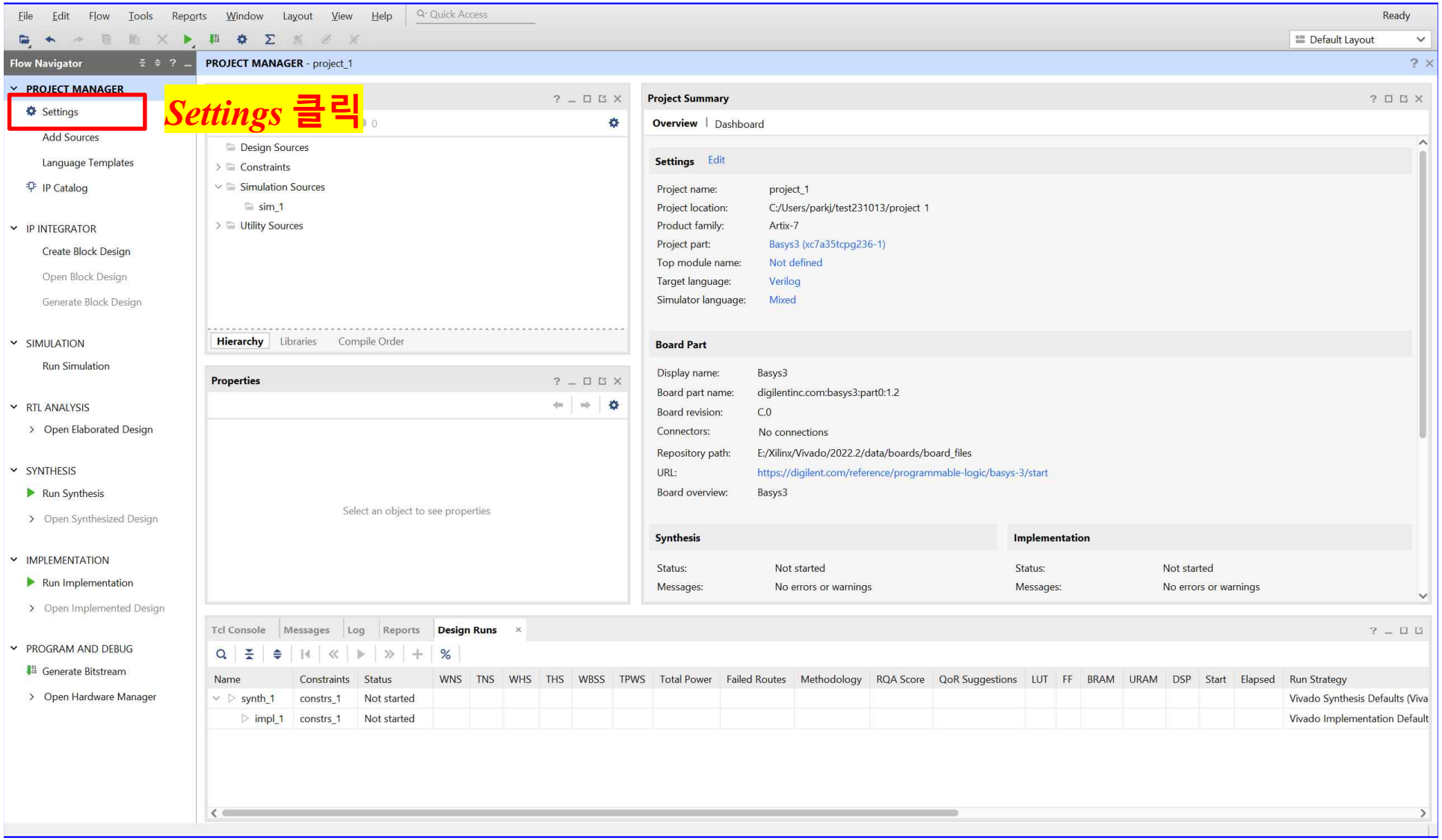
AMBA and AXI Connect Example

PWM Control

[Create Block Design]

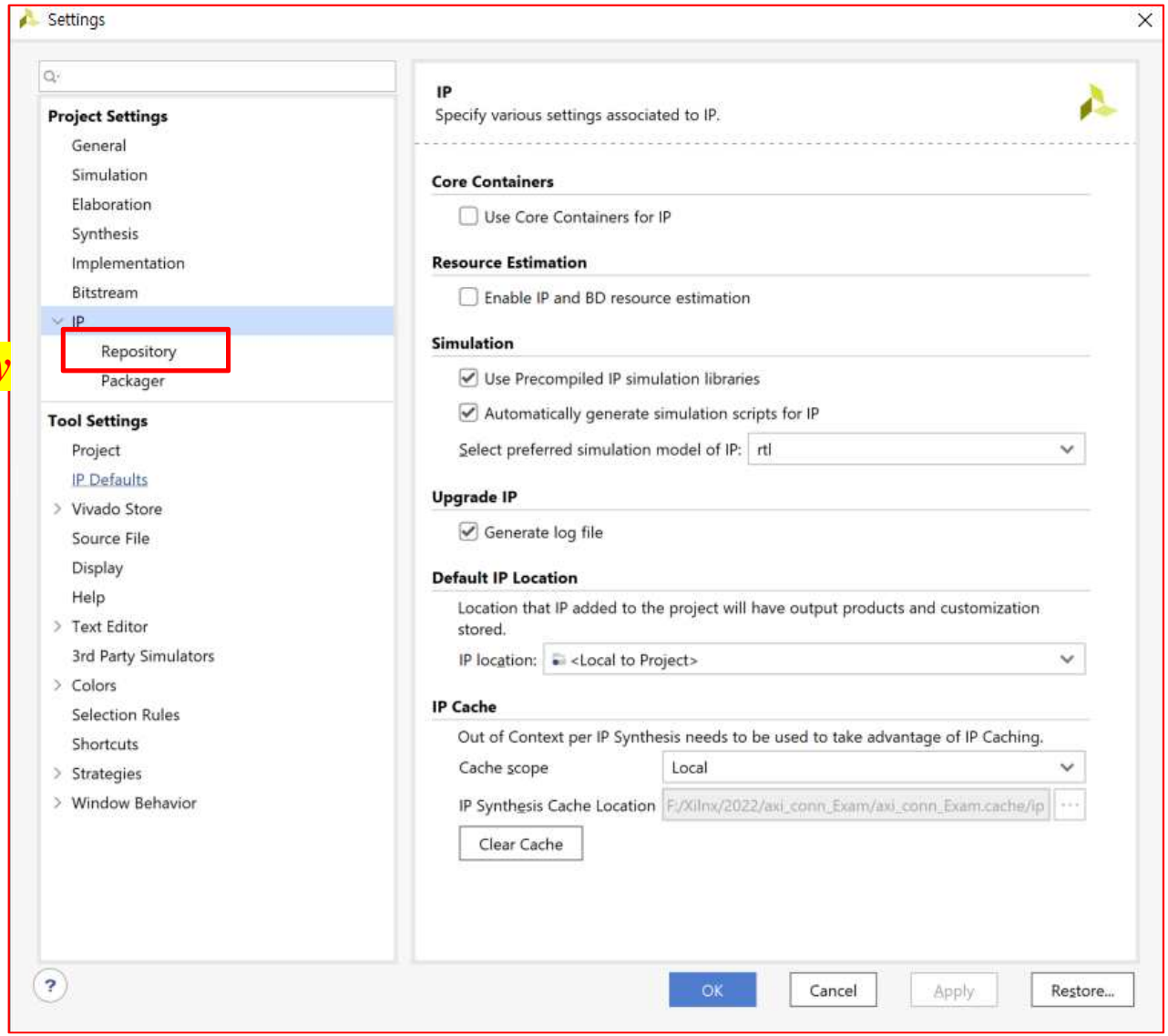
➢ AXI Connect ➔ PWM Control

➢ AXI Connect ➔ PWM Control

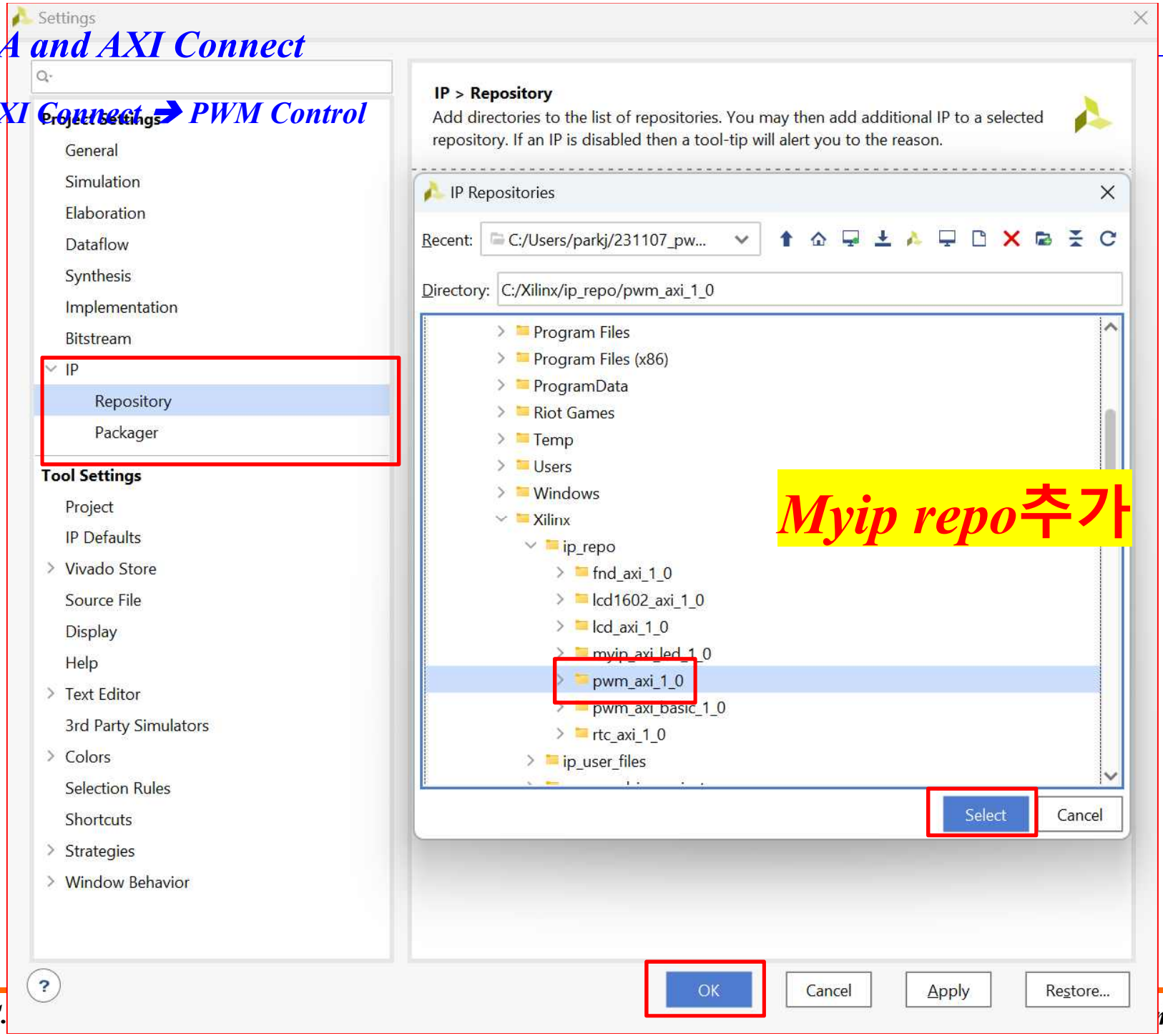


- AXI Connect ➔ PWM Control

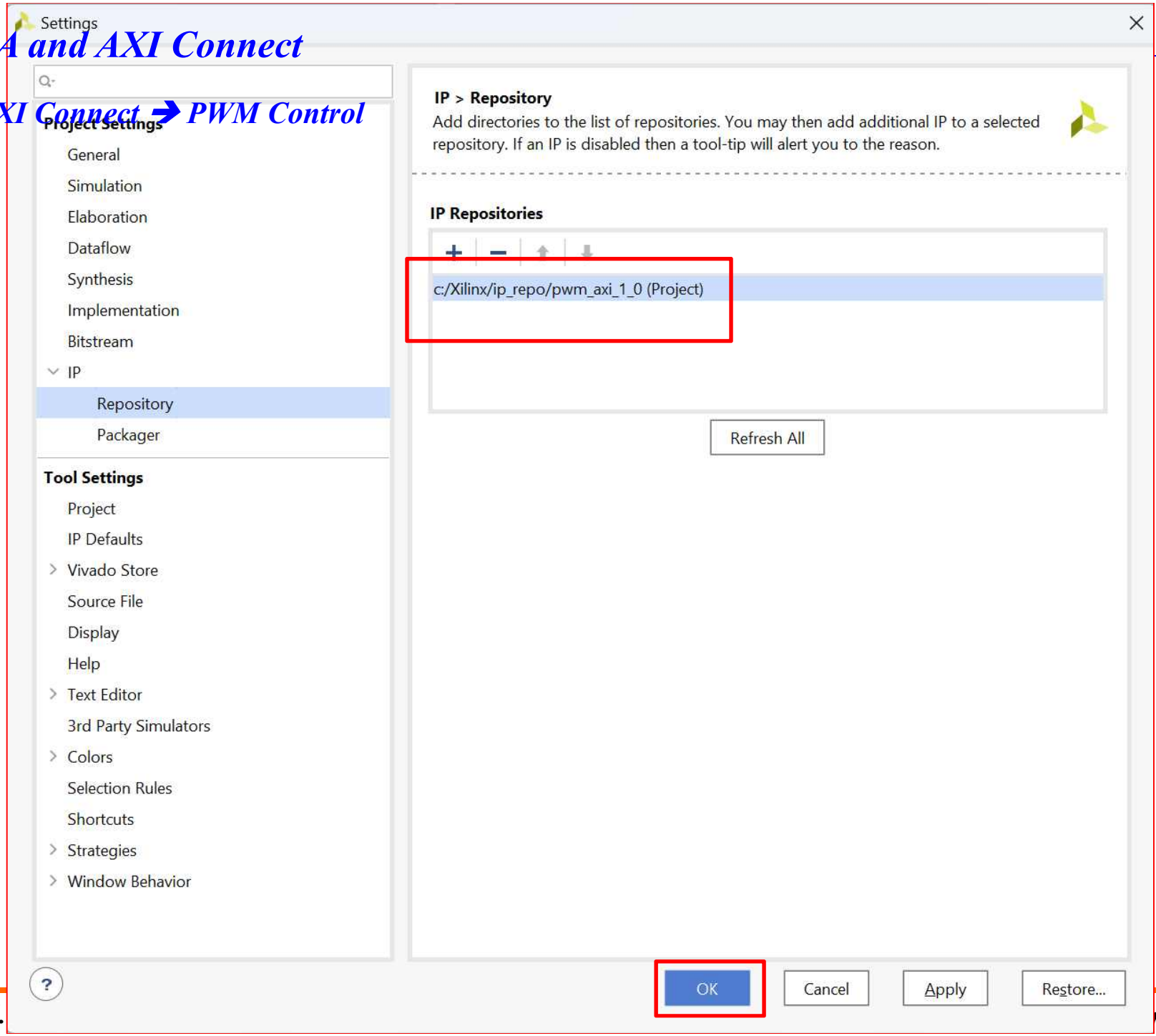
Repository



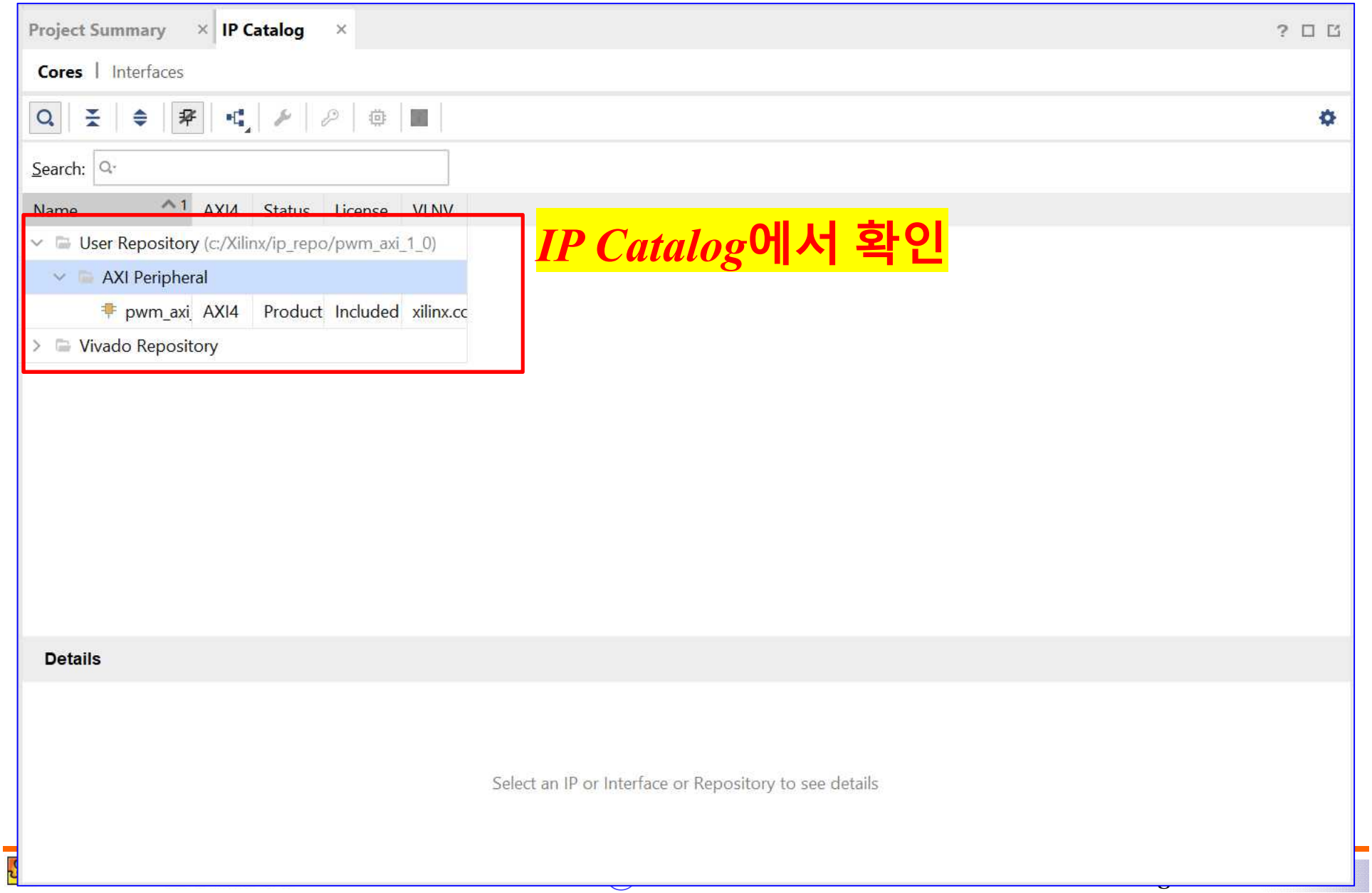
- AXI Connect ➔ PWM Control



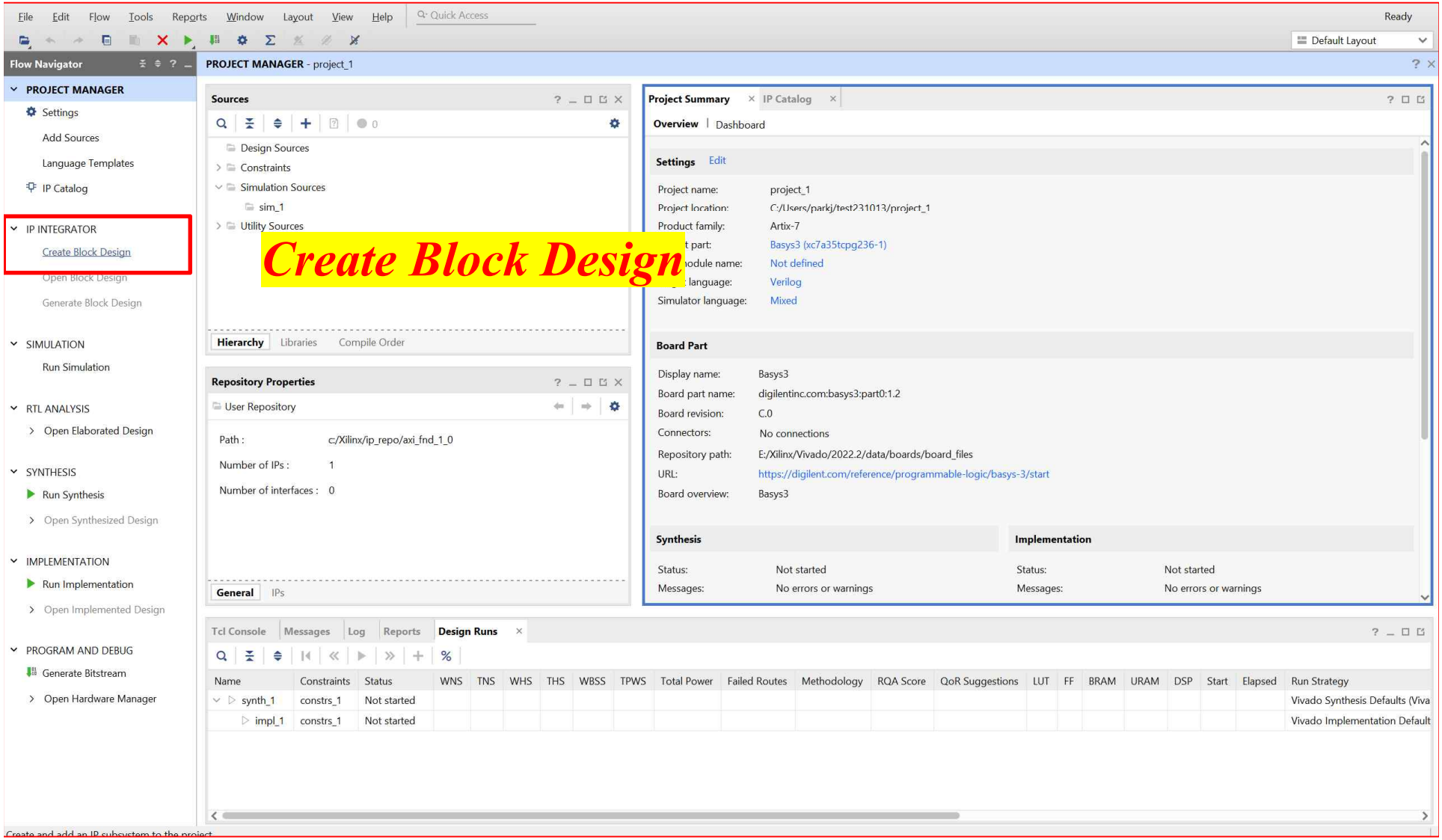
- AXI Connect ➔ PWM Control



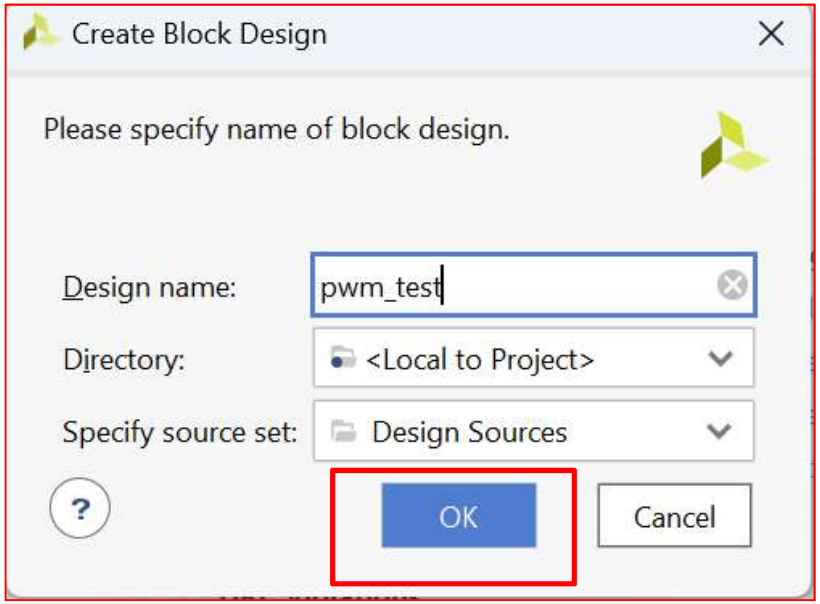
- ***AXI Connect ➔ PWM Control***



➢ AXI Connect ➔ PWM Control

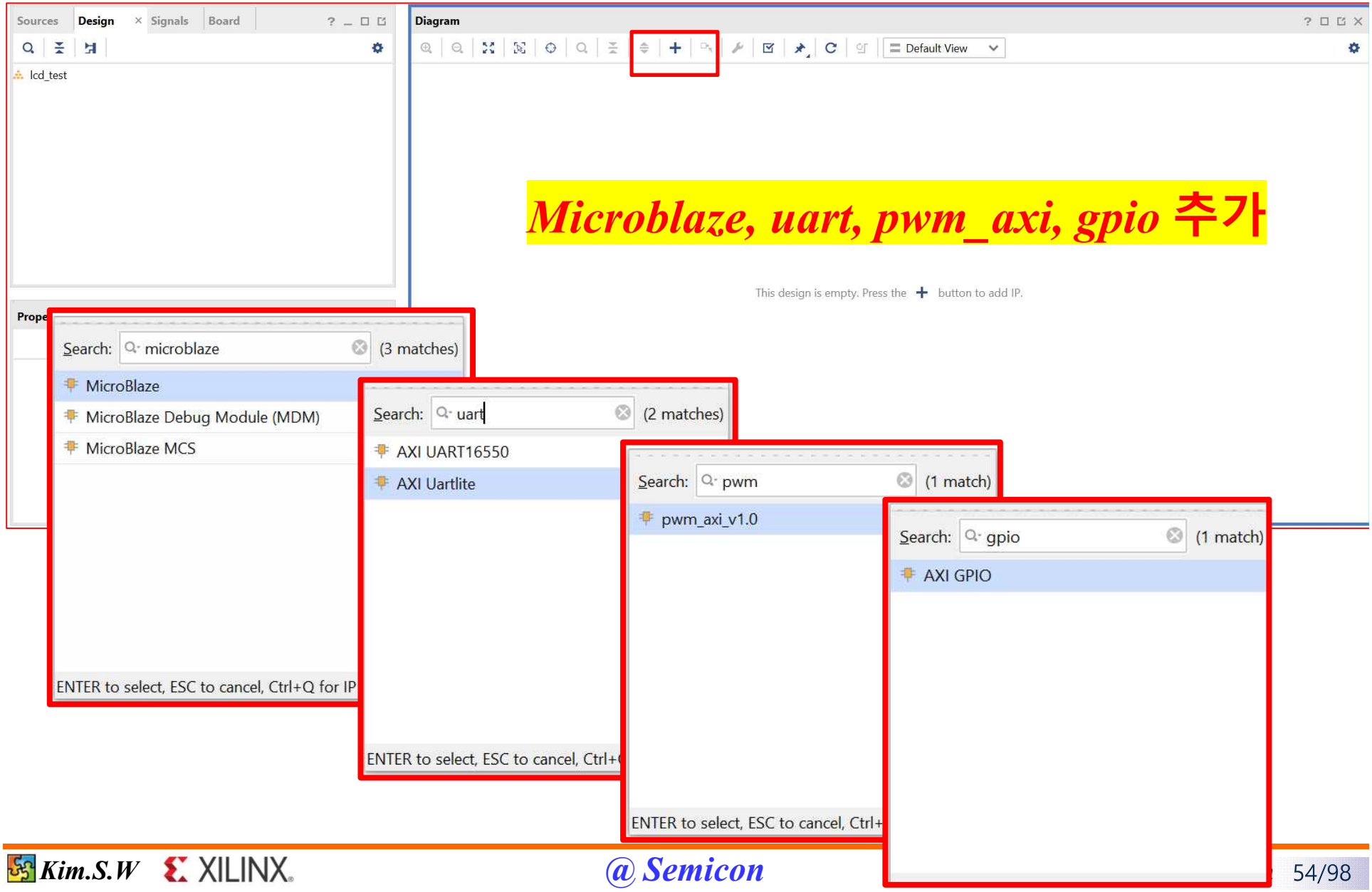


- AXI Connect ➔ PWM Control

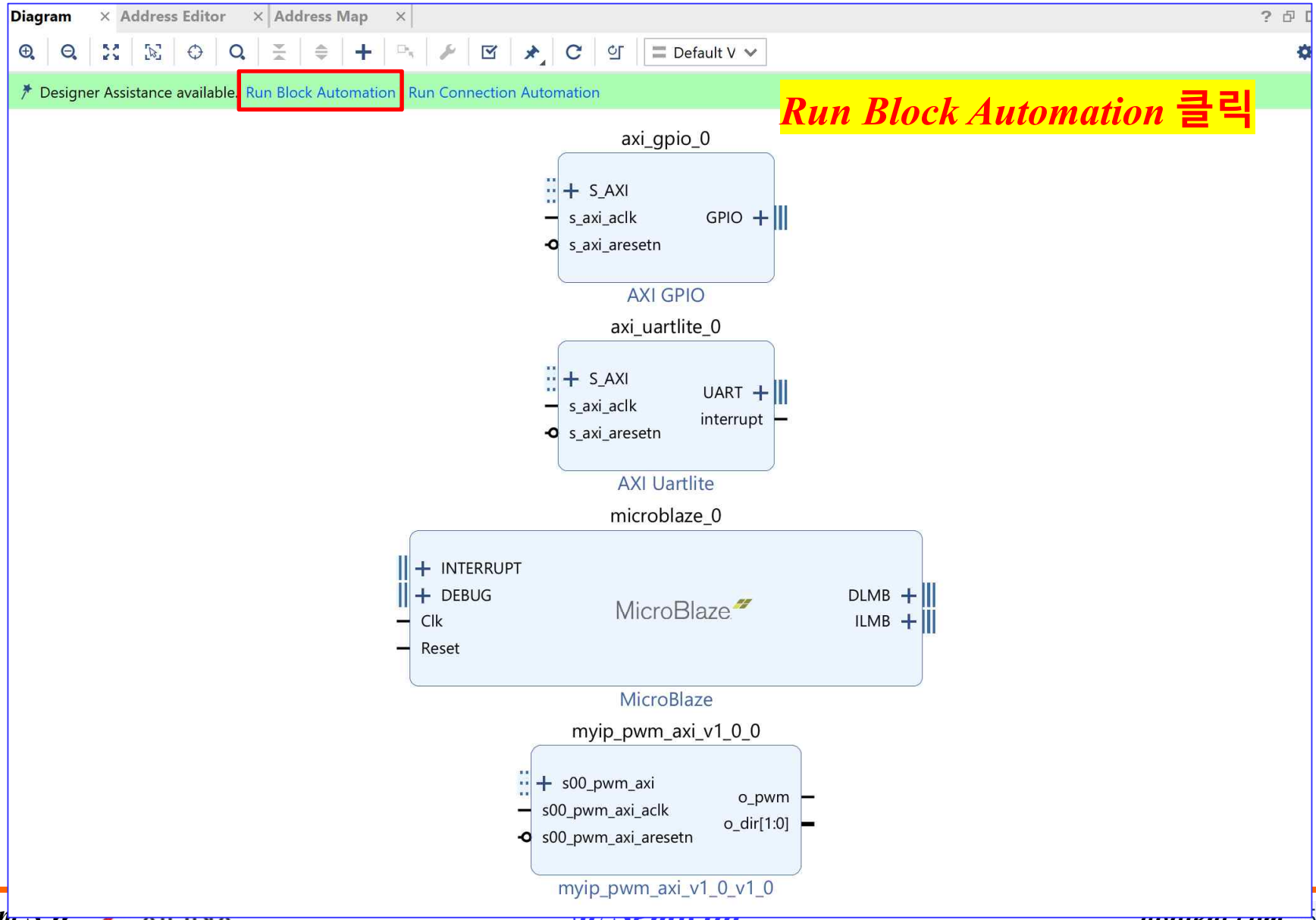


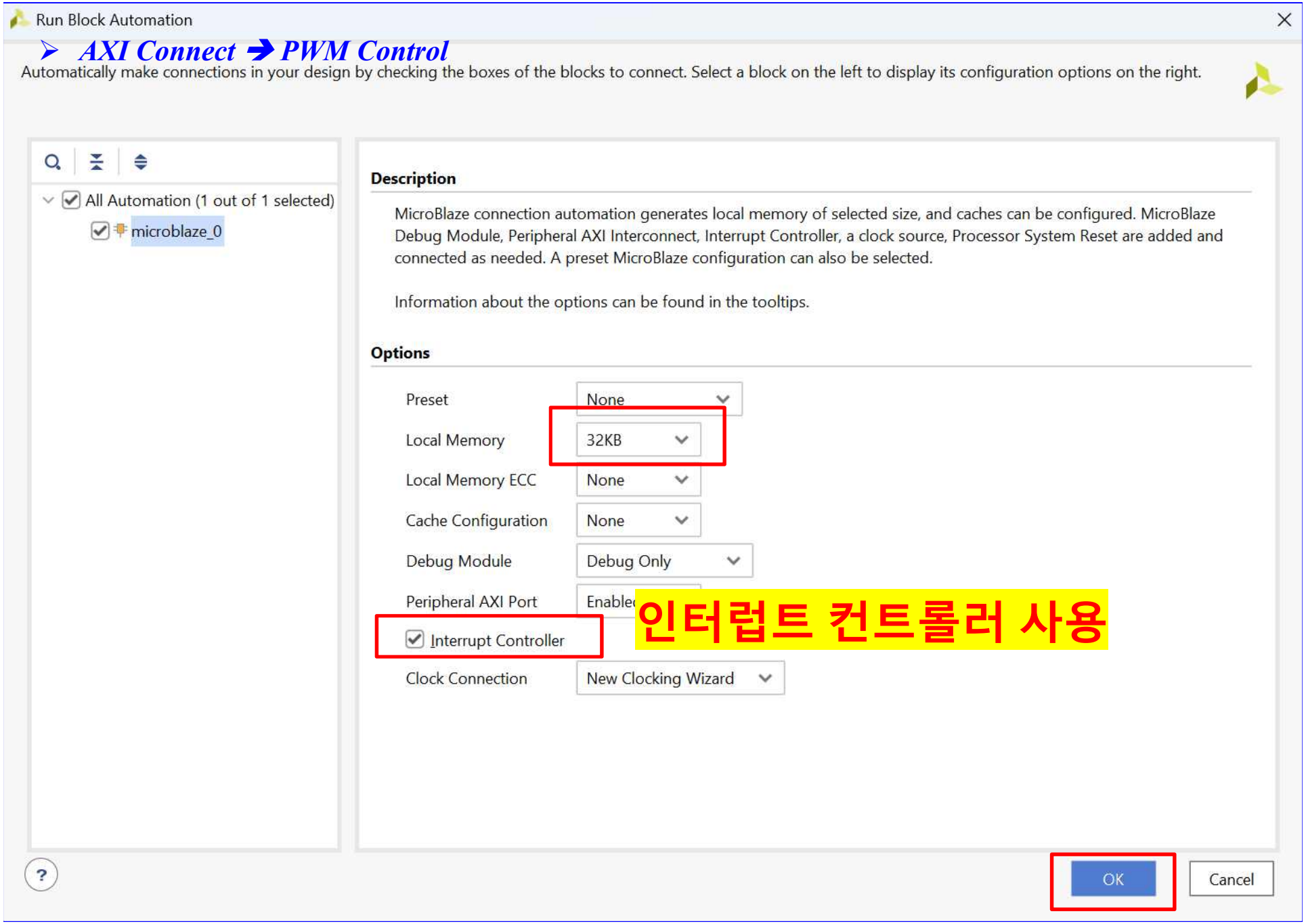
이름 설정 후 OK

➢ AXI Connect ➔ PWM Control

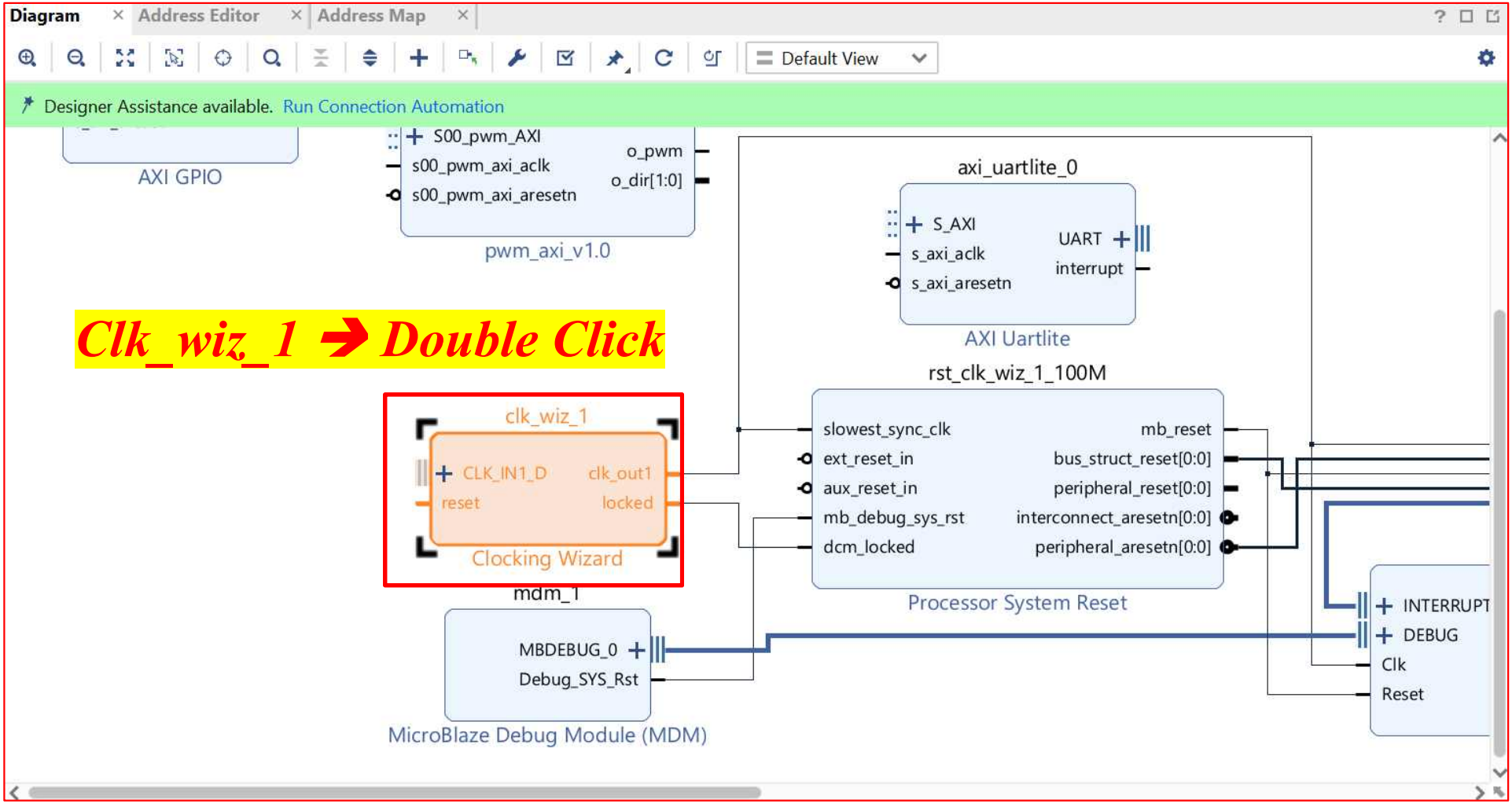


➢ AXI Connect ➔ PWM Control

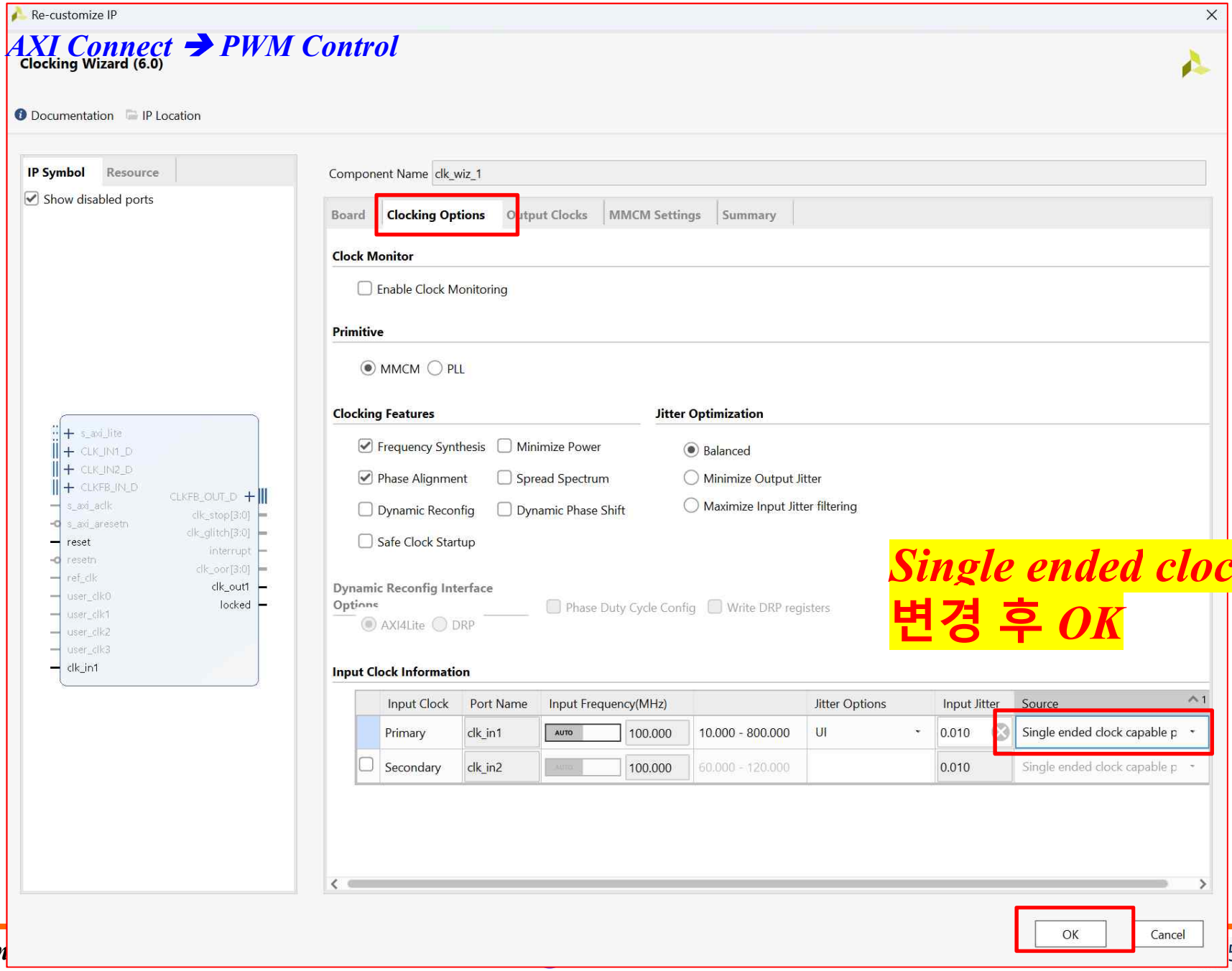



Run Block Automation
➢ AXI Connect ➔ PWM Control
Automatically make connections in your design by checking the boxes of the blocks to connect. Select a block on the left to display its configuration options on the right.
All Automation (1 out of 1 selected)
microblaze_0
Description
MicroBlaze connection automation generates local memory of selected size, and caches can be configured. MicroBlaze Debug Module, Peripheral AXI Interconnect, Interrupt Controller, a clock source, Processor System Reset are added and connected as needed. A preset MicroBlaze configuration can also be selected.
Information about the options can be found in the tooltips.
Options
Preset
None
Local Memory
32KB
Local Memory ECC
None
Cache Configuration
None
Debug Module
Debug Only
Peripheral AXI Port
Interrupt Controller
인터럽트 컨트롤러 사용
Clock Connection
New Clocking Wizard
OK
Cancel

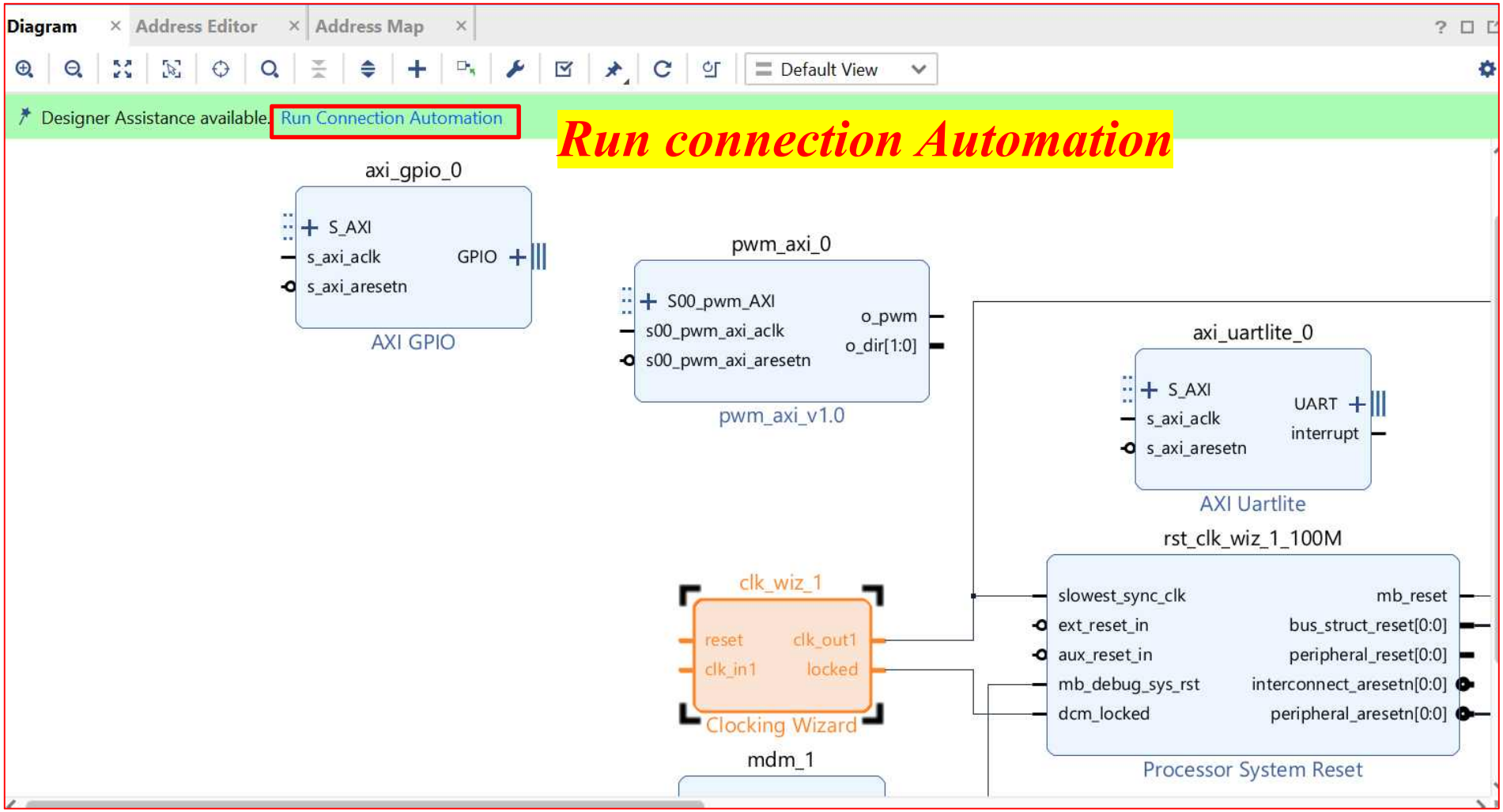
➢ AXI Connect ➔ PWM Control

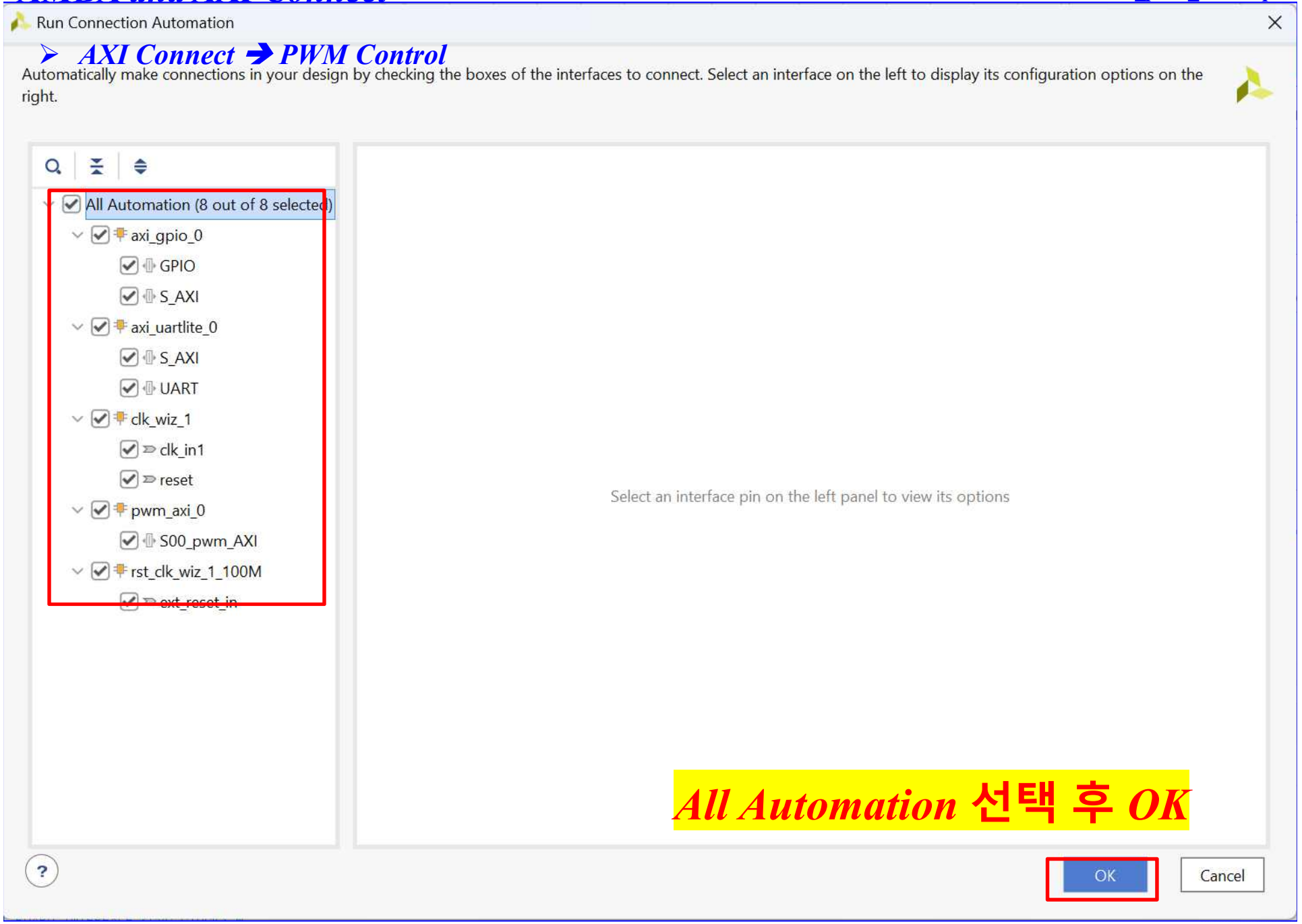


➢ AXI Connect ➔ PWM Control

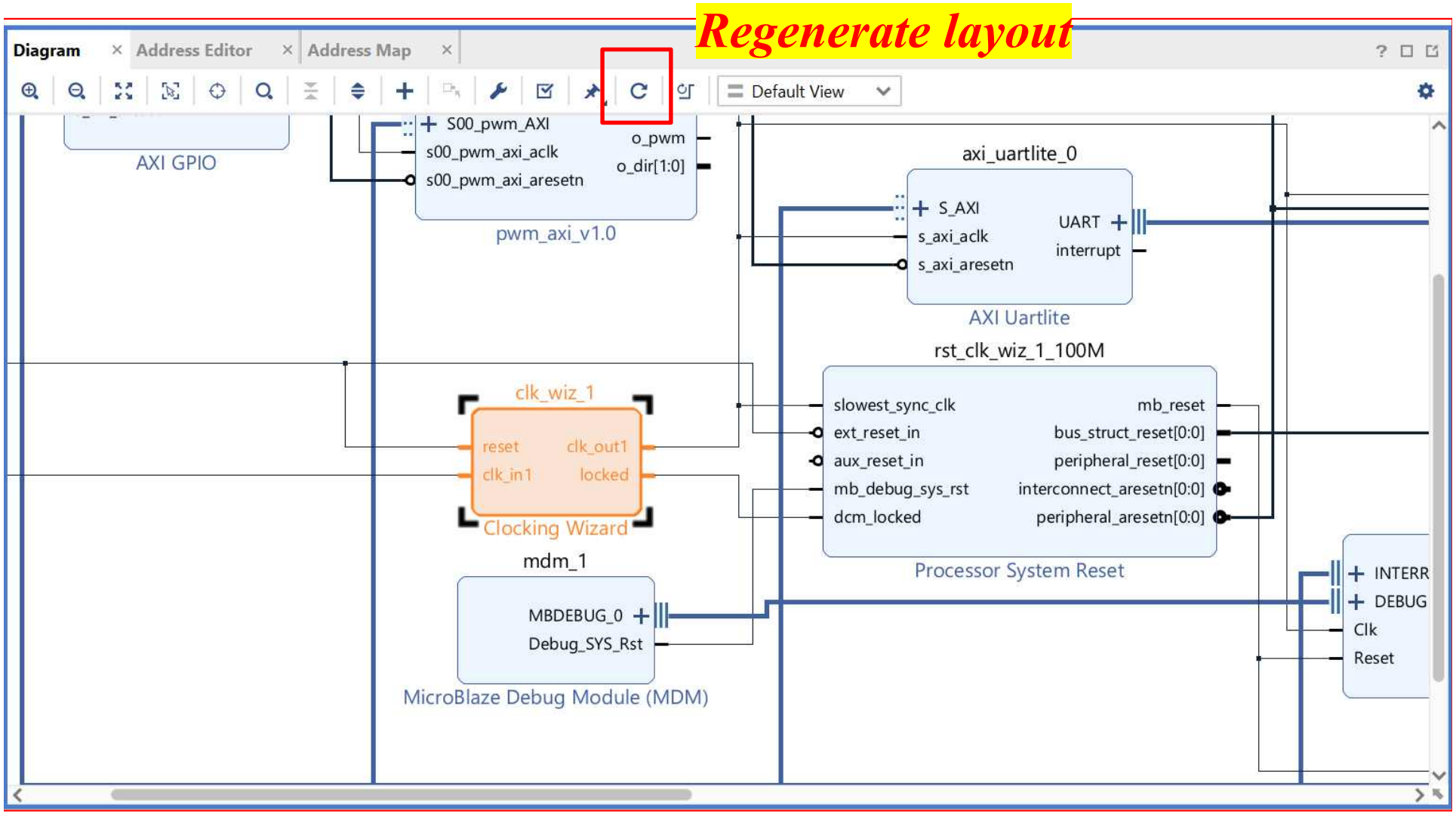


➢ AXI Connect ➔ PWM Control

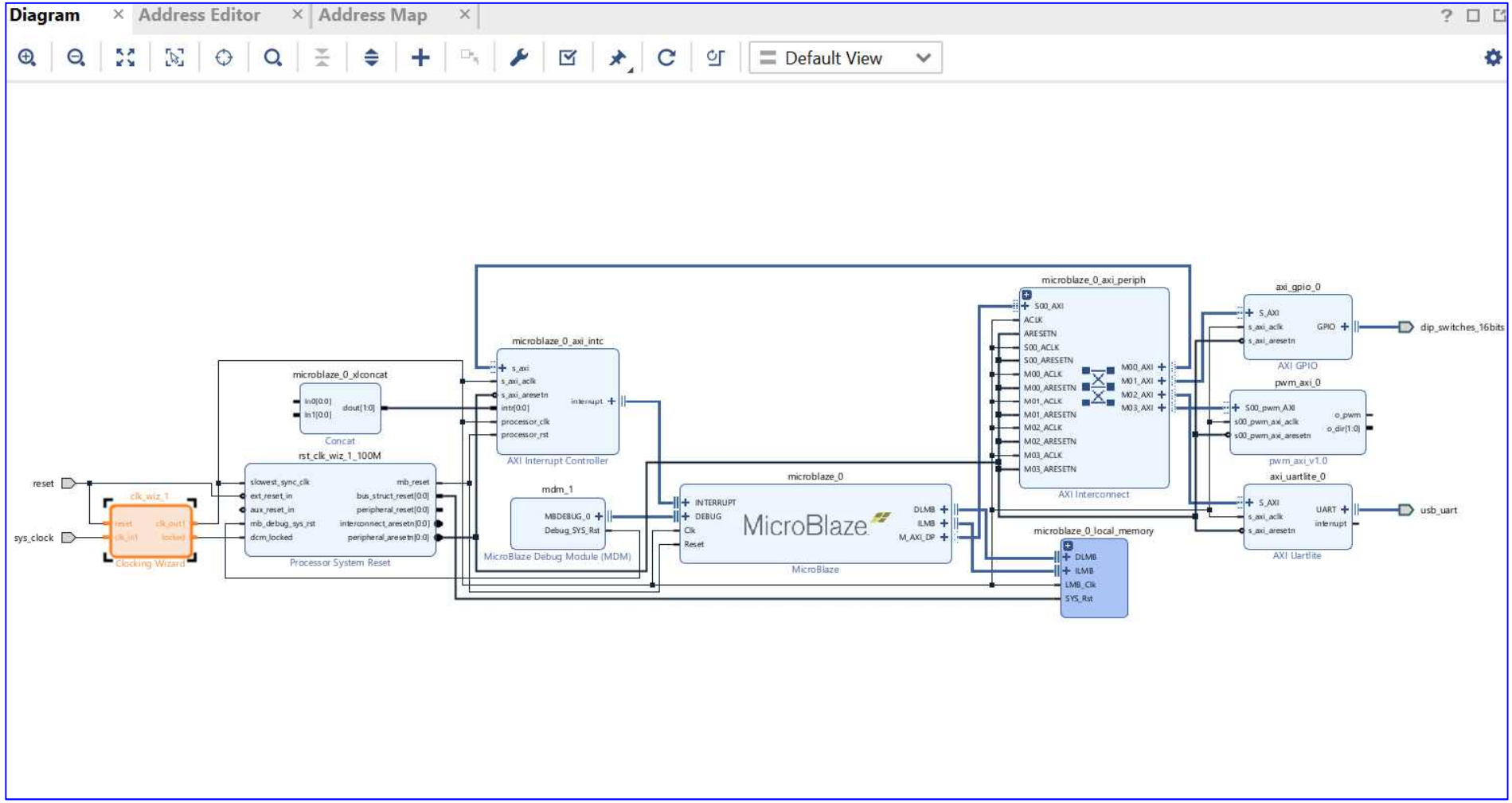



Run Connection Automation
➢ AXI Connect ➔ PWM Control
Automatically make connections in your design by checking the boxes of the interfaces to connect. Select an interface on the left to display its configuration options on the right.
All Automation (8 out of 8 selected)
axi_gpio_0
GPIO
S_AXI
axi_uartlite_0
S_AXI
UART
clk_wiz_1
clk_in1
reset
pwm_axi_0
S00_pwm_AXI
rst_clk_wiz_1_100M
ext_reset_in
Select an interface pin on the left panel to view its options
All Automation 선택 후 OK
OK
Cancel

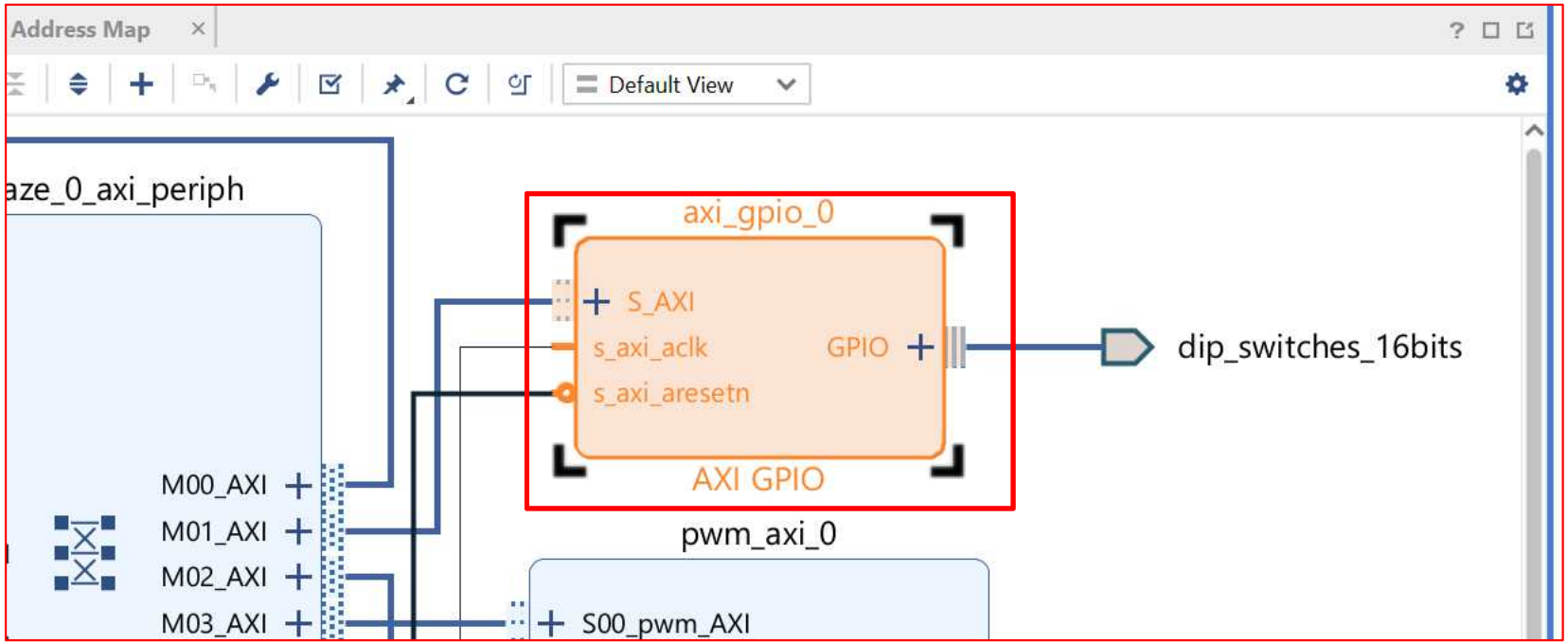
➢ AXI Connect ➔ PWM Control



- ***AXI Connect ➔ PWM Control***

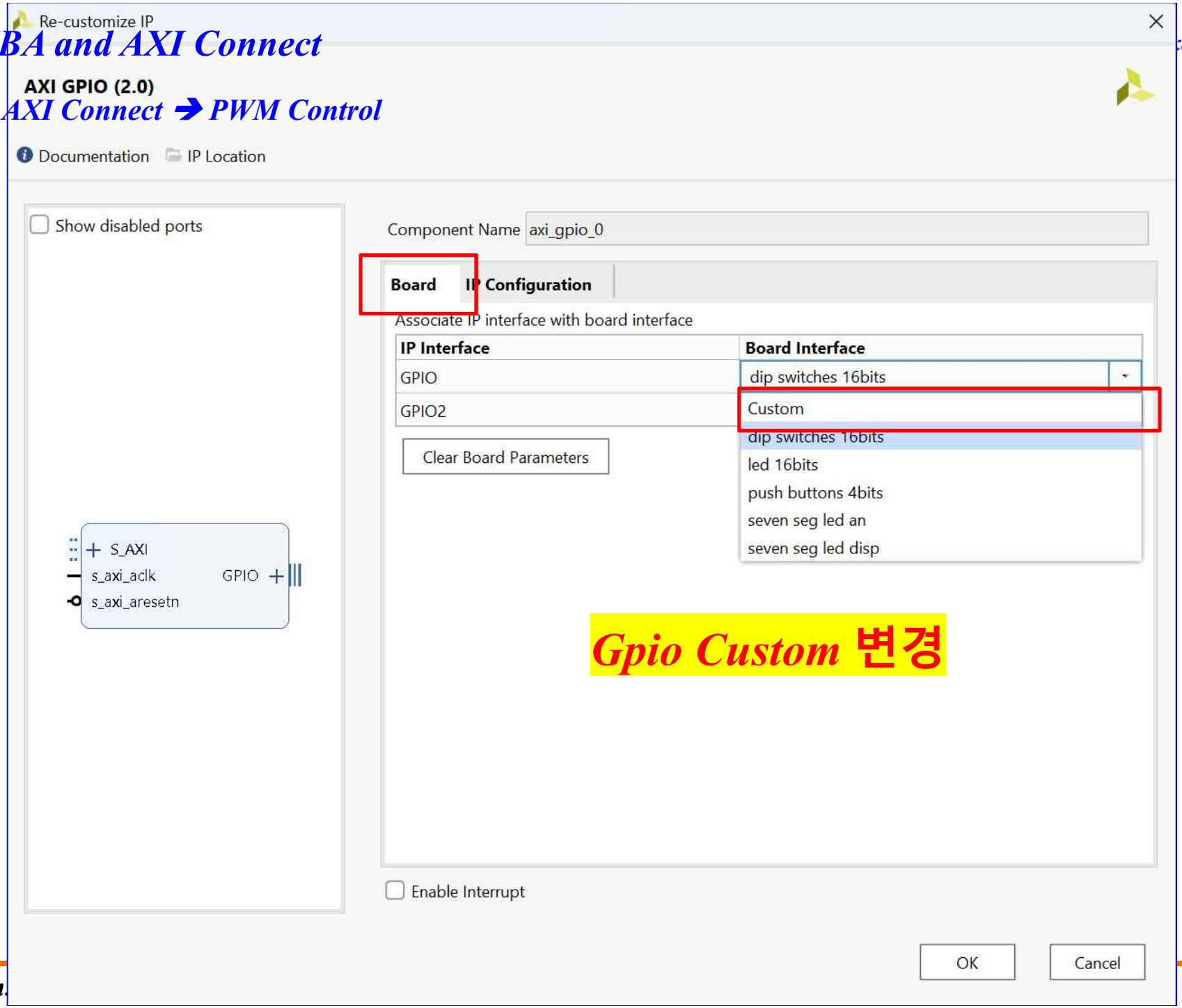


- ***AXI Connect ➔ PWM Control***

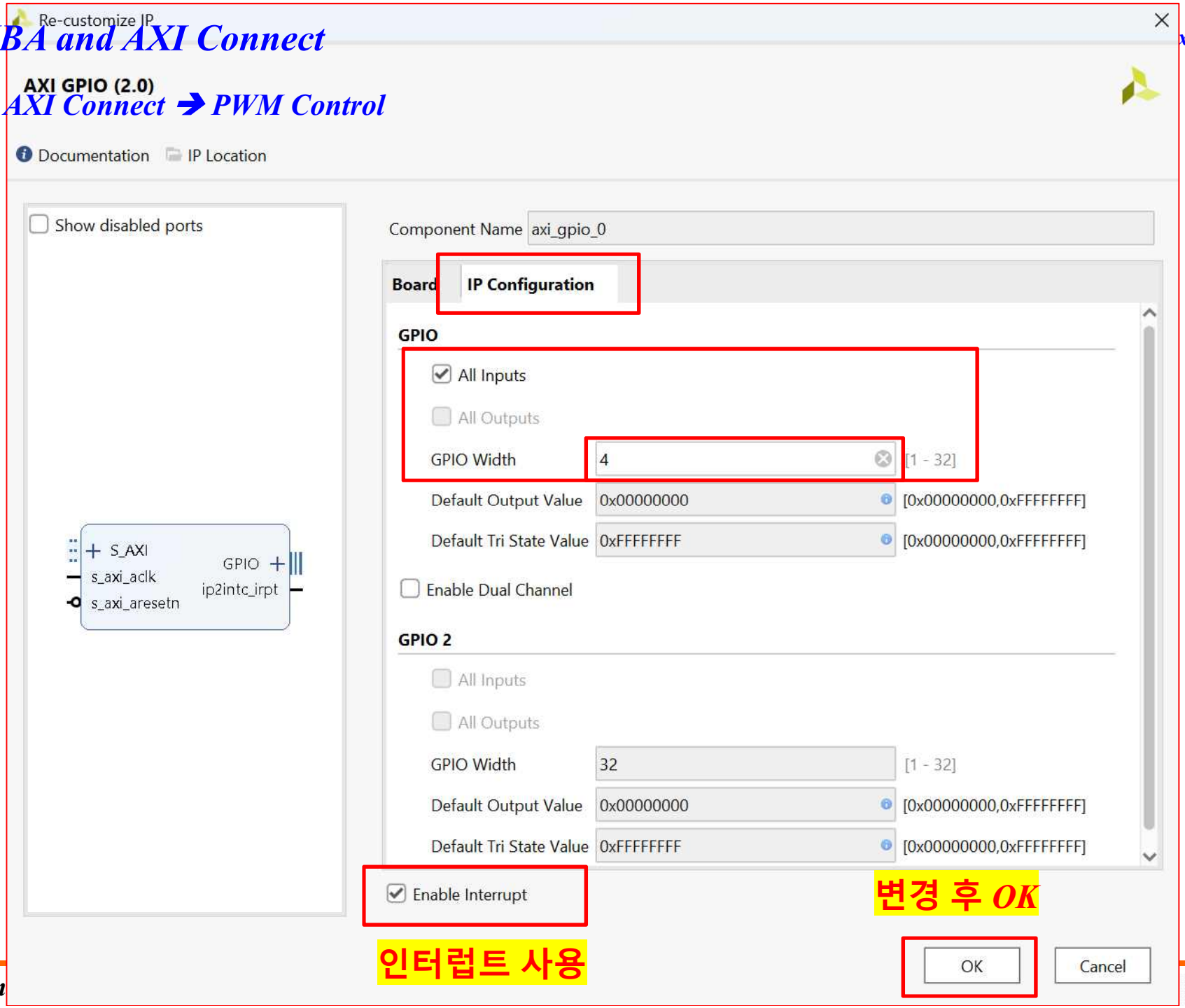


Gpio 더블클릭 후 수정

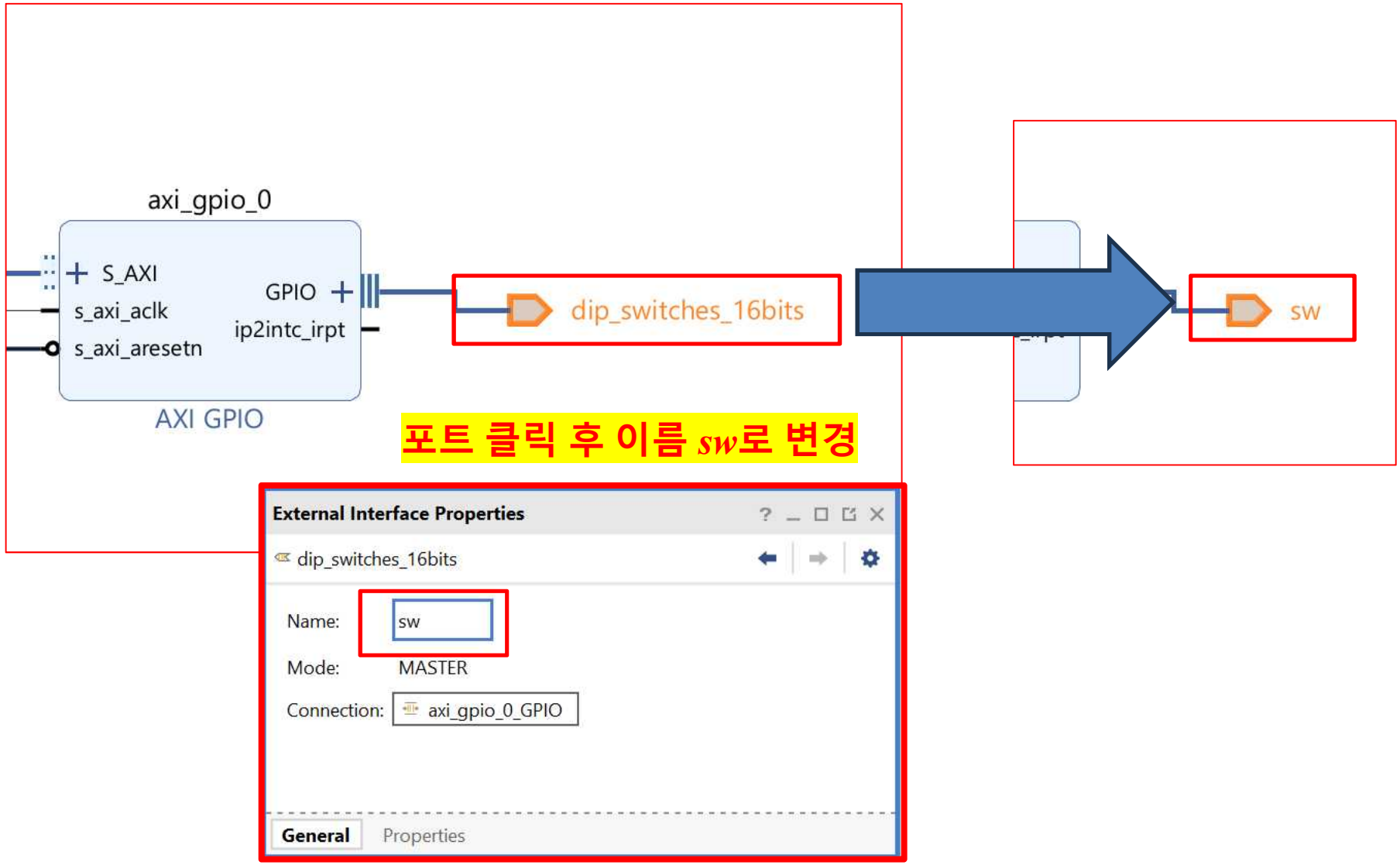
- AXI Connect ➔ PWM Control



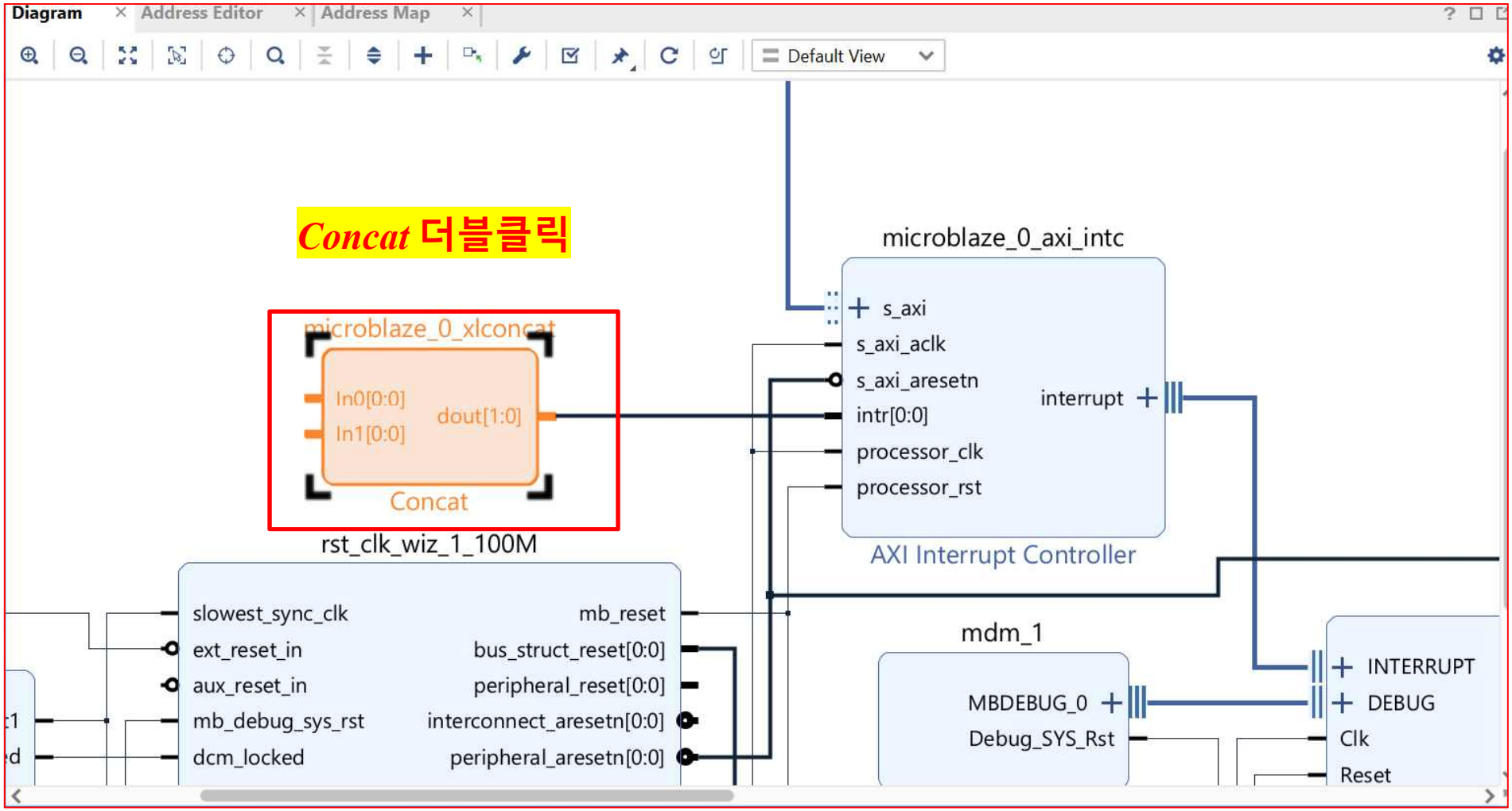
- AXI Connect ➔ PWM Control



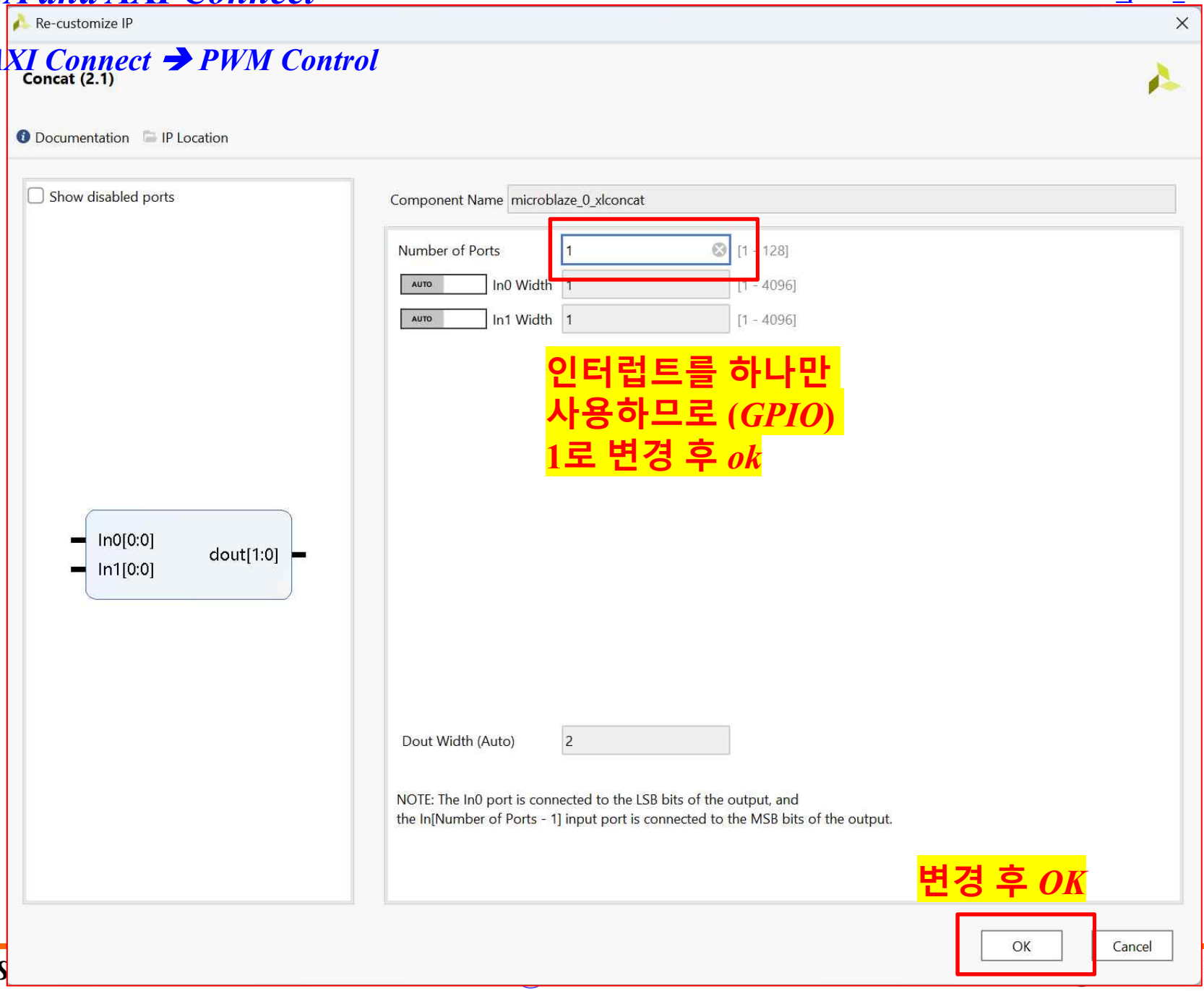
➢ AXI Connect ➔ PWM Control



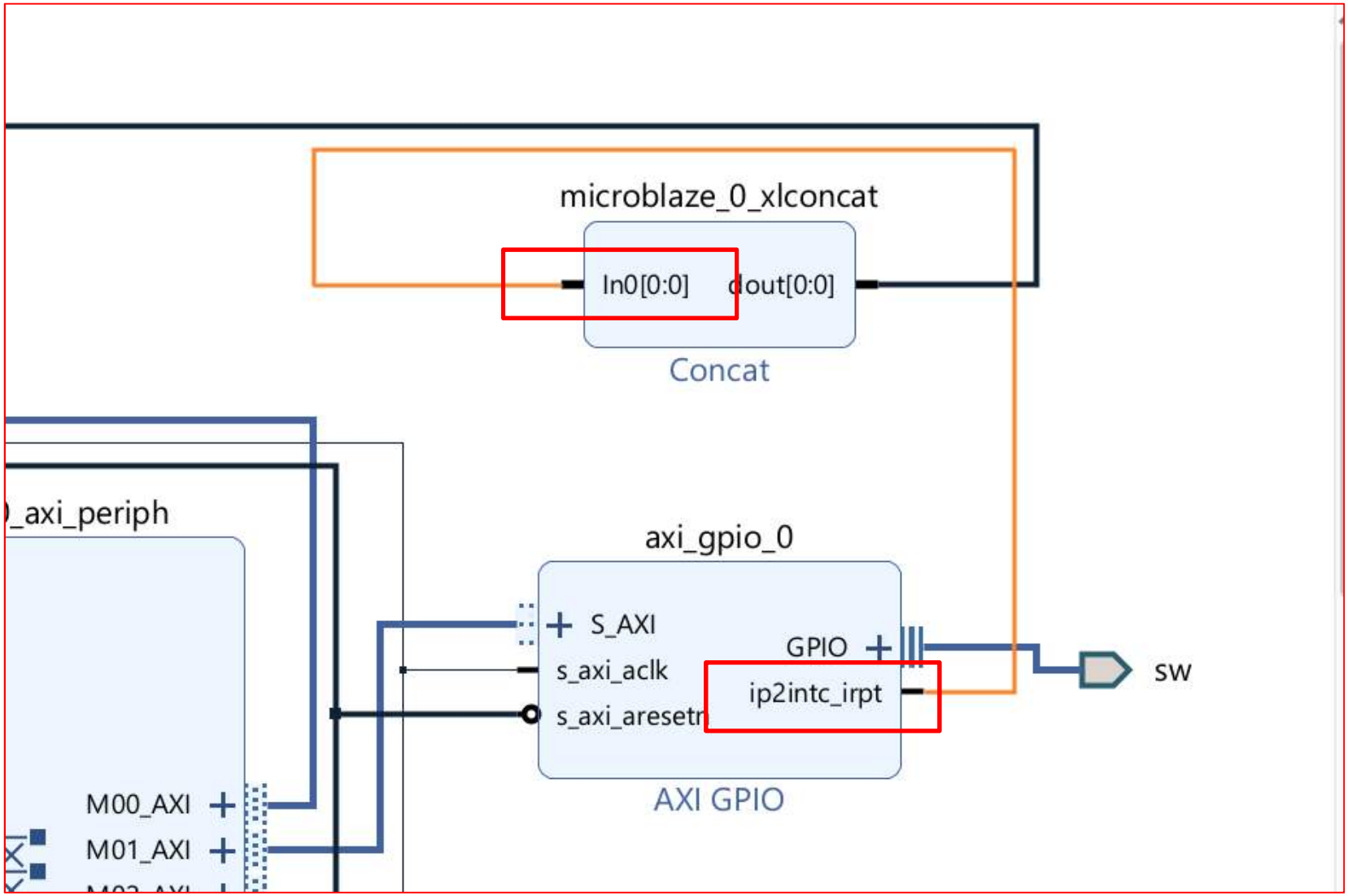
➢ AXI Connect ➔ PWM Control



- AXI Connect ➔ PWM Control

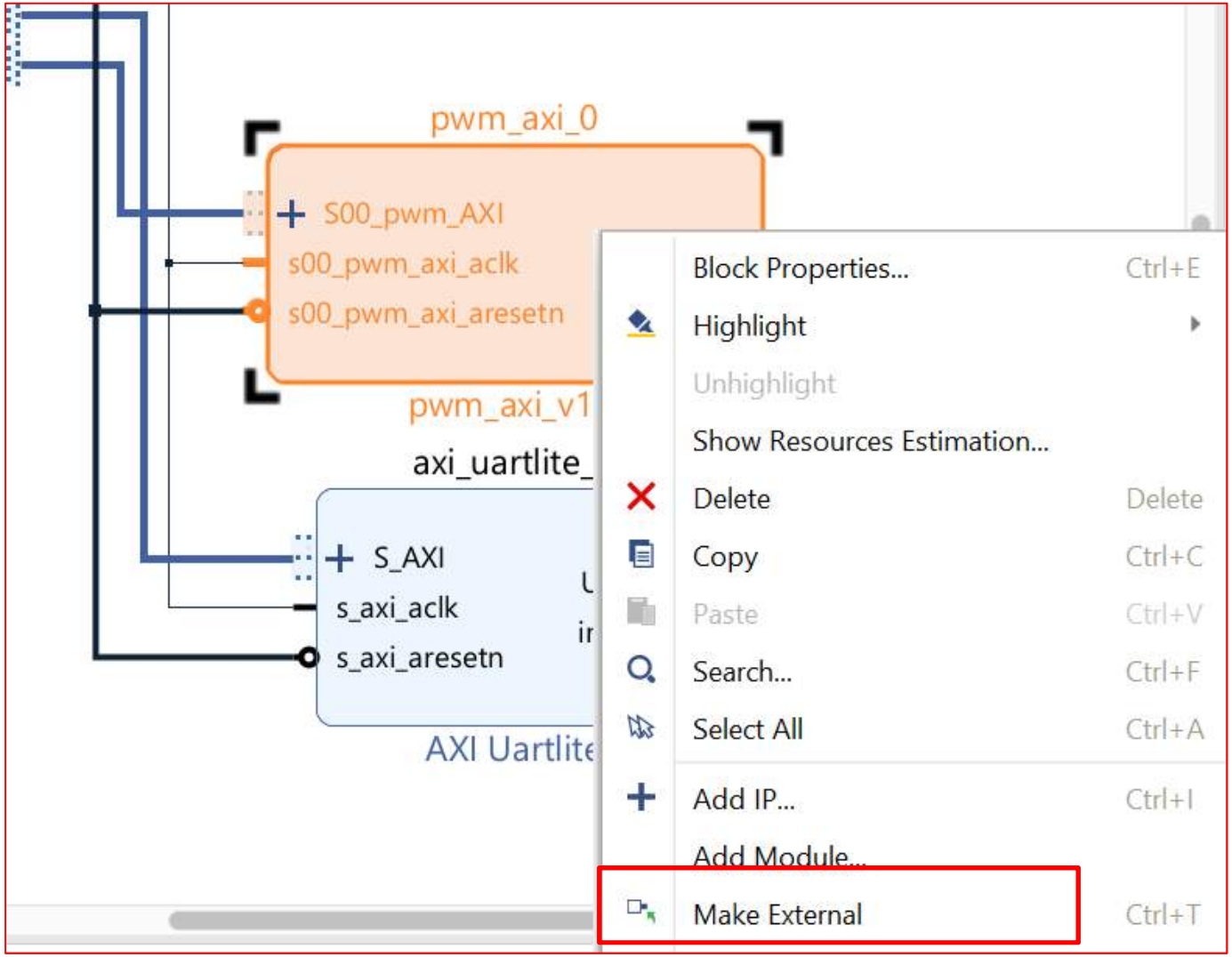


- AXI Connect ➔ PWM Control



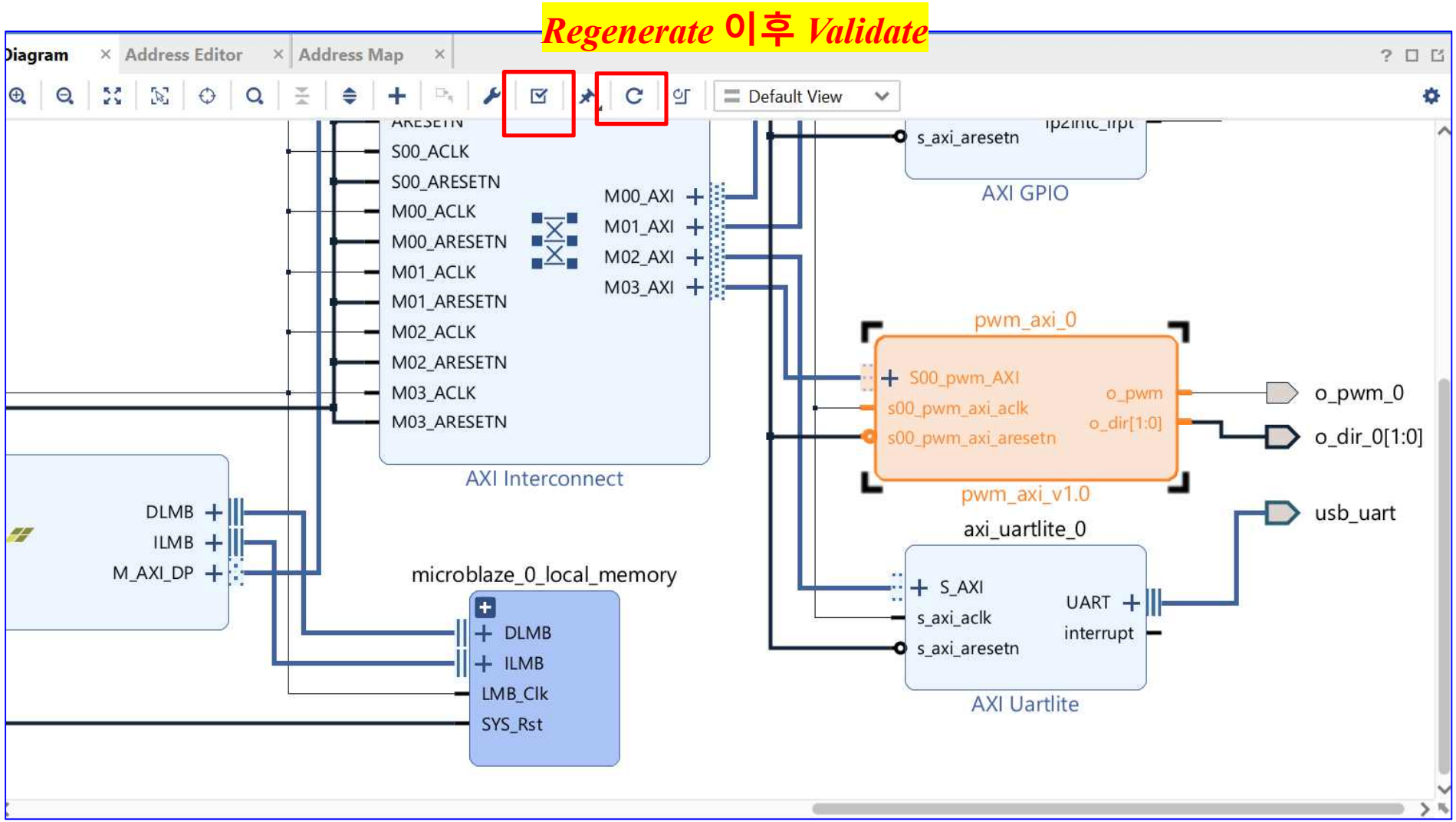
Concat과 GPIO의 인터럽트 연결

- AXI Connect ➔ PWM Control

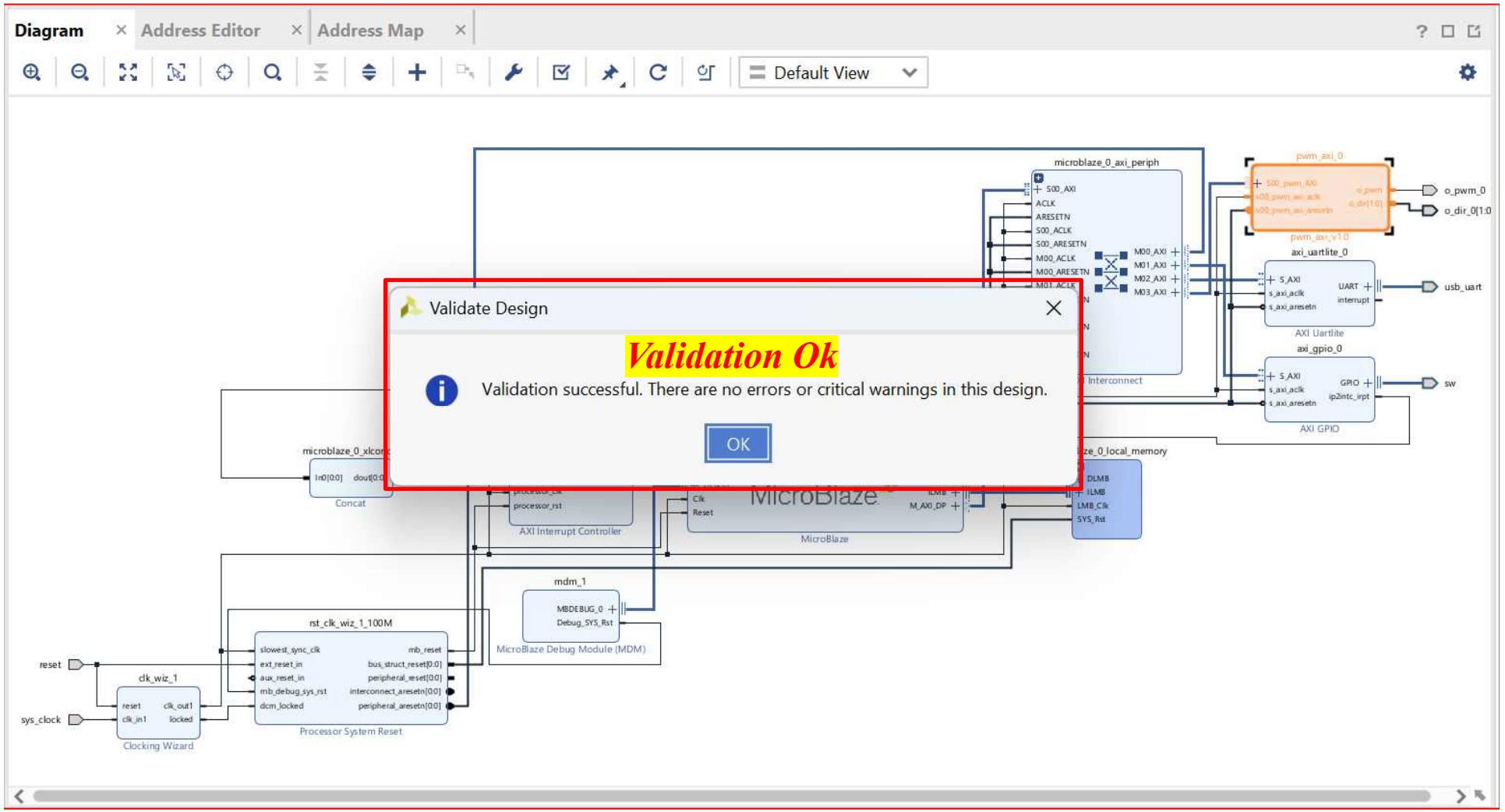


PWM 우클릭 후 Make External

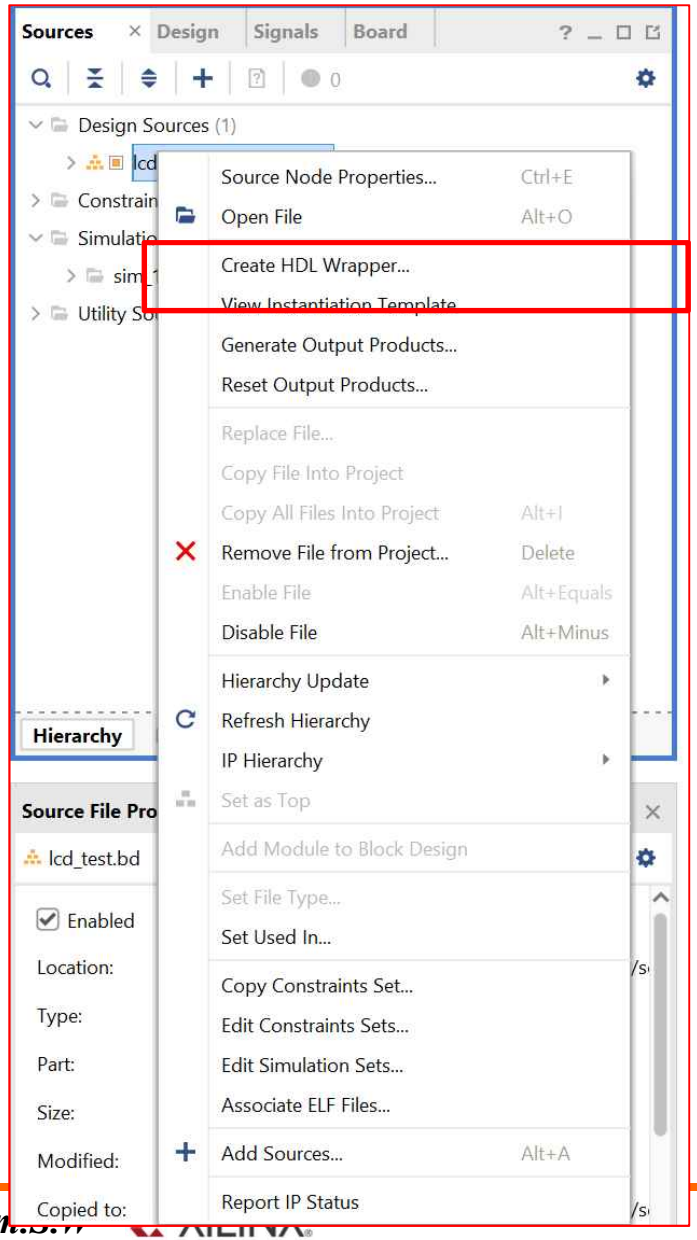
➢ AXI Connect ➔ PWM Control

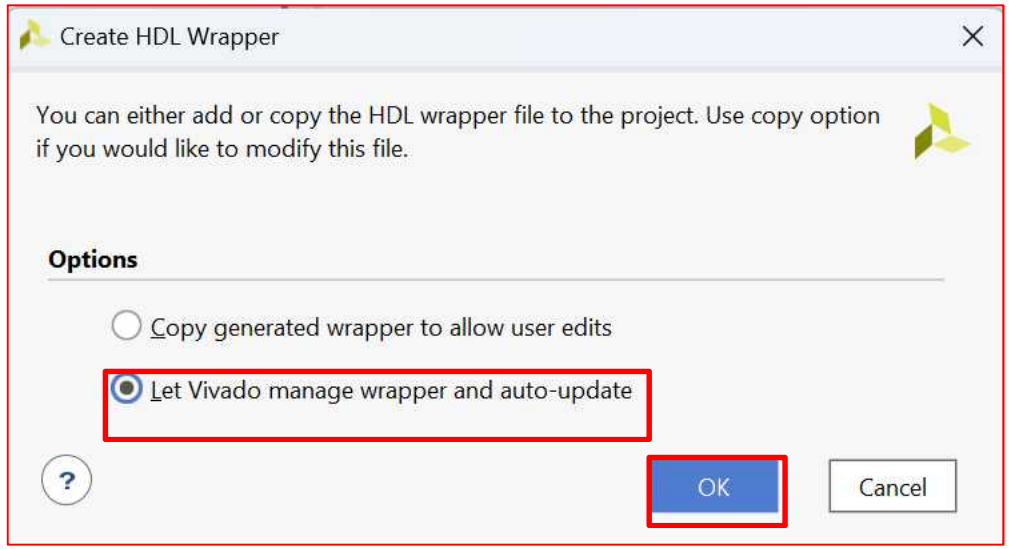


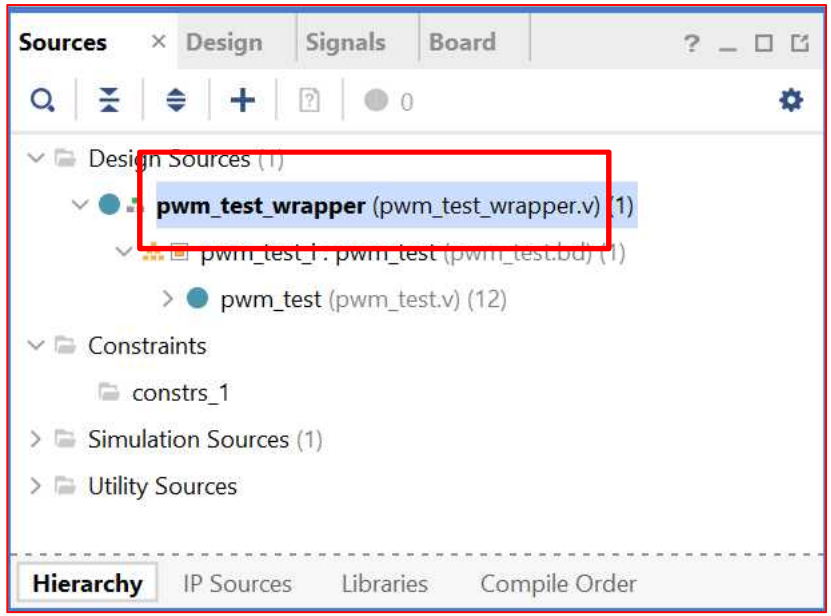
➢ AXI Connect ➔ PWM Control



➢ AXI Connect ➔ PWM Control

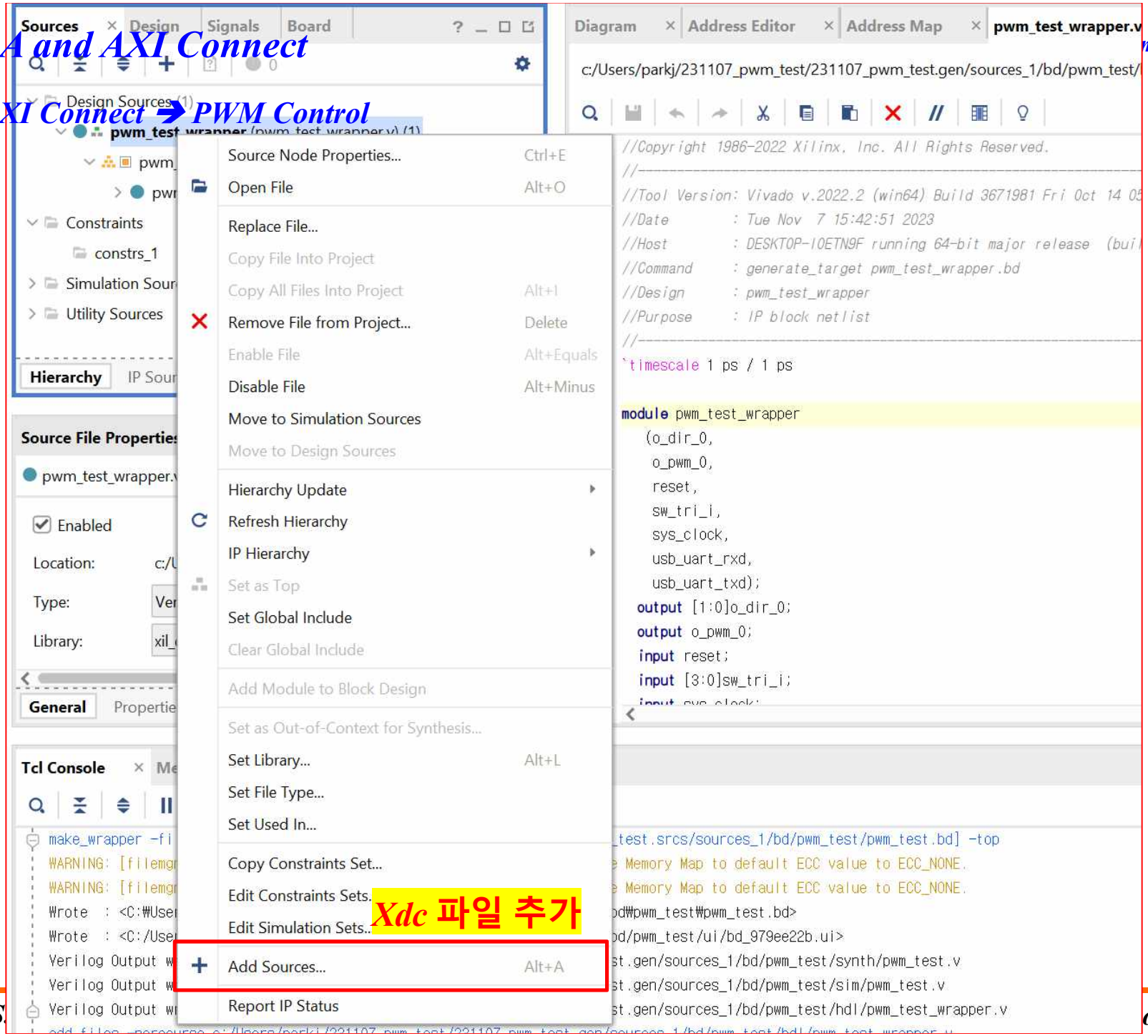






AMBA and AXI Connect

➢ AXI Connect ➔ PWM Control



- AXI Connect ➔ PWM Control

Basys-3-Master.xdc

axi_pwm_exam.xpr

```
1.  ## This file is a general .xdc for the Basys3 rev B board
2.  ## To use it in a project:
3.  ## - uncomment the lines corresponding to used pins
4.  ## - rename the used ports (in each line, after get_ports) according to the top level signal names in the
    project

5.  ## Clock signal
6.  set_property -dict { PACKAGE_PIN W5   IOSTANDARD LVCMOS33 } [get_ports sys_clock]
7.  create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports sys_clock]

8.  ## Switches
9.  set_property -dict { PACKAGE_PIN V17   IOSTANDARD LVCMOS33 } [get_ports {sw_tri_i[0]}]
10. set_property -dict { PACKAGE_PIN R2    IOSTANDARD LVCMOS33 } [get_ports {reset}]

11. ##Buttons
12. set_property -dict { PACKAGE_PIN U18   IOSTANDARD LVCMOS33 } [get_ports {sw_tri_i[3]}]
13. set_property -dict { PACKAGE_PIN W19   IOSTANDARD LVCMOS33 } [get_ports {sw_tri_i[2]}]
14. set_property -dict { PACKAGE_PIN T17   IOSTANDARD LVCMOS33 } [get_ports {sw_tri_i[1]}]

15. ##Pmod Header JC
16. set_property -dict { PACKAGE_PIN K17   IOSTANDARD LVCMOS33 } [get_ports {o_dir_0[0]}];#Sch
    name = JC1
17. set_property -dict { PACKAGE_PIN M18   IOSTANDARD LVCMOS33 } [get_ports {o_dir_0[1]}];#Sch
    name = JC2
18. set_property -dict { PACKAGE_PIN N17   IOSTANDARD LVCMOS33 } [get_ports {o_pwm_0}];#Sch
    name = JC3

19. ##USB-RS232 Interface
20. set_property -dict { PACKAGE_PIN B18   IOSTANDARD LVCMOS33 } [get_ports usb_uart_rxd]
21. set_property -dict { PACKAGE_PIN A18   IOSTANDARD LVCMOS33 } [get_ports usb_uart_txd]

22. ## Configuration options, can be used for all designs
23. set_property CONFIG_VOLTAGE 3.3 [current_design]
24. set_property CFGBVS VCCO [current_design]

25. ## SPI configuration mode options for QSPI boot, can be used for all designs
26. set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]
27. set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]
28. set_property CONFIG_MODE SPIx4 [current_design]
```

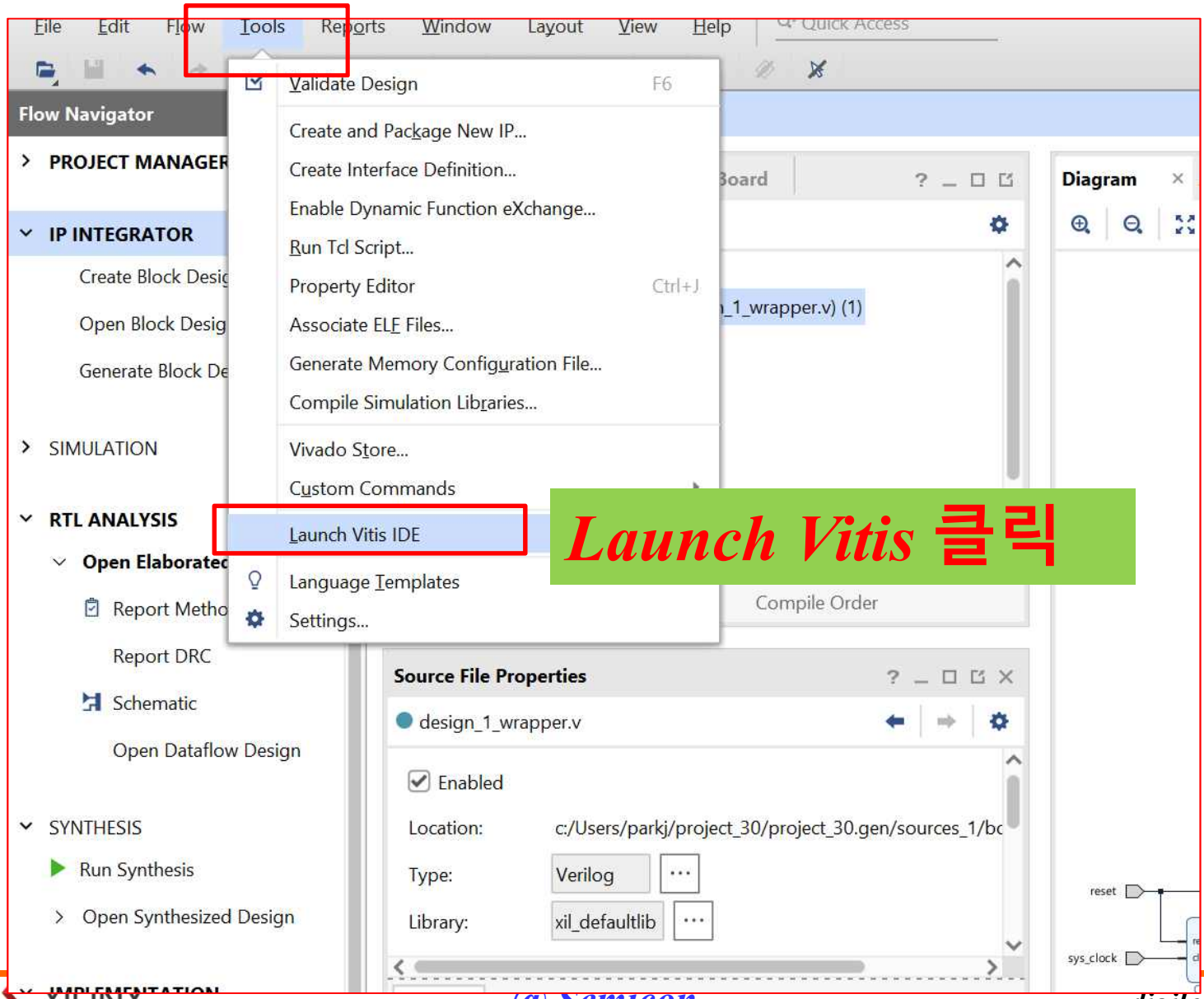
➢ AXI Connect ➔ PWM Control

- Create Block Design
- Blaze Uart & MyIP 추가 및 연결
- sys_clock & reset 설정
- MyIP Make External
- XDC Constraints 설정
- Bitstream
- Export Hardwar

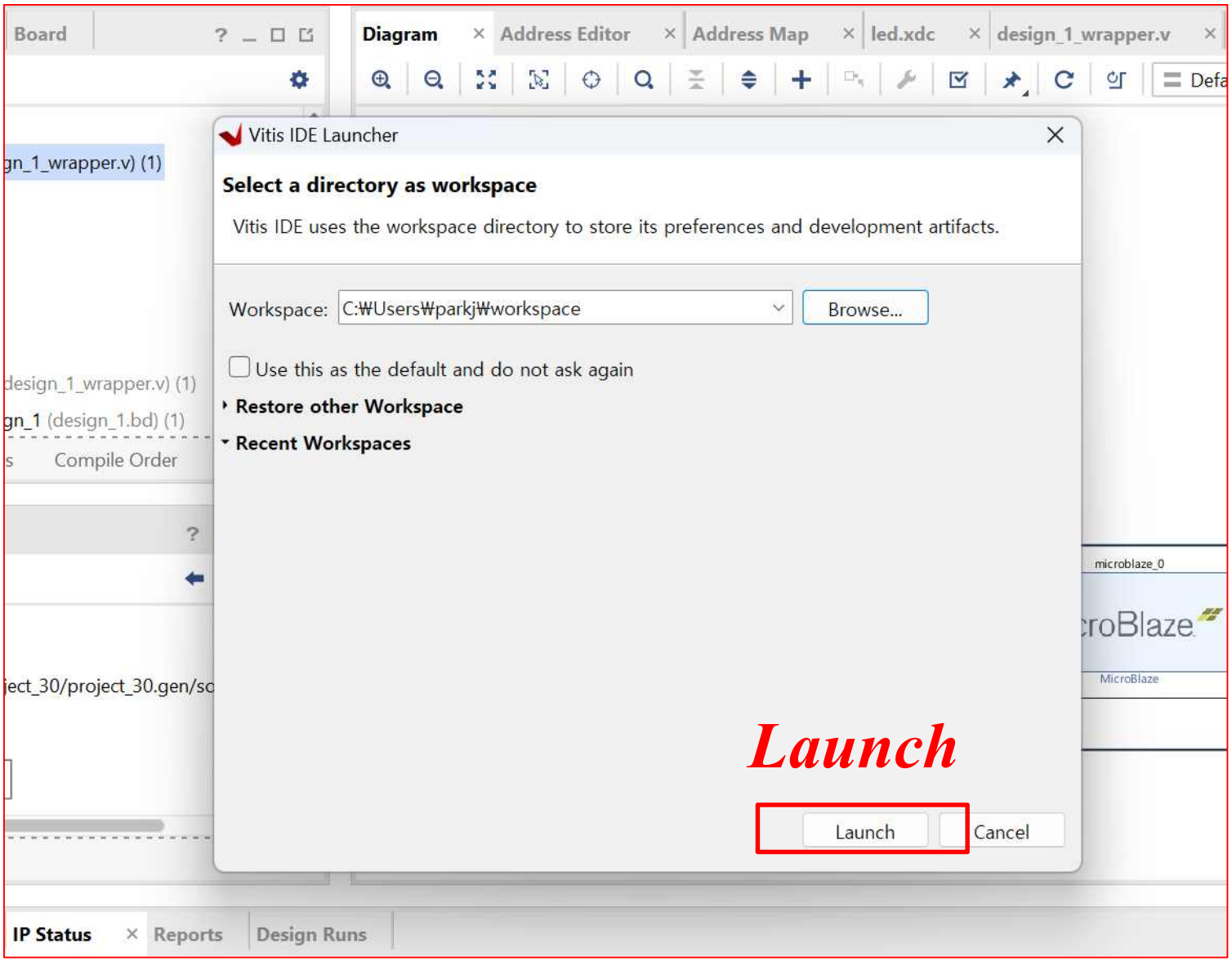
AMBA and AXI Connect

✓ ***Vitis Application SW***

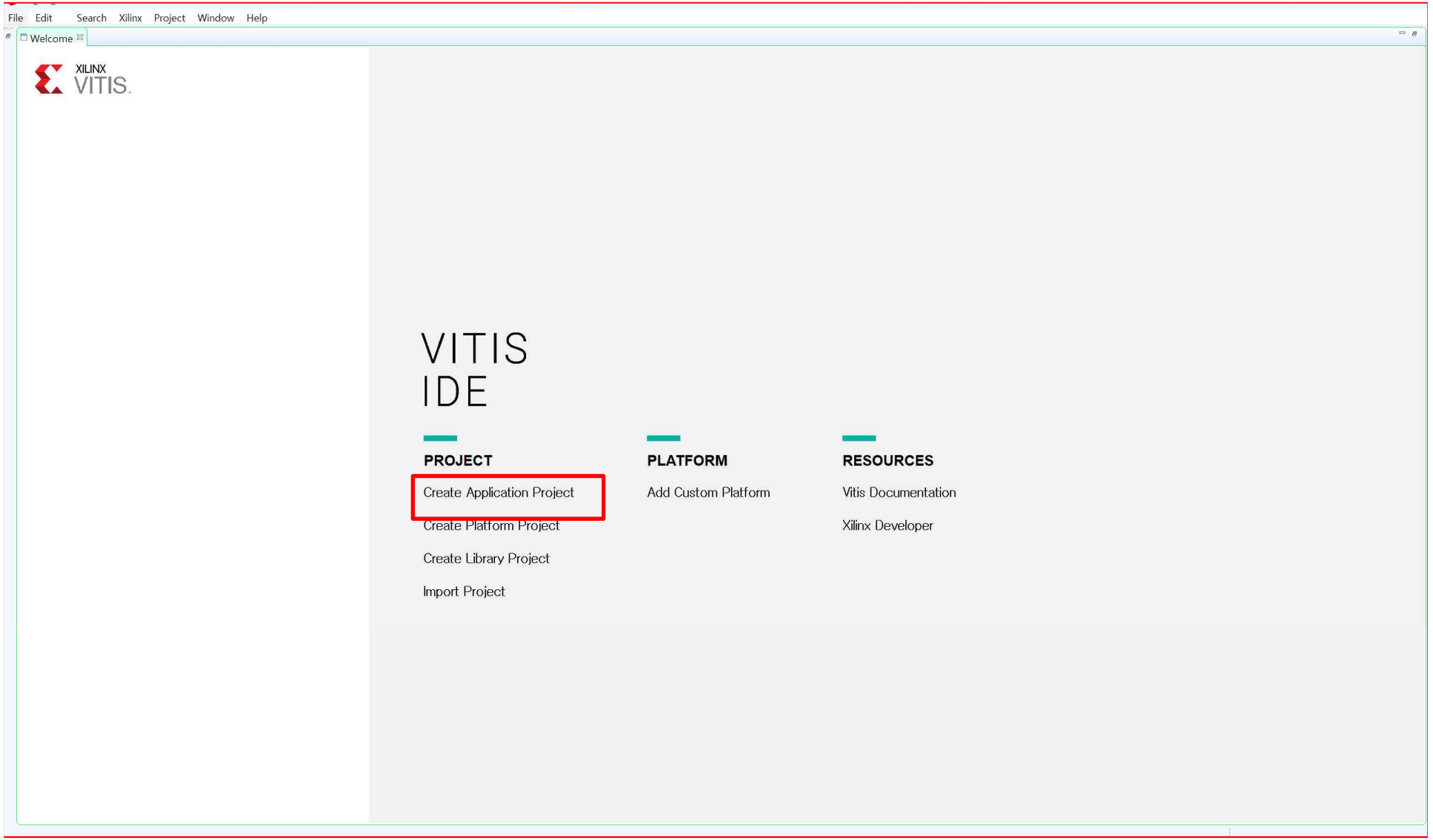
➢ AXI Connect ➔ PWM Control



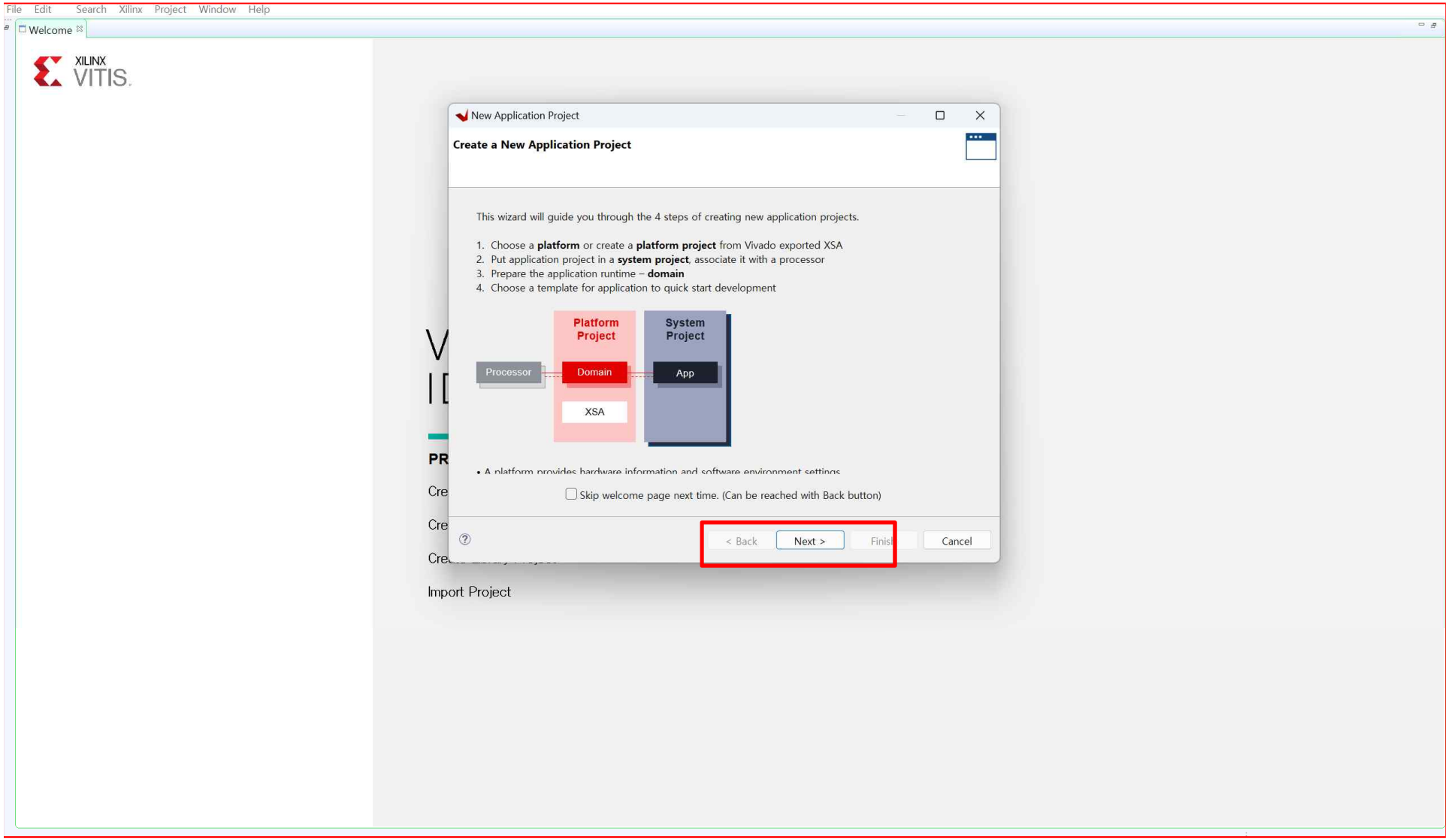
➢ AXI Connect ➔ PWM Control



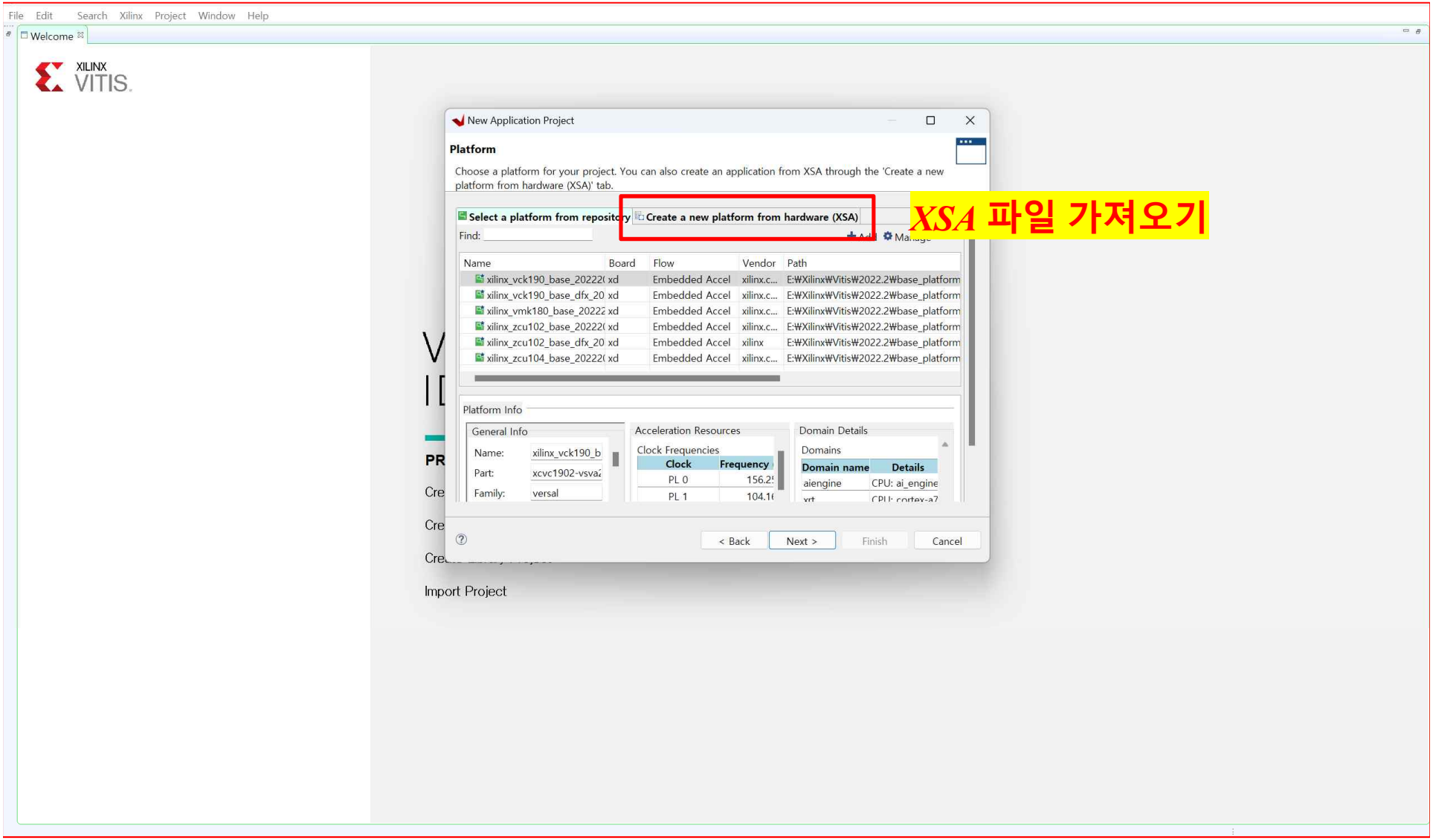
➢ AXI Connect ➔ PWM Control



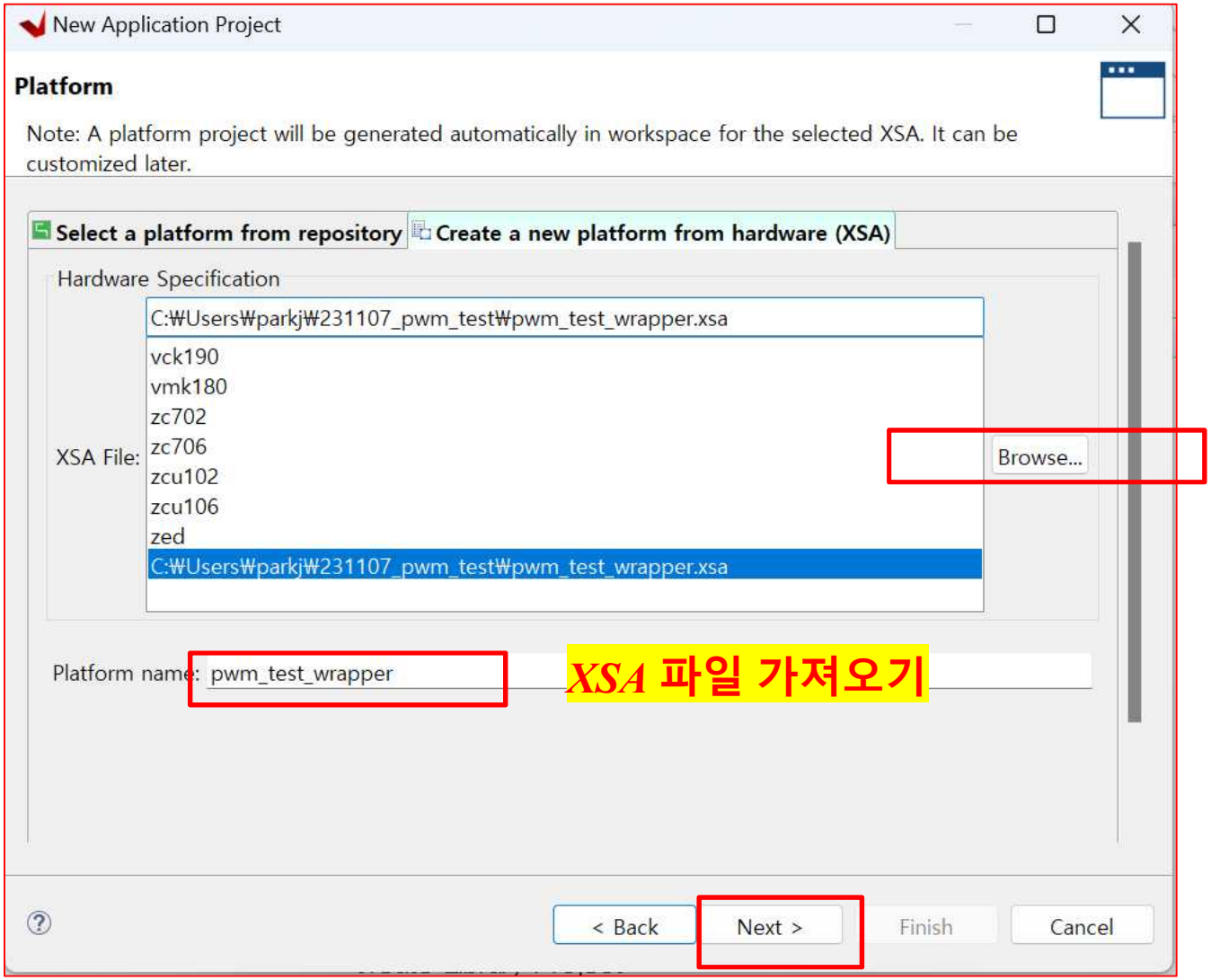
➢ AXI Connect ➔ PWM Control



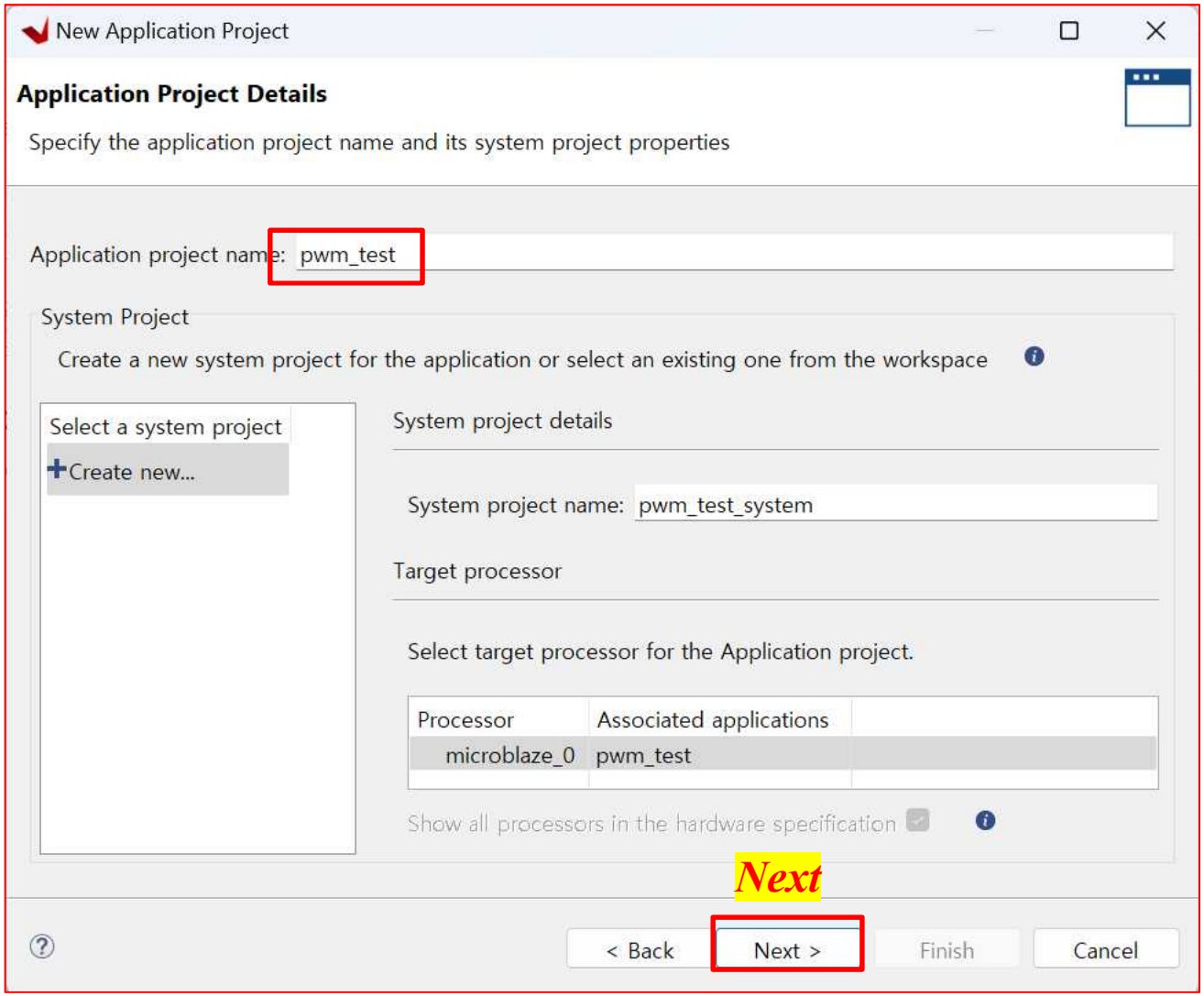
➢ AXI Connect ➔ PWM Control



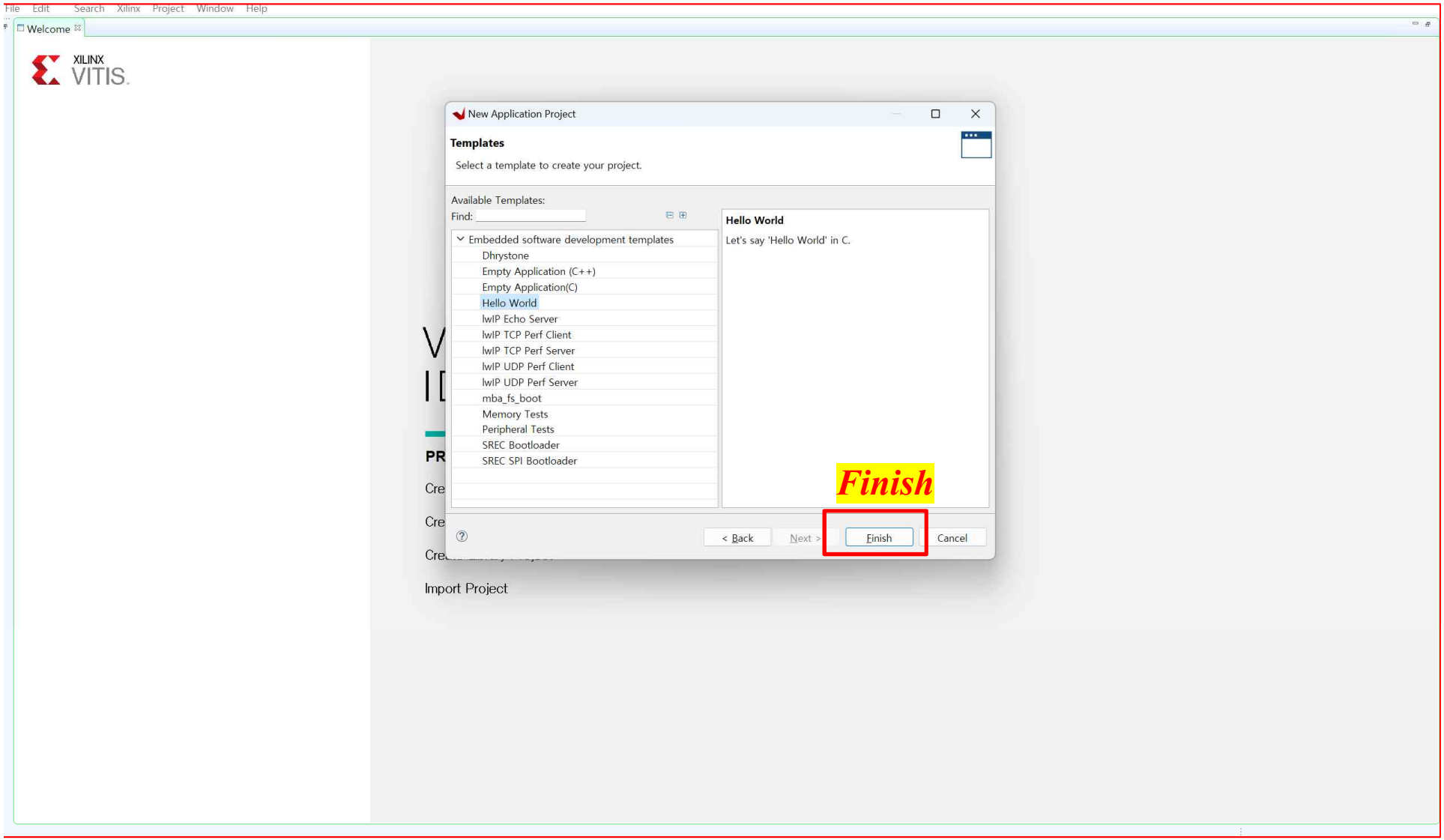
➢ AXI Connect ➔ PWM Control



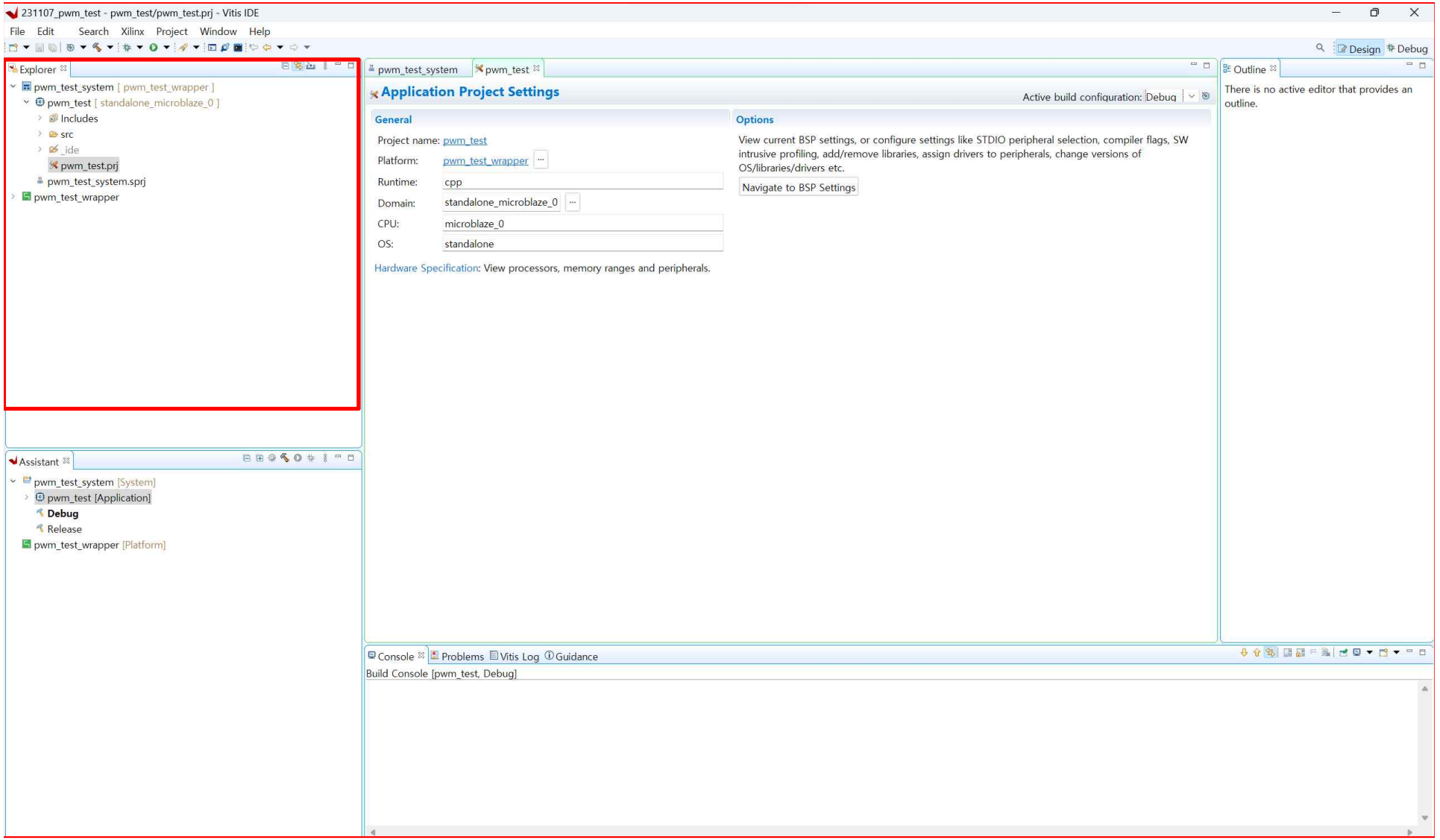
➢ AXI Connect ➔ PWM Control



➢ AXI Connect ➔ PWM Control

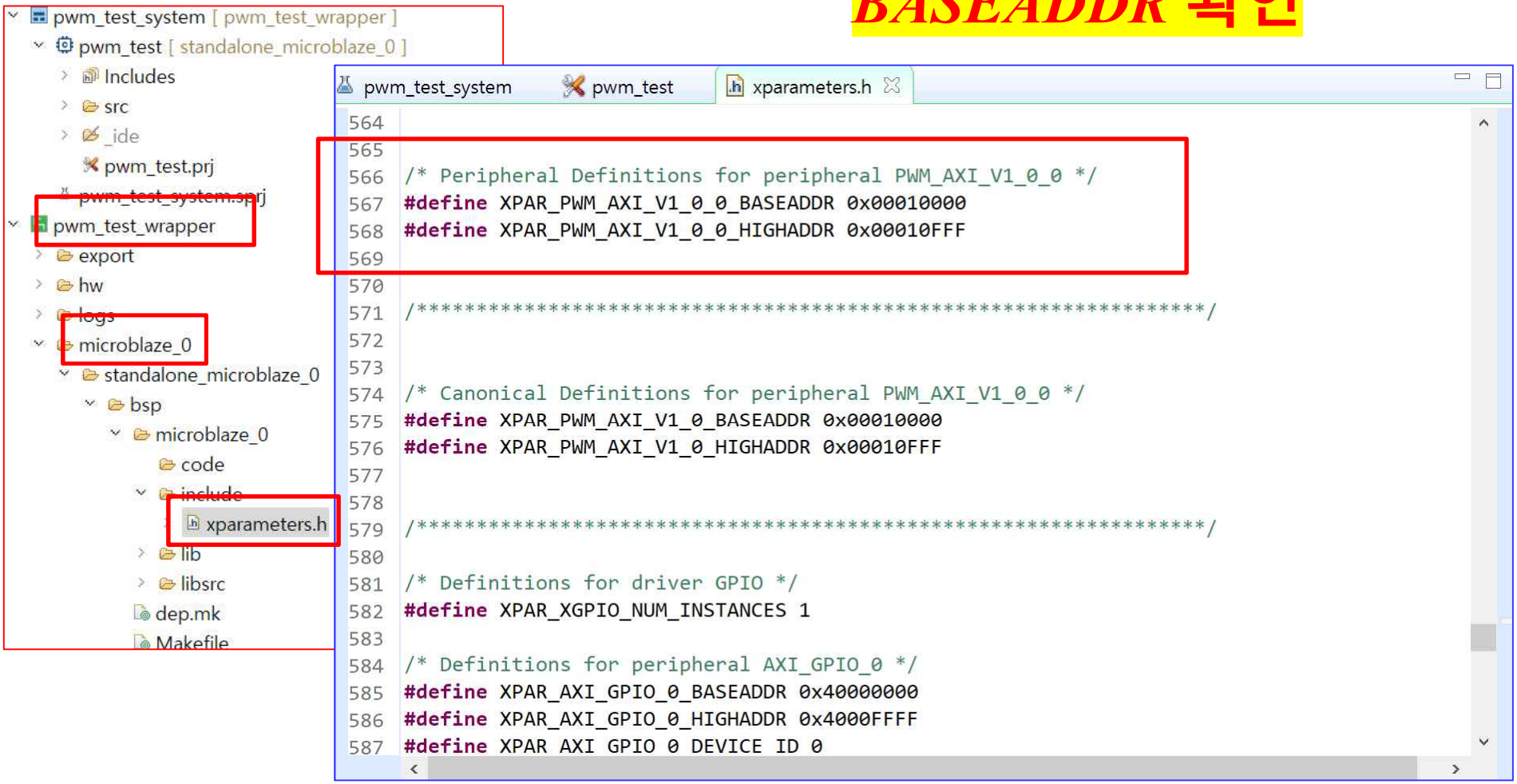


➢ AXI Connect ➔ PWM Control

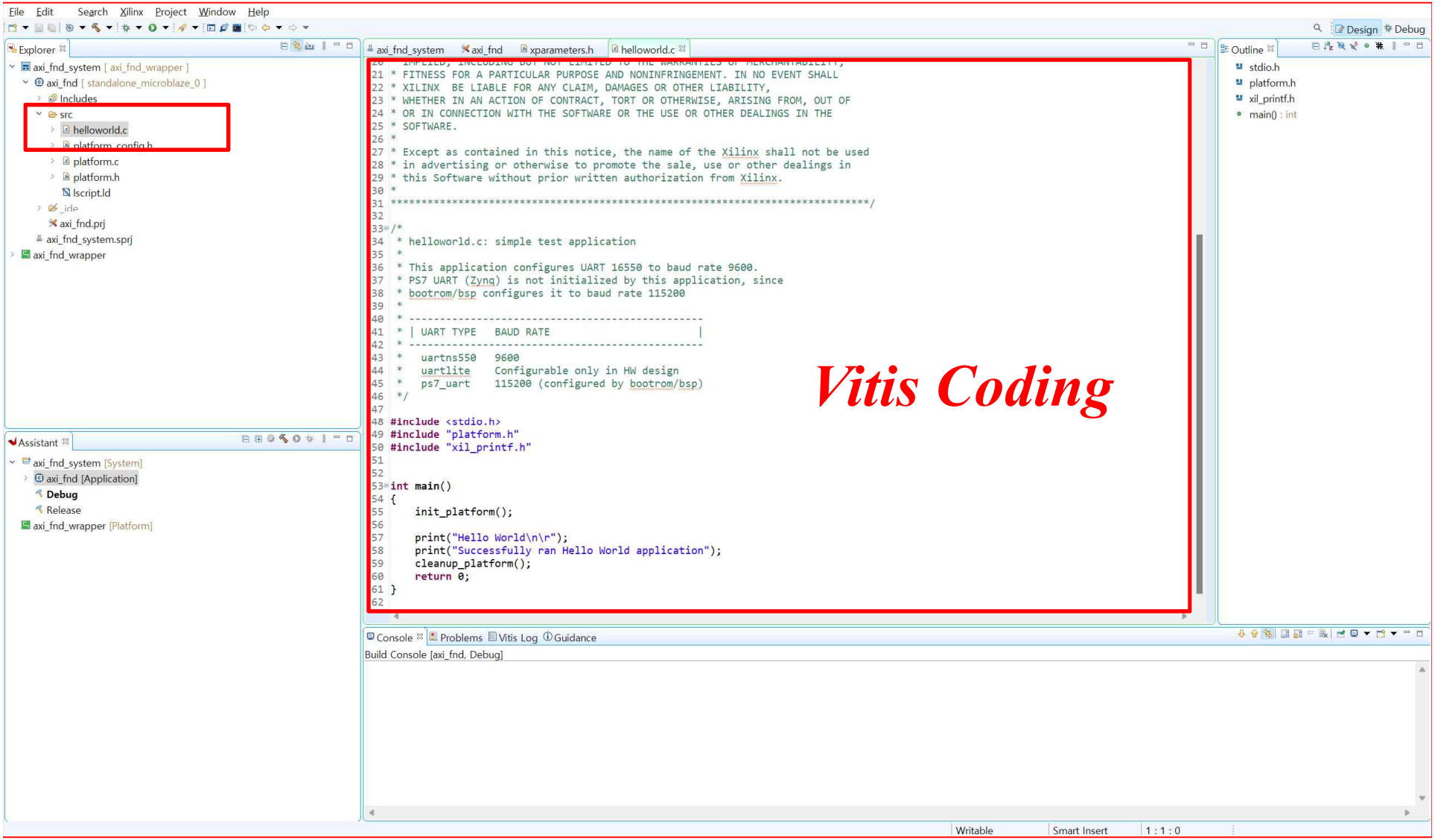


➢ AXI Connect ➔ PWM Control

BASEADDR 확인



➢ AXI Connect ➔ PWM Control



➢ AXI Connect ➔ PWM Control

Helloworld.c (1)

```
#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"
#include "xparameters.h"
#include "xgpio.h"
#include "sleep.h"
#include "xil_exception.h"
#include "xintc.h"

#define AXI_PWM0_BASEADDR   0x44A00000
#define PWM0_enable   AXI_PWM0_BASEADDR + 0x00
#define PWM0_dir   AXI_PWM0_BASEADDR + 0x04
#define PWM0_ocr   AXI_PWM0_BASEADDR + 0x08

/* definitions for btn0, AXI_GPIO_1 */
#define GPIO_BTN0_DEVICE_ID    XPAR_GPIO_0_DEVICE_ID
#define BTN0_CHANNEL1        1
#define BTN0_INT_MSK         XGPIO_IR_CH1_MASK /* Channel 1
Interrupt Mask */

/* definitions for Interrupt */
#define INTC_DEVICE_ID        XPAR_INTC_0_DEVICE_ID
#define INTC_BTN0_INT_VEC_ID   XPAR_INTC_0_GPIO_0_VEC_ID

XGpio    GPIO_BTN;       /* The instance of the btn */
XIntc Intc;
```

주소 확인

PWM IP 참고

```
assign i_en = slv_reg0;
assign i_dir = slv_reg1;
assign i_ocr = slv_reg2;
// User logic ends
```

Helloworld.c (2)

```
void Gpio1Handler(void *CallBackRef);

void interrupt_init(void)
{

   XIntc_Initialize(&Intc, INTC_DEVICE_ID);
   XIntc_Connect(&Intc, INTC_BTN0_INT_VEC_ID,
(XInterruptHandler)Gpio1Handler, &GPIO_BTN);

   XIntc_Start(&Intc, XIN_REAL_MODE);
   XIntc_Enable(&Intc, INTC_BTN0_INT_VEC_ID);
}
void exception_init(void)
{
   Xil_ExceptionInit();
   Xil_ExceptionRegisterHandler(XIL_EXCEPTION_ID_INT,
(Xil_ExceptionHandler)XIntc_InterruptHandler, &Intc);
   Xil_ExceptionEnable();
}
void gpio_init(void)
{
  XGpio_Initialize(&GPIO_BTN,  GPIO_BTN0_DEVICE_ID);

  XGpio_InterruptEnable(&GPIO_BTN, BTN0_CHANNEL1);

  XGpio_InterruptGlobalEnable(&GPIO_BTN);
}
```

➢ AXI Connect ➔ PWM Control

Helloworld.c (3)

```
u8 direction=0b01;
u8 duty=0;
u8 RunFlag=0;

void Gpio1Handler(void *CallbackRef)
{
  XGpio *GpioPtr = (XGpio *)CallbackRef;



      if ((XGpio_DiscreteRead(&GPIO_BTN, BTN0_CHANNEL1) & (1<<1)))
                {

        if(duty>=80)
        duty=80;
        else
        duty=duty+20;
                  Xil_Out32(PWM0_ocr,duty);
        usleep(300000);
                }

        ///
        if ((XGpio_DiscreteRead(&GPIO_BTN, BTN0_CHANNEL1) & (1<<2)))
                {
        if(duty==0)
        duty=0;
        else
        duty=duty-20;
         Xil_Out32(PWM0_ocr,duty);
             usleep(300000);
                  }
```

- ❖ *u8 direction* : 모터의 방향 제어를 위한 변수
- ❖ *u8 duty* : 모터의 방향 제어를 위한 변수
- ❖ *u8 RunFlag* : 모터의 방향 제어를 위한 변수

- ❖ *GPIO SW1*: 듀티비 증가
- ✓ 스위치를 누를 때마다 듀티비가 20씩 증가
- ✓ 듀티비가 80이 되면 스위치를 눌러도 듀티비가 증가하지 않고 80으로 유지

- ❖ *GPIO SW2*: 듀티비 감소
- ✓ 스위치를 누를 때마다 듀티비가 20씩 감소
- ✓ 듀티비가 0이 되면 스위치를 눌러도 듀티비가 감소하지 않고 0으로 유지

➢ AXI Connect ➔ PWM Control

Helloworld.c (4)

```
  if ((XGpio_DiscreteRead(&GPIO_BTN,
BTN0_CHANNEL1) & (1<<3))){

  if(RunFlag==1)
  {
  RunFlag=0;
  Xil_Out32(PWM0_enable,0);
  Xil_Out32(PWM0_ocr,0);
  usleep(300000);
  }

  else if (RunFlag==0)
  {
  RunFlag=1;
      Xil_Out32(PWM0_enable,1);
      Xil_Out32(PWM0_ocr,duty);
      usleep(300000);
  }

  }


  XGpio_InterruptClear(GpioPtr, BTN0_CHANNEL1);
/* Clear the Interrupt */
}
```

❖ GPIO SW3: RUN / STOP

✓ RUN Flag가 1일때 (모터 동작상태)

➔ 모터를 정지시키고, RUN Flag=0

✓ RUN Flag가 0일때 (모터 정지상태)

➔ 모터를 동작시키고, RUN Flag=1

➢ AXI Connect ➔ PWM Control

Helloworld.c (5)

```
int main()
{

   init_platform();
   interrupt_init();
   gpio_init();
   exception_init();
   print("Hello World\n\r");
   print("PWM RUN \n\r");
   cleanup_platform();


   while(1){

     if ((XGpio_DiscreteRead(&GPIO_BTN, BTN0_CHANNEL1) & (1<<0)))
            Xil_Out32(PWM0_dir,0b10);
       else

            Xil_Out32(PWM0_dir,0b01);

     usleep(300000);


   }
   return 0;
}
```

❖ **Main 함수**

✓ 앞에서 정의한 init 함수들 실행.

❖ **GPIO SW0: 정/역회전 제어**

✓ SW0 토글시 모터드라이버 *IN1*,*IN2*에 신호 10을 인가

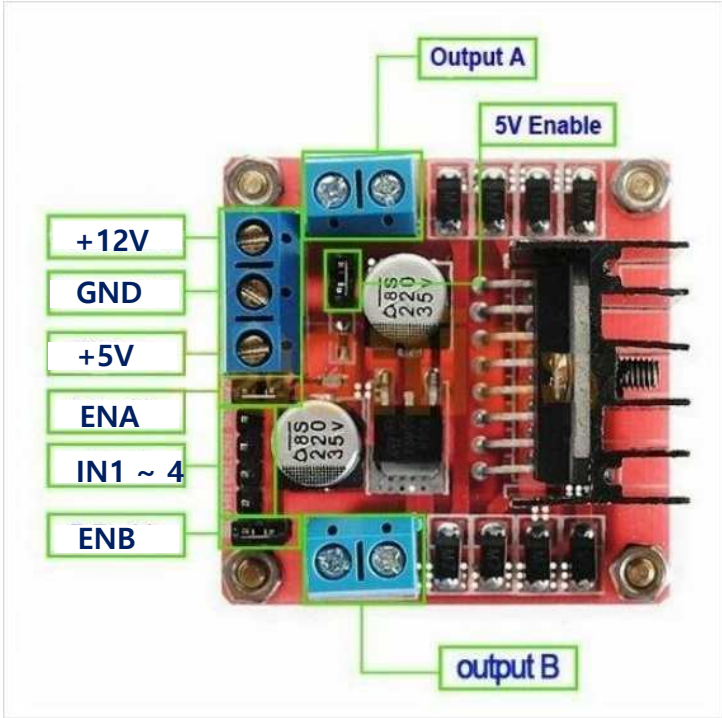
✓ 기본 상태에서는 모터드라이버 *IN1*,*IN2*에 신호 01을 인가

➢ 모터 구동을 위한 모듈 : *L298N*(모터 드라이버)[1/2]

- *L298N* 기능 : 1. DC, 스텝 모터의 방향 및 속도 제어
  2. H-브리지 구조: 모터를 양방향으로 제어가능. 즉, 모터의 회전방향을 제어하거나 정지
  3. 두개의 모터를 독립적으로 제어. 두개의 출력 핀 제공(Output A, Output B)
  4. 주로 PWM신호를 사용하여 모터 속도를 조절

- *L298N* 핀 구성 :

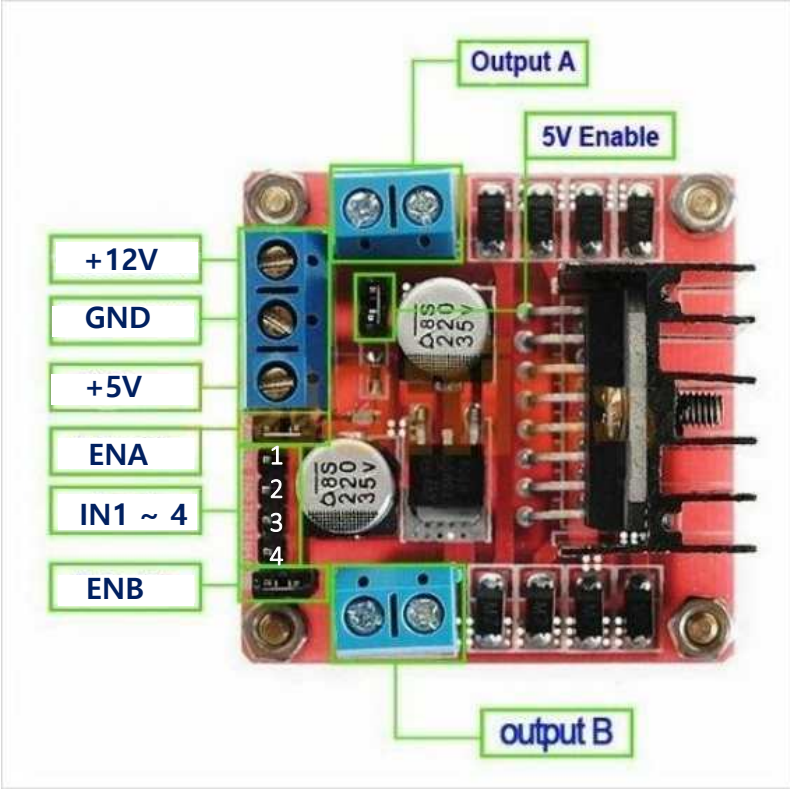
1. *IN1, IN2* : 첫 번째 모터의 회전 방향 제어
2. *IN3, IN4* : 두 번째 모터의 회전 방향 제어
3. *OUTA* : 첫 번째 모터의 출력 핀(모터와 연결)
4. *OUTB* : 두 번째 모터의 출력 핀(모터와 연결)
5. *ENA* : 첫 번째 모터의 속도 제어를 위한 PWM 입력 핀
6. *ENB* : 두 번째 모터의 속도 제어를 위한 PWM 입력 핀
7. *+12V* : 모터 전원 입력(+5V ~ 35V)
8. *+5V* : 내부 논리 회로 전원 입력 (+5V)
9. *GND* : 공통 접지



➢ 모터 구동을 위한 모듈 : *L298N*(모터 드라이버)[2/2]

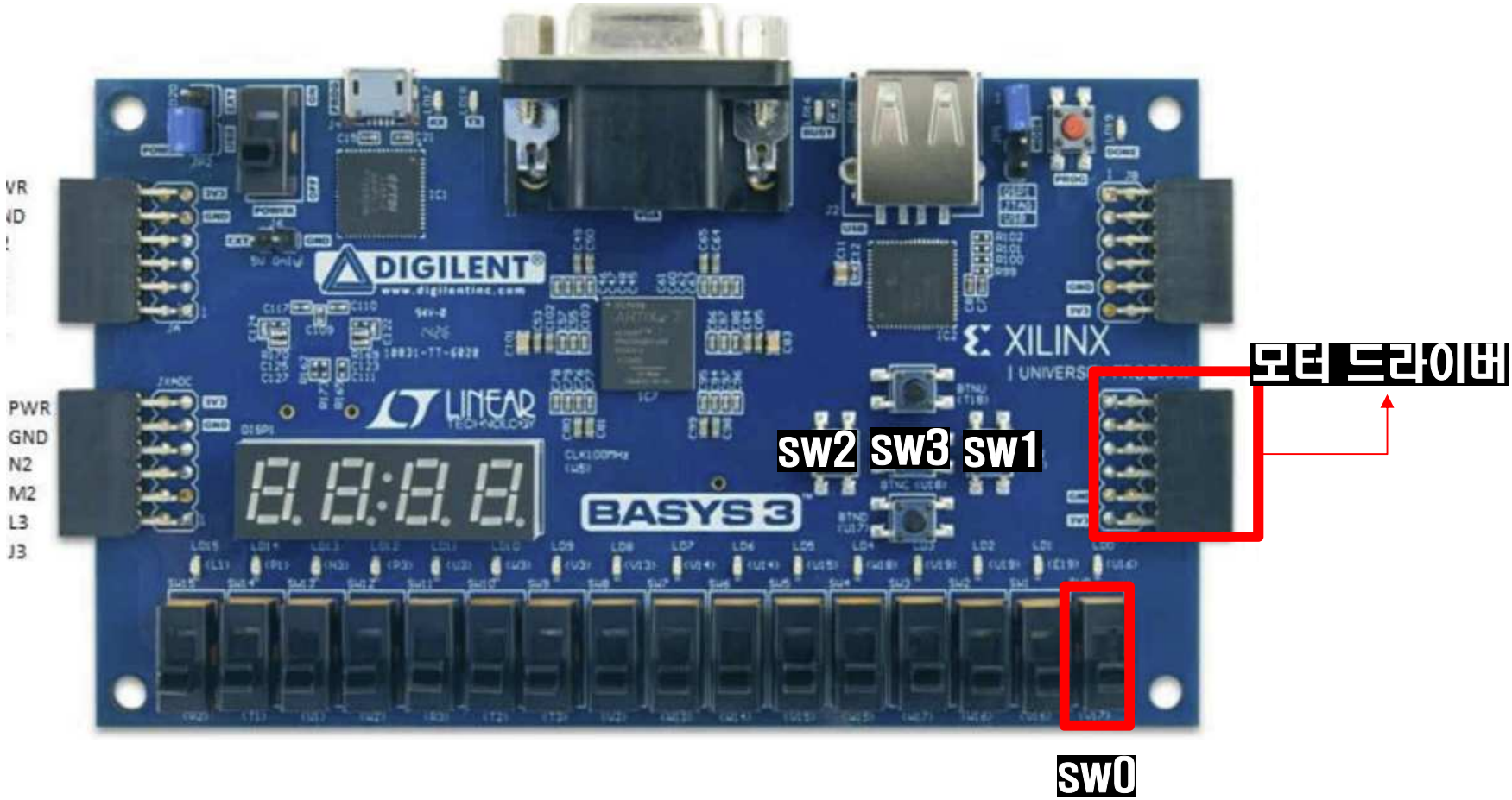
- 모터 구동을 위한 결선

1. ***IN1, IN2***: ***DC MOTER*** 결선(***DC MOTOR***방향제어)
2. ***OUTA***: ***DC MOTER*** 결선 (***MOTOR*** 출력 핀)
3. ***ENA*** : ***DC MOTER*** 결선(속도제어)
4. ***+12V*** : 외부전원 +5V인가
5. ***+5V*** : 결선X
6. ***GND*** : 외부전원과 ***BASYS3***을 공통 접지



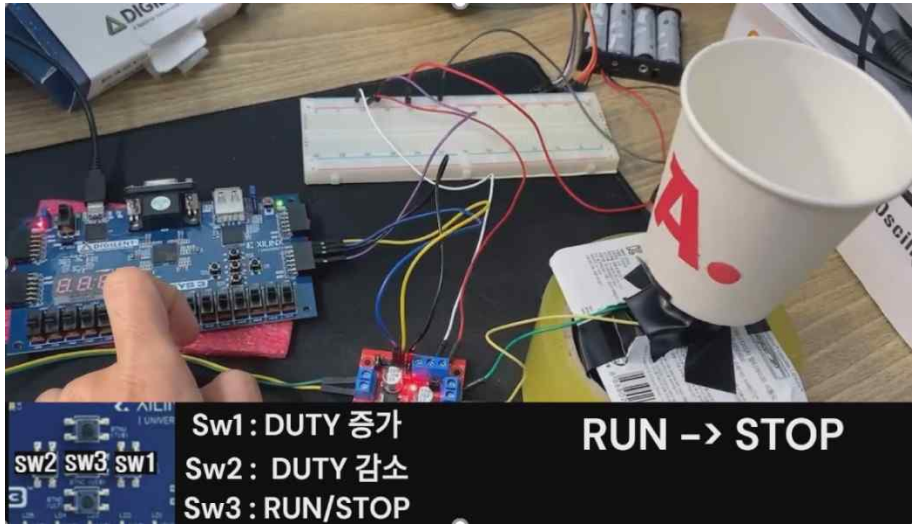
➢ Design Goal

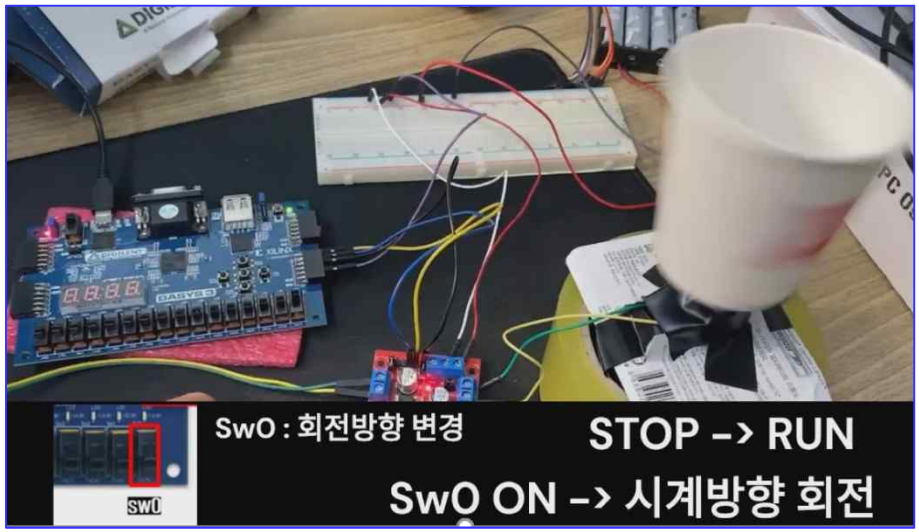
- GPIO를 활용한 PWM Control 실습

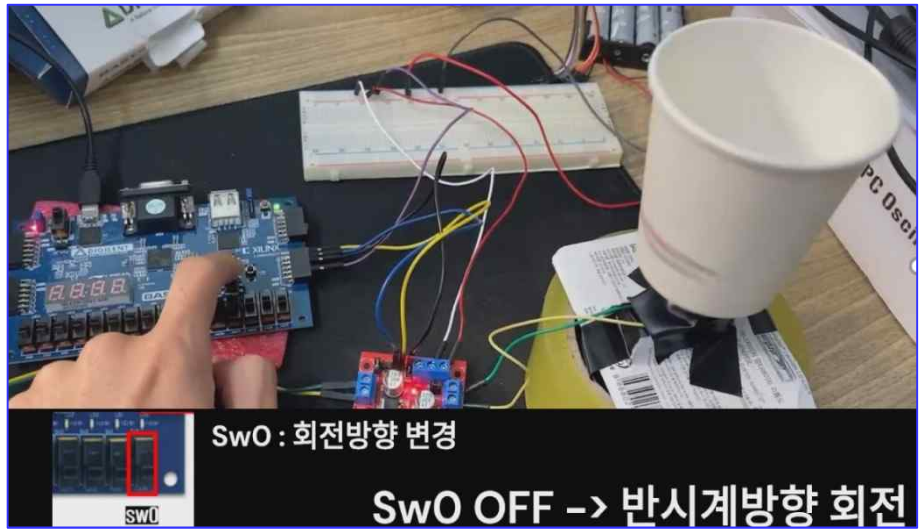


- SW0 : 기본상태에서 0b01, 토글일때 0b10
- SW1 : PWM 듀티비 증가 SW2 : PWM 듀티비 감소 SW3 : RUN / STOP

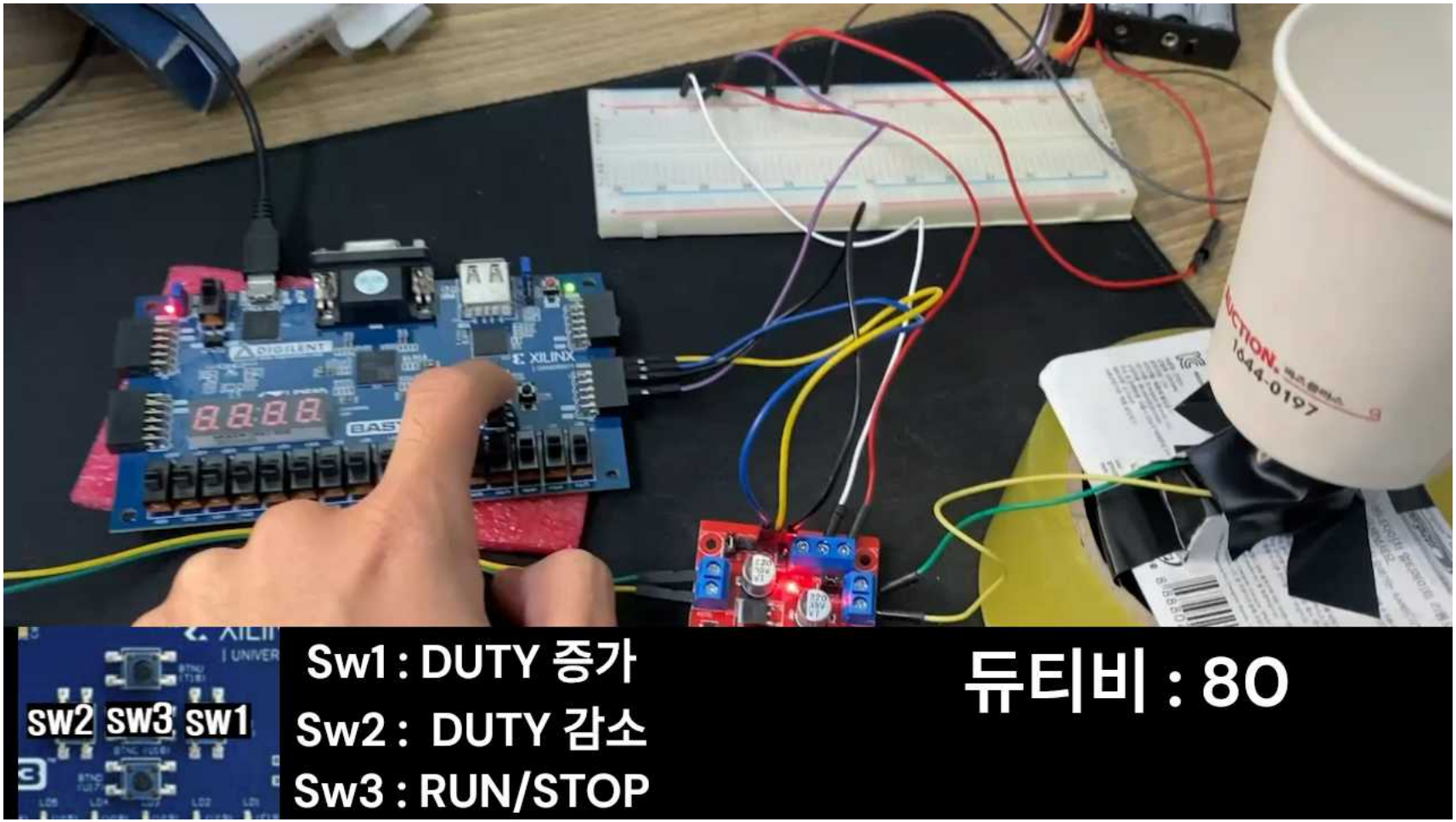
➢ *AXI Connect ➔ PWM Control ➔ Program Device and Implementation Result*

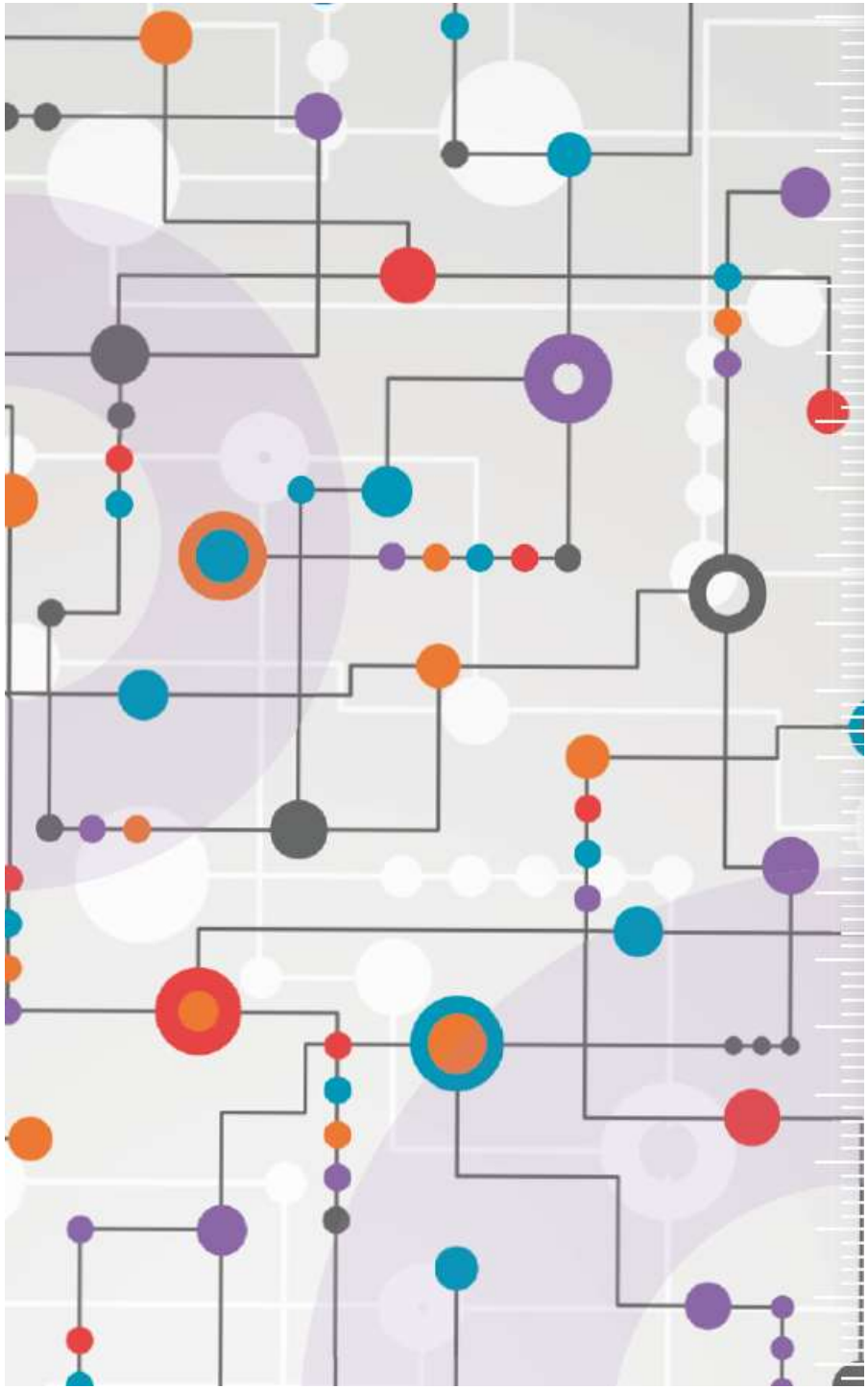






- *AXI Connect ➔ PWM Control ➔ Program Device and Implementation Result*





수고하셨습니다.